

Learning in Firm-to-Firm Trade*

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Abstract

We study how firms' productivity growth depends on the productivity of their suppliers and buyers and examine the aggregate implications of this form of learning. Using data on firm-to-firm transactions in Belgium, we document evidence consistent with learning from both suppliers and buyers, particularly from the most productive ones. To study the implications of this form of learning in general equilibrium, we develop a model in which firm productivity growth is shaped by the productivity of an endogenously formed network of suppliers and buyers, where firms bargain over the surpluses from trade and learning. Learning between suppliers and buyers increases bilateral trade volumes: suppliers charge lower markups when they learn more from buyers, and buyers have higher demand when they learn more from suppliers. We estimate a small open economy model with firms learning from both domestic and foreign trading partners. Quantitatively, we find that the value of learning is equivalent to 19 percent of aggregate consumption. Welfare gains from reductions in international trade participation costs are underestimated by over 7 percent without learning. Gains from trade are amplified when foreign partners are more productive through productivity catch-up and network upgrading, but are dampened when less productive foreign firms crowd out domestic learning and weaken incentives to form productive relationships.

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1 Introduction

Firms learn from one another by exchanging production-related ideas, a mechanism long recognized as a key driver of productivity growth. A growing body of evidence shows that such learning opportunities are not “in the air” but instead occur within networks of trading partners (Alfaro-Urena et al. 2022, Amiti et al. 2024, Fadeev 2023). The direction and intensity of learning thus depend critically on network structure. However, little is known about the magnitude of such learning and its implications in general equilibrium. How do learning and network structure shape aggregate productivity and welfare? What are the aggregate implications of supplier-buyer learning in endogenous networks?

Using detailed data on firm-to-firm transactions in Belgium, we find that firms’ productivity grows faster over time when they connect with productive suppliers and buyers. To establish this, we consider two complementary empirical approaches. First, we estimate regressions of firm productivity growth on exposure to the productivity of suppliers and buyers. Productivity growth is positively associated with the average productivity of a firm’s suppliers and buyers, with stronger correlations for exposure to relatively better partners in the productivity distribution. Second, we implement an event-study design, following Alfaro-Urena et al. (2022) and Amiti et al. (2024), to examine the productivity trajectories of firms that connect with suppliers or buyers in the top productivity quintile for the first time. We focus on the heterogeneous gains across firms with different pre-event productivity levels. The results show that initially less productive firms experience substantial productivity gains from connecting with top partners, both in the short and long run, while the estimates for initially more productive firms are muted. Taken together, the evidence supports the view that firms learn from both suppliers and buyers — more from the most productive partners than others.

Motivated by these findings, we develop a dynamic general equilibrium model of firm learning in networks. We formalize learning as a generalized dynamic process for Hicks-neutral productivity that depends on the productivity of a firm’s suppliers and buyers. The model features two novel elements.

First, learning and network structure interact with each other. On the one hand, the productivity process, or learning technology, depends flexibly on the entire distribution of suppliers’ and buyers’ productivity, weighted by trade shares. Therefore, firm learning is directly shaped by the network structure. On the other hand, the network itself is endogenous. Firms choose both the extensive and intensive margins of their relationships in a forward-looking, learning-aware manner, internalizing expected gains from trade and learning. In this way, network formation is in turn shaped by learning opportunities.

Second, firms negotiate with their suppliers and buyers over bilateral quantities and trade volumes following a Nash-in-Nash bargaining protocol. Bargaining internalizes both

gains from trade and gains from learning, departing from the common assumption that learning benefits are external to firms. A key theoretical prediction is that contract terms not only reflect supplier cost and buyer demand but also embed the value of learning in the relationship. Specifically, suppliers charge lower markups when they learn more from buyers, while buyers demand more inputs when they learn more from suppliers. Learning within relationships improves bilateral contract terms in a way that leads to more bilateral trade. The resulting improvements raise firm-level productivity directly and aggregate productivity indirectly as they accumulate through the network.

Our quantitative model represents Belgium, a small open economy with a high degree of trade openness. Given Belgium’s extensive participation in international trade, domestic firms likely derive substantial surpluses from both trade and learning through interactions with foreign suppliers and buyers. In addition to forming domestic networks endogenously, firms can also establish trading relationships with foreign partners after incurring a fixed cost of entering the foreign market, as in [Melitz \(2003\)](#). The learning technology depends on the productivity of both domestic and foreign partners and captures potential substitution between these two sources of learning. As in domestic firm-to-firm interactions, domestic firms bargain with foreign suppliers and buyers, so that bilateral contract terms with foreign partners also reflect learning within those relationships.

Our framework creates new mechanisms for international technology diffusion that amplify the gains from trade. Domestic firms can learn from foreign suppliers and buyers, with learning opportunities increasing in the productivity gap between domestic and foreign firms. Intuitively, when foreign suppliers and buyers are more productive than domestic firms, they serve as sources of knowledge that help domestic firms catch up to the global technological frontier. Recognizing this potential for learning, firms have additional incentives — beyond lower foreign input costs or higher foreign demand — to connect with foreign partners, with these productive efficiency gains transmitted through endogenous adjustments in the domestic network. Under bargaining, contract terms reflect the value of learning within each relationship: domestic firms have higher demand for inputs from foreign suppliers and set lower markups to foreign buyers, with both effects increasing in their productivity gap relative to foreign partners. Following reductions in trade costs, these mechanisms endogenously strengthen links between domestic and foreign firms and, consequently, accelerate international productivity convergence.

We quantify the model using a two-step procedure. In the first step, we recover productivity using data on sales to domestic final demand and labor share of total variable costs. The model-implied productivity measure is not confounded by heterogeneity related to supplier costs, buyer demand, and relationship-specific contract terms. Recovering productivity from a structural relationship is similar to the approach of [Grieco et al. \(2016\)](#)

and [Dhyne et al. \(2023\)](#). We then use the productivity series to estimate parameters in the learning technology, conditional on the observed network. This step relies on a standard timing assumption: the current network structure and contract terms are determined prior to the realization of future productivity shocks ([De Loecker and Syverson 2021](#)). We estimate a submodular learning technology in which firms disproportionately learn from their most productive suppliers and buyers, consistent with the reduced-form evidence. We also estimate the productivity of foreign suppliers and buyers and find that the productivity of the average foreign firm lies at the 96th percentile of the domestic productivity distribution. Given the submodular learning technology, this implies that foreign partners are key sources of knowledge for domestic firms that trade with them.

In the second step, we estimate the parameters governing the extensive and intensive margins of the network using the simulated method of moments. Specifically, we target average partner productivity and the probability of participating in international trade across the productivity distribution, which capture firms' exposure to domestic and foreign productivity at different productivity levels. We also target moments on trade shares with domestic and foreign partners, across the productivity distribution and in the aggregate, which capture the intensity of firms' trading relationships. Our estimated model successfully reproduces positive assortative matching in productivity between firms and their partners, as well as productivity-based selection into international trade, and it fits the targeted moments reasonably well.

We show that the estimated model can replicate the event study patterns, even though they were not targeted in estimation. In the model, an event is defined as being exposed to the most productive partners in a given period but not in the previous one. We simulate differential productivity trajectories for treated firms with different pre-event productivity levels and compare them to their empirical counterparts. Consistent with the data, the estimated model predicts that initially less productive firms experience stronger and more persistent productivity growth after the event. Quantitatively, the simulated long-run productivity effects across different treated firm groups align with the empirical estimates. These results indicate that the estimated model captures key patterns of productivity dynamics, supporting its empirical relevance.

We use the estimated model to quantify the importance of learning in shaping firm behavior and aggregate outcomes. To this end, we consider a counterfactual scenario in which firms do not learn from their suppliers or buyers. In the absence of learning, aggregate welfare falls by about 19 percent. The welfare losses are even larger for firm owners, reaching 25 percent. Firms not only experience slower productivity growth but also invest in and match with less productive suppliers and buyers. Decomposing firm-owner welfare change using a first-order approximation, we find that changes in productivity growth and network

structure each account for about 11 percentage points of the welfare loss, suggesting that learning-induced network adjustment is an equally important channel as productivity growth in shaping firm performance. These findings highlight that learning influences firm outcomes at multiple stages of operation and plays a quantitatively important role in determining both firm-level and aggregate productivity.

We then apply the model to study the effects of reducing international trade participation costs. Specifically, we consider a counterfactual in which the unconditional mean of the fixed cost distribution for international trade is lowered by 10 percent. In response, aggregate welfare rises by 1.18 percent. When learning is ignored, aggregate welfare increases by only 1.09 percent, which is 7.6 percent less than the baseline response, indicating that learning plays a quantitatively important role in shaping the gains from trade. Varying the magnitude of the shock, we find that learning amplifies the gains from trade and contributes to more than 7 percent of the total gains for a wide range of positive shocks we consider. Intuitively, because foreign suppliers and buyers are more productive than Belgian firms on average, they serve as key sources of knowledge. Learning therefore becomes more valuable as the economy increasingly relies on productive foreign partners for both trading and learning.

How would foreign innovation and productivity growth affect the Belgian economy? To answer this question, we examine how the gains from trade depend on the productivity of foreign firms. Specifically, we vary the level of foreign productivity while holding foreign input costs and demand shifters fixed at their baseline values, which ensures that differences in gains from trade arise solely from heterogeneous learning from foreign partners. We find that gains from trade increase substantially when foreign firms are more productive: learning accounts for more than 40 percent of the welfare gains from reducing international trade participation costs when foreign productivity increases by one log point relative to the baseline — roughly corresponding to the productivity of U.S. firms in our sample. Conversely, as foreign partners become less productive, the gains from trade diminish. In this case, exposure to unproductive foreign firms crowds out learning from productive domestic partners and weakens firms’ incentives to invest in productive relationships. As a result, when foreign firms are sufficiently unproductive, opening up to trade in a learning economy can reduce welfare gains to levels below those observed in an otherwise identical economy without learning.

Related Literature. By integrating learning from suppliers and buyers into a dynamic model of network formation, this paper contributes to the growing literature on endogenous firm-to-firm networks.¹ Our framework builds on [Demir et al. \(2024\)](#), where networks are

¹We treat networks as endogenous rather than exogenous. Many studies have examined the implications of exogenous firm-to-firm networks for the transmission of shocks. Examples include [Hulten \(1978\)](#), [Baqaee and Farhi \(2019\)](#), [Baqaee and Farhi \(2020\)](#), [Baqaee and Farhi \(2024\)](#), which focus on the implications for

formed through directed search and matching,² and extends their model to incorporate learning dynamics and bargaining between suppliers and buyers. A few papers study endogenous network dynamics, focusing on adjustment frictions (Lim 2018, Huneus 2018, Fontaine et al. 2023) or firm age (Aekka and Khanna 2024). We instead emphasize productivity dynamics and their interactions with endogenous network structure. Models with endogenous networks and bargaining have been studied by Grossman et al. (2024) and Acemoglu and Tahbaz-Salehi (2025), where firms bargain only over static gains from trade.³ In our framework, firms also bargain over dynamic gains from learning, generating novel predictions for bilateral markup and demand. To our knowledge, this is the first quantitative model that incorporates endogenous networks, productivity dynamics, and bargaining.

By studying learning in firm-to-firm networks, this paper contributes to the large literature on knowledge diffusion across firms. A substantial body of research has shown that diffusion is an important driver of firm productivity growth (Syverson 2011).⁴ One strand of this literature detects knowledge flows through patent citation networks, while another examines diffusion through international channels such as importing, exporting, and foreign direct investment.⁵ By contrast, our study focuses on diffusion through domestic and international firm-to-firm trade networks, exploiting the unique features of our data. Closely related are Alfaro-Urena et al. (2022) and Amiti et al. (2024), who use event studies to estimate productivity spillovers along such networks, and Iyoha (2023), who uses the control function approach in production function estimation. We consider learning from a broader category of firms, including productive domestic ones, and study the general equilibrium implications of knowledge diffusion through domestic firm-to-firm trade networks in the context of a novel structural model.

With bargaining embedded in endogenous firm-to-firm networks, our framework also contributes to the theoretical literature on knowledge diffusion and aggregate growth. Eaton and Kortum (1996), Kortum (1997), and Lucas (2009) model growth through random meeting

aggregate efficiency, and Dhyne et al. (2022), Huneus et al. (2021), Dhyne et al. (2025), Komatsu and Dhyne (2023), which study implications for labor market outcomes.

²The search-and-matching approach has also been adopted by Miyauchi (2024), Arkolakis et al. (2025), and Aekka and Khanna (2024), focusing on random matching between suppliers and buyers. Alternative approaches to endogenous network formation include models with selection through fixed costs (Lim 2018, Huneus 2018, Bernard et al. 2022, Dhyne et al. 2023) and models in which firms choose from a set of available production techniques (Oberfield 2018, Boehm and Oberfield 2020, Acemoglu and Azar 2020, Eaton et al. 2022, Panigrahi 2021).

³Models of bargaining with exogenous networks have been studied by Horn and Wolinsky (1988), Collard-Wexler et al. (2019), and Alvarez et al. (2023).

⁴We focus on knowledge diffusion rather than knowledge creation, and on productivity improvements rather than other forms of learning, such as learning about new products (Rachapalli 2021) or uncertainty about demand. In addition, we study learning between firms that trade at arm’s length, as opposed to related-party trade (Atalay et al. 2014, Kleinman 2022).

⁵For the former, see Acemoglu et al. (2016), Liu and Ma (2021). For the latter, see Javorcik (2004), Aw et al. (2011), De Loecker (2013), Setzler and Tintelnot (2021).

between agents. [Alvarez et al. \(2013\)](#) and [Buera and Oberfield \(2020\)](#) further demonstrate the role of selection in shaping exposure to knowledge and growth. Our model departs from the literature by incorporating endogenous firm-to-firm networks through which knowledge flows, which introduces the notion of non-random meeting and firm-specific learning. In addition, the literature mostly assumes that knowledge diffuses serendipitously, where learning gains are external to the firm.⁶ Our model instead allows firms to internalize these gains through bargaining and shows that the resulting bilateral markup and demand effects are an important source of productivity gains. We demonstrate quantitatively that these features are central for predicting the effects of trade shocks and policies.⁷

The rest of the paper is organized as follows. Section 2 describes the data and provides motivating evidence on firm learning from suppliers and buyers. Section 3 introduces the model and derives its theoretical predictions. Section 4 outlines the quantification strategy and discusses key equilibrium properties. Section 5 applies the estimated model to conduct two counterfactual exercises. The first quantifies the importance of learning in shaping firm behavior and aggregate outcomes, while the second examines how learning influences the welfare effects of reductions in international trade participation costs and how these effects depend on the productivity of foreign firms. Section 6 concludes.

2 Motivating Evidence

In this section, we present reduced-form evidence on firm learning from suppliers and buyers. Section 2.1 describes the data and sample construction. We then provide two sets of results. Section 2.2 uses a peer effect regression framework to examine how firm productivity growth correlates with the productivity of their suppliers and buyers. Section 2.3 complements these results with an event-study design that explores firm’s productivity trajectory before and after connecting with top suppliers or buyers in terms of their productivity for the first time.

2.1 Data

Data Sources. Our analyses draw on three administrative data sources from Belgium for the period 2002-2019. The first is the Business-to-Business (B2B) transactions database,

⁶[Fadeev \(2023\)](#) show that patent citations are concentrated among suppliers and buyers, suggesting that firms exercise some control when sharing know-how with trading partners. [Choi and Shim \(2024\)](#) develops a model in which technology adoption fees are determined through Nash bargaining.

⁷This paper also connects to research on learning between workers within narrowly defined teams or geographic units. [Jarosch et al. \(2021\)](#) and [Herkenhoff et al. \(2024\)](#) examine learning among coworkers within firms. [Crews \(2023\)](#) and [Lhuillier \(2023\)](#) study learning within cities and its implications for growth and inequality. This literature highlights that aggregate human capital depends crucially on learning technology and equilibrium sorting. We extend these insights to the context of arm’s-length trade, focusing on learning between firms.

which includes the universe of the firm-to-firm transactions that are no less than 250 euros between VAT-liable firms in each available year. The B2B database provides a comprehensive characterization of the Belgian domestic production networks at the extensive and intensive margins. The second is the annual account database, which includes a wide array of firm-level balance sheet variables (such as sales, employment, and input costs) and demographics (such as industry and location). The third is the international trade database, which includes firm-level imports and exports at the country-product level. These three datasets are merged through unique anonymized firm identifiers.

Sample Selection. We impose several sample restrictions. First, we keep the firm-year observations with at least one full-time-equivalent employee, positive labor costs, and positive total sales. Second, following [De Loecker et al. \(2014\)](#), we keep the firms with tangible assets of more than 100 euros and positive total assets in at least one observable year. Lastly, we drop the firms in public and financial sectors. Table A1 in Appendix A reports summary statistics of our analysis sample.

2.2 Regression Framework

We start by exploring the relationship between firms' productivity growth and the productivity of their suppliers and buyers. Our regression design follows [Jarosch et al. \(2021\)](#) and [Herkenhoff et al. \(2024\)](#), who study the relationship between workers' wage growth and coworker wages. We adapt their framework to the context of learning between firms by estimating the following regression:

$$Y_{jt+h} = \alpha Y_{jt} + \beta^S Y_{jt}^S + \beta^B Y_{jt}^B + \gamma \mathbf{X}_{jt} + \eta_{ind(j)t} + \eta_{loc(j)t} + \varepsilon_{jt+h}. \quad (1)$$

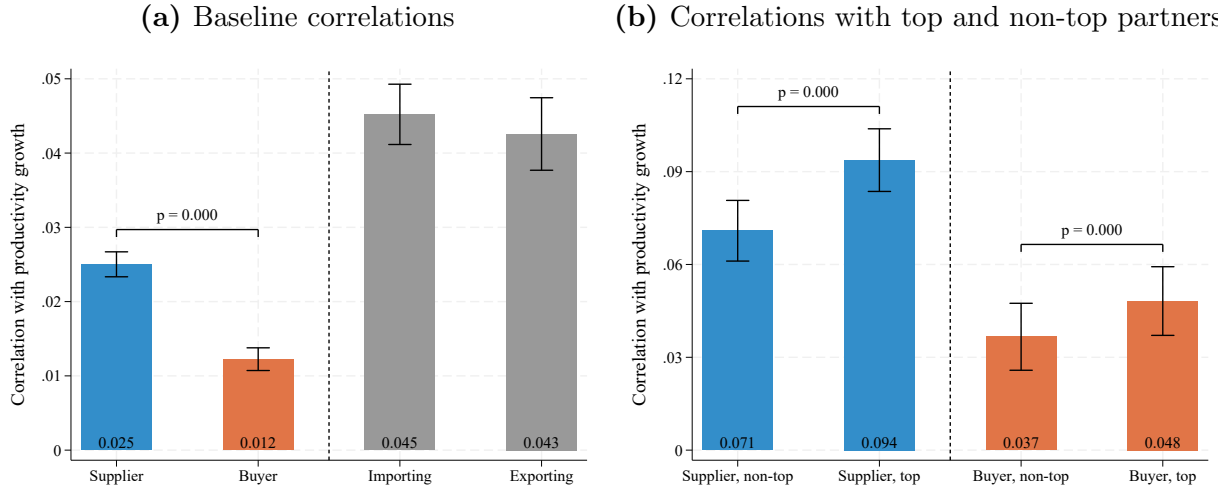
The main outcome is the productivity of firms, which we treat as their stock of knowledge. Our baseline measure of productivity is log value added per worker,⁸ though we consider alternative measures as robustness checks, discussed below. Equation (1) projects the future productivity of firm j in year $t+h$, Y_{jt+h} , where $h \geq 1$, on its current productivity, Y_{jt} , as well as the average productivity of its current suppliers Y_{jt}^S and buyers Y_{jt}^B . The productivity of each supplier or buyer is weighted by its trade share, which constitutes an intuitive measure of exposure.⁹ The coefficients of interest, β^S and β^B , capture the correlation between firm's future productivity and exposure to supplier and buyer productivity. To facilitate compar-

⁸Value added is computed from total sales minus total intermediate input costs, which include input purchases from domestic suppliers and imports.

⁹Specifically, each supplier's productivity is weighted by the firm's expenditure share on inputs from that supplier relative to its total expenditures on all suppliers. Similarly, each buyer's productivity is weighted by the firm's sales share to that buyer relative to its total sales to all buyers.

ison, we standardize Y_{jt}^S and Y_{jt}^B such that β^B and β^S are associated with per standard deviation change in exposure. The vector $\mathbf{X}_{jt}$ includes additional controls such as dummies for importing, exporting, and R&D.¹⁰ We also include industry-year fixed effects, $\eta_{ind(j)t}$, and location-year fixed effects, $\eta_{loc(j)t}$.¹¹ The error term ε_{jt+h} captures residual variation in productivity growth not explained by the regressors. We restrict the sample to firms with at least one supplier and one buyer in a year to ensure that they have non-zero exposure to both suppliers and buyers.

Figure 1: Results from estimating equations (1) and (2)



Notes: The figure plots the results from estimating equation (1) or (2). Productivity is measured as log value added per worker. Exposure measures to supplier and buyer productivity are standardized. The 95th confidence intervals are indicated by the error bars. The sample includes firms with at least one supplier and one buyer in a year. Additional results are reported in Table A2 in Appendix A.

Figure 1a plots the results from estimating equation (1) with a baseline horizon of $h = 1$, clustering standard errors at the firm level. We find that productivity growth is positively correlated with exposure to partner productivity and is more sensitive to supplier productivity than to buyer productivity. A one-standard-deviation increase in exposure to supplier and buyer productivity is associated with 2.5 percent and 1.2 percent higher productivity in the following year.

To explore which types of partners matter more for productivity growth, we adapt equa-

¹⁰The R&D dummy is constructed from the annual accounts. It equals one if a firm reports positive intangible assets or R&D expenditures in a given year.

¹¹Industries are classified according to four-digit NACE codes. Locations are classified according to two-digit NIS codes (arrondissement).

tion (1) to allow for heterogeneous exposure to different groups of partners:

$$Y_{jt+h} = \alpha Y_{jt} + \sum_{g \in G} \beta^{S,g} Y_{jt}^{S,g} + \sum_{g \in G} \beta^{B,g} Y_{jt}^{B,g} + \gamma \mathbf{X}_{jt} + \eta_{ind(j)t} + \eta_{loc(j)t} + \varepsilon_{jt+h}. \quad (2)$$

In our preferred specification, we split firms into two groups: $G = \{\text{top}, \text{non-top}\}$, where top firms are those in the top quintile of the productivity distribution across firms and years. $Y_{jt}^{S,g}$ and $Y_{jt}^{B,g}$ denote the trade-share-weighted exposure to suppliers and buyers in group g , while $\beta^{S,g}$ and $\beta^{B,g}$ captures the group-specific sensitivity. As before, exposure measures are standardized for each group. Figure 1b plots the estimates of equation (2). While productivity growth is positively correlated with exposure to both groups, it is more sensitive to exposure to top partners (the difference in estimated coefficients between top and non-top partners is statistically significant). Figure A1 in Appendix A presents the coefficients from a specification with G defined by productivity quintiles, confirming that this pattern persists.

Additional Results. As a robustness check, we estimate equations (1) and (2) including firm fixed effects. This addresses the potential concern that estimated coefficients are confounded by unobserved time-invariant firm characteristics that are correlated with both productivity growth and exposure to partner productivity. For instance, firms with better management practices may both connect with more productive partners and achieve higher productivity growth themselves. The results, reported in columns (3) and (4) of Table A2 in Appendix A, show that the patterns are qualitatively similar to those without fixed effects. This suggests that our findings are not driven by unobserved time-invariant firm characteristics.

In Appendix A, we provide additional results of estimating variations of equations (1) and (2). First, we vary the horizon h to examine the correlation between exposure and long-run growth. The results, reported in Table A3 in Appendix A, show that the coefficients increase with the horizon, indicating that higher exposure to partner productivity — particularly from top suppliers — is associated with stronger productivity growth over time.

Second, we explore heterogeneity by characteristics of firms themselves and their partners. Table A4 in Appendix A reports estimates of equation (2) by firm productivity quintile. The estimates reveal substantial heterogeneity across the productivity distribution. Overall, productivity growth tends to be more sensitive to supplier productivity than to buyer productivity, and more sensitive to top partners than to non-top partners, indicating that our results are stable across firms with different productivity. Table A5 in Appendix A further reports the results for young firms (no more than five years old) and old firms. As firms age, their productivity growth becomes more sensitive to supplier productivity and less sensitive

to buyer productivity, suggesting a shift in the source of learning over the firm lifecycle.

We also consider alternative definitions of partner groups G . As shown in Table A6 in Appendix A, among a firm’s top partners, productivity growth is more sensitive to exposure to partners that are older, have longer relationships with the firm, participate in international trade, or conduct R&D. These results are consistent with firms learning more from top partners who tend to be more knowledgeable.

Lastly, we examine the robustness of our results to alternative measures of productivity. Specifically, we estimate productivity using a range of methods, including the control function approach such as Olley and Pakes (1996), Levinsohn and Petrin (2003), Wooldridge (2009), Akerberg et al. (2015), as well as a model-consistent approach.¹² Table A7 in Appendix A shows that our main findings are robust across these alternative measures.

2.3 Event Study

In this section, we provide complementary evidence on firm learning through an event-study design, following Alfaro-Urena et al. (2022) and Amiti et al. (2024). The event is defined as connection for the first time with at least one top trading partner, who, as suggested by the results in Section 2.2, are the primary sources of learning. Top firms are defined as those whose average productivity across all years is at the top quintile of the productivity distribution. We consider first-time connection with top suppliers and buyers as two separate events.¹³

Unlike Alfaro-Urena et al. (2022) and Amiti et al. (2024), who compare the productivity growth of treated firms with that of untreated firms, our analysis focuses on heterogeneous effects among treated firms with different pre-event productivity levels. Intuitively, less productive firms may benefit more from connecting with top partners than more productive ones. To examine this, we group treated firms into quintiles based on their productivity prior to the event and estimate the following equation:

$$Y_{jt} = \sum_{g \in G} \sum_{s=-5}^5 \beta_s^g D_{jt,s}^g + \delta_j + \eta_{ind(j)t} + \eta_{loc(j)t} + \varepsilon_{jt}. \quad (3)$$

Here, Y_{jt} is the productivity of firm j in year t as defined in Section 2.2. $D_{jt,s}^g$ is an indicator for whether treated firm j belongs to group $g \in G = \{1, 2, 3, 4, 5\}$ and year t is s years relative to the event. We consider an event window spanning five years before and after the event.

¹²For the control function approach, the estimation procedure is adapted to account for potential learning from suppliers and buyers, following De Loecker (2013) and Iyoha (2023). For the model-consistent approach, see Section 4.2.

¹³For firms that experience a supplier (buyer) event, they may also experience a buyer (supplier) event either before or after the supplier (buyer) event.

Observations outside this window are censored at $s = -5$ or $s = 5$. Equation (3) further includes firm fixed effects, δ_j , industry-year fixed effects, $\eta_{ind(j)t}$, and location-year fixed effects, $\eta_{loc(j)t}$. The error ε_{jt} is clustered at the firm level. The regression sample includes only treated firms whose events occur at least four years after entry (or the beginning of the sample) and at least four years before exit (or the end of the sample), ensuring sufficient observations both before and after the event.

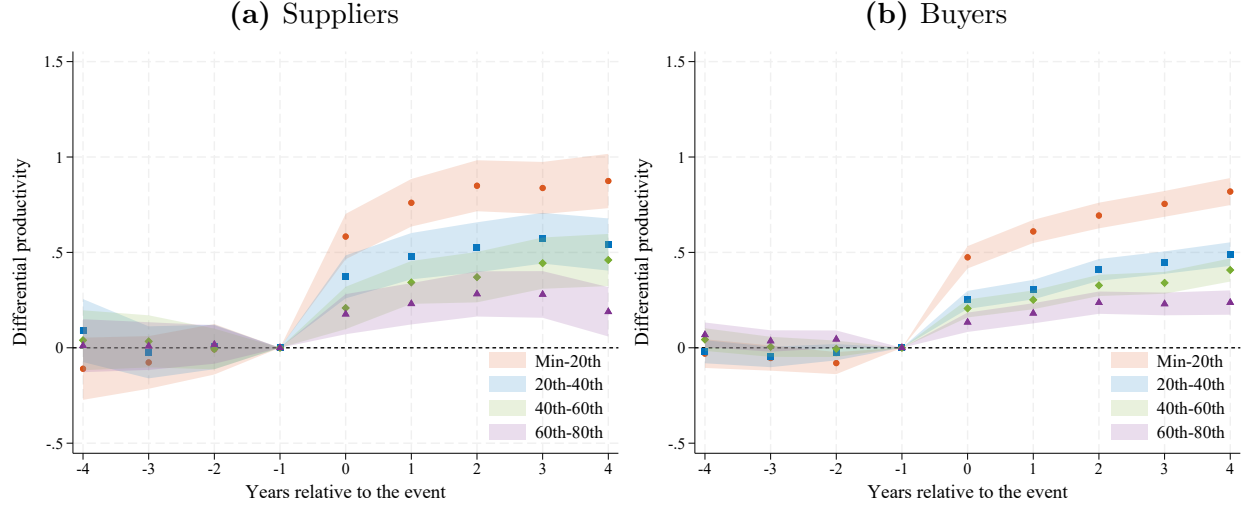
The coefficient for event year $s = -1$ is normalized to zero within each group g , such that all effects are measured relative to the year before the event. Our focus is on the differential patterns of β_s^g across groups g . We let the group of treated firms in the highest quintile of initial productivity (i.e., $g = 5$) be the reference group and compute the differences in estimates between each group and the reference group in each event year (i.e., $\beta_s^g - \beta_s^5$). Identification relies on the standard parallel trend assumption: firms from different groups are not on differential productivity paths relative to the reference group prior to the event.

Figure 2a plots the estimated coefficients for firms that connect with top suppliers for the first time. The results suggest that relative to firms in the top quintile of initial productivity, less productive firms see sizable productivity gains after starting sourcing from top suppliers. For treated firms in the bottom quintile of initial productivity, productivity rises by about 58 percent in the first year of connection and by about 87 percent four years later. In contrast, initially more productive firms show less productivity responses. These findings suggest that productivity growth increases with the productivity gap between the top supplier and the buyer. Notice that there is no pre-trend in productivity for any group, which suggests no strong correlations between productivity and connection with top suppliers prior to the event.

Figure 2b shows the results for firms who connect with top buyers for the first time. The pattern of productivity growth is similar to that in Figure 2a. For firms in the bottom quintile of initial productivity, productivity rises by about 47 percent in the first year of connection and by about 82 percent four years later, relative to firms in the top quintile of initial productivity. As before, these gains diminish as the initial productivity of firms increases. The results in Figure 2 are consistent with learning from both suppliers and buyers, where the least productive firms tend to benefit disproportionately more from top partners.

Alternative Explanations. While it is reassuring that there is no pre-trend in productivity prior to the event, several alternative mechanisms could still account for some of the patterns documented above. First, the coefficient in event year $s = 0$ may partly reflect reverse causality, in which firms receive a positive productivity shock and then select into connections with top partners. In the most conservative scenario, where all of the productiv-

Figure 2: Productivity after first-time connection with top partners



Notes: The figure plots the results from estimating equation (3) for firms that connect with top suppliers (Panel (a)) or buyers (Panel (b)) for the first time. Top firms are those whose average productivity across all years is at the top quintile of the productivity distribution. Treated firms are grouped into quintiles based on their productivity before the event. Firms in the top quintile of initial productivity are the reference group. Estimates for firms in lower quintiles are normalized relative to the estimates for the reference group. Scattered points represent the mean coefficients, and the shaded areas represent the 95th confidence intervals. Underlying estimates are reported in Table A8 in Appendix A.

ity gains in $s = 0$ are attributed to reverse causality, the persistent productivity growth for less productive firms after $s = 0$ suggests that these firms may still learn from their partners. Second, the estimates might pick up improvements in matching in addition to productivity growth. We re-estimate equation (3) using our model-consistent productivity measure, which is purged of supplier and buyer quality under certain assumptions (see Section 4.2). As shown in column (6) of Tables A9 and A10 in Appendix A, the results are similar to those in Figure 2, suggesting that the estimates are not driven by improvements in matching. Lastly, the results could be explained by mean reversion in productivity. However, the absence of pre-trend rules out mean reversion before the event. While mean reversion could explain a spike at $s = 0$, it is inconsistent with the persistent growth thereafter. Therefore, the results cannot be fully explained by alternative mechanisms, supporting learning as the primary interpretation.

Summary. The empirical findings are consistent with firms learning from both suppliers and buyers, particularly from the most productive ones. In the following sections, we develop and estimate a model to study the aggregate implications of such learning. These findings will guide the functional form of the learning technology and serve as validation for the estimated model.

3 Model

In this section, we develop a general equilibrium model of productivity dynamics, endogenous network formation, and bargaining. The key feature of the model is the interaction between firms' productivity growth and the endogenous network structure. On the one hand, a firm's future productivity depends on the productivity distribution of its suppliers and buyers (i.e., the network structure). On the other hand, the network itself is endogenous, as firms make forward-looking, learning-conscious decisions that account for potential learning opportunities. As a result, productivity and the network jointly evolve over time. Moreover, we assume trade terms between suppliers and buyers are determined through Nash-in-Nash bargaining (Horn and Wolinsky 1988). Here, relationship-specific surpluses encompass both static gains from trade and dynamic gains from learning. This departs from the common assumption that learning gains are external to firms and is natural in our setting, where learning arises locally through firm-to-firm interactions.

3.1 Environment

Time is discrete. The economy is at the steady state, which allows us to drop time subscripts. There is no aggregate uncertainty. To simplify exposition, we present the model of a closed economy to highlight the novel features of the model. In Section 3.6, we extend the model to a small open economy that nests in the domestic structure and quantify it to the Belgian economy in Section 4.

Workers. There is a mass L of homogeneous workers. Each worker is endowed with one unit of labor that she supplies inelastically. The labor market is competitive, and each worker receives a wage w . Workers use their labor income to purchase final goods for consumption, and they do not save or borrow. Workers have linear instantaneous utility and discount future utility at rate β .

Final Good Firms. There is a representative final good firm, indicated by H . The final good firm produces output using intermediate inputs with a CES technology given by

$$Y = \left(\int_{\Omega} m(\omega, H)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}},$$

where $m(\omega, H)$ is the quantity of intermediate inputs produced by firm $\omega \in \Omega$ used by firm H , and $\sigma > 1$ is the elasticity of substitution. The ordering of the two arguments of $m(\omega, H)$ indicates the flow of goods, where the first argument indicates the origin and the second

argument indicates the destination. The final good market is competitive, and each unit of output is sold at price P .

Intermediate Good Firms. There is a unit mass of entrepreneurs who set up firms to produce intermediate goods. We use entrepreneurs and firms interchangeably to refer to intermediate good firms. Each firm produces a differentiated variety, indexed by $\omega \in \Omega$. Firms are heterogeneous in their Hicks-neutral total factor productivity (TFP or knowledge), denoted $z(\omega) \in \mathbb{Z} \subset \mathbb{R}_+$.¹⁴ Varieties also differ by their quality $q(\omega) \in \mathbb{Q} \subset \mathbb{R}_+$, which is endogenous to the firms. Firms generate profits, which are used by entrepreneurs to purchase final goods for consumption. As with workers, entrepreneurs have linear instantaneous utility. Firms are destroyed at an exogenous rate δ in each period, such that the discount rate for entrepreneurs is $\beta(1 - \delta)$. Destroyed firms are replaced by an equal mass of entrants that draw productivity from a given distribution, $h_e(z)$.

Firm ω has a network of suppliers and buyers, denoted as $S(\omega)$ and $B(\omega)$, respectively. The firm hires workers and sources inputs from suppliers in $S(\omega)$ to produce output, determined by a production technology. Output is then sold to final good firms and buyers in $B(\omega)$. The firm also learns from its suppliers and buyers. Learning takes time, such that the knowledge sourced from the firm's current trading partners is reflected in its productivity in the next period. The extent to which the firm learns from each trading partner is governed by a learning technology, which specifies how the firm's future productivity depends on the distribution of productivity of its suppliers and buyers as well as its own productivity in the current period.

The network of a firm is endogenous. We adopt the directed search and matching approach of network formation following [Demir et al. \(2024\)](#). Specifically, the firm ω chooses the quality $q(\omega)$ of its products. The product markets are segmented by quality. Firms with quality q match with other suppliers and buyers with the same quality. Unlike the static setup in [Demir et al. \(2024\)](#), quality choice in our framework is forward-looking and learning-conscious such that firms account for not only the static gains from trade but the dynamic gains from learning when deciding on quality and their network. For simplicity, we assume trading relationships last one period, and firms form new networks in the next period.¹⁵

¹⁴For model tractability, we abstract from richer heterogeneity among firms such as industry, position in the supply chain, location, and ownership (e.g., multinational firms from different countries). These different types, however, can be parsimoniously treated as different segments in the productivity distribution.

¹⁵We abstract from persistence in individual links, which entails tracking a higher-dimensional state space over time. As discussed below, trading partners of the same type, whether old or new, are indifferent to the firm in terms of payoff. Our model is able to predict persistence in the type of links that are expected to be formed through persistence in productivity. For example, more productive firms are more likely to choose a higher quality and connect with other productive partners on average, even though the realized network for

Trade terms between suppliers and buyers are determined through bargaining. For tractability, we follow the Nash-in-Nash bargaining protocol as in [Horn and Wolinsky \(1988\)](#) and [Grossman et al. \(2024\)](#), where each supplier-buyer pair negotiates a contract that specifies a quantity and a transfer (the unit price is then determined), taking the negotiation outcomes of all other pairs as given.¹⁶ Importantly, surpluses from bargaining encompass gains from both trade and learning with each trading partner. This departs from the common assumption that learning gains are external to firms but is natural in our setting, as firms interact locally with their suppliers and buyers and are likely to act strategically to internalize how much they learn from each other. For the remainder of this section, we lay out the technology of firms and characterize their decisions.

3.2 Production and Learning Technology

Production Technology. The production technology of a firm ω takes a standard nested-CES form. The outer nest aggregates the bundle of intermediate inputs $M(\omega)$ and labor $l(\omega)$:

$$y(\omega) = z(\omega) \left(\alpha^{\frac{1}{\rho}} M(\omega)^{\frac{\rho-1}{\rho}} + l(\omega)^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \quad (4)$$

where $z(\omega)$ is the Hick-neutral productivity, $\alpha > 0$ is the weight of intermediate input bundle relative to labor (whose weight is normalized to one), and $\rho > 1$ is the elasticity of substitution between intermediate inputs and labor. The intermediate input bundle $M(\omega)$ is another CES aggregator of inputs from suppliers:

$$M(\omega) = \left(n_S(\omega)^{-\varphi} \int_{S(\omega)} \phi(q(\omega))^{\frac{1}{\sigma}} m(\omega', \omega)^{\frac{\sigma-1}{\sigma}} d\omega' \right)^{\frac{\sigma}{\sigma-1}}, \quad (5)$$

where $m(\omega', \omega)$ is the quantity of inputs from supplier ω' , and $\phi(q(\omega))$ is the quality shifter. $\sigma > 1$ is the elasticity of substitution across inputs from suppliers in $S(\omega)$. We denote $n_S(\omega) = |S(\omega)|$ as the mass of suppliers, where $|\cdot|$ represents the cardinality operator. The term $n_S(\omega)^{-\varphi}$ controls the degree of love-of-variety across inputs. The aggregator is the standard [Dixit and Stiglitz \(1977\)](#) aggregator when $\varphi = 0$, whereas setting $\varphi = \frac{1}{\sigma}$ removes the love-of-variety effect.

Learning Technology. Firms are also endowed with a learning technology that governs how future productivity depends on its current productivity, as well as the productivity

a given firm may vary across periods.

¹⁶Equivalently, firms can negotiate a quantity and a unit price, after which the transfer is determined, or they can negotiate a unit price and a transfer, after which the quantity is determined.

distribution of its current suppliers and buyers. We postulate the following functional form for the learning technology:

$$\log z_+(\omega) = \underbrace{\lambda_O + \gamma \log (\lambda_S \bar{z}_S(\omega)^\zeta + \lambda_B \bar{z}_B(\omega)^\zeta + (1 - \lambda_S - \lambda_B) z(\omega)^\zeta)}_{\log \bar{z}_+(\omega)}^{\frac{1}{\zeta}} + \xi_+(\omega), \quad (6)$$

where $z_+(\omega)$ denotes the future productivity of firm ω , $\log \bar{z}_+(\omega)$ is the mean component that depends on current productivity, and $\xi_+(\omega)$ is an i.i.d. shock distributed $Normal(0, \varsigma_\xi)$ that is realized in the next period. Let $g(z_+; \log \bar{z}_+(\omega))$ be the distribution of future productivity for firm ω . Within $\log \bar{z}_+(\omega)$, λ_O captures the common productivity growth across all firms. The key term on the right-hand side of equation (6) is the second term that reflects how future productivity depends on the current productivity of suppliers, buyers, and the firm itself. Analogous to the production technology, the learning technology takes a nested-CES structure. The outer nest aggregates a bundle of supplier productivity $\bar{z}_S(\omega)$, a bundle of buyer productivity $\bar{z}_B(\omega)$, and firm's own productivity $z(\omega)$. λ_S and λ_B are the exposure to supplier and buyer productivity, respectively. ζ governs the degree of substitutability across different sources of knowledge. γ determines the returns to scale of the learning technology or the persistence of current knowledge. The bundle of supplier (buyer) productivity is another CES aggregator of the productivity from different suppliers (buyers):

$$\bar{z}_S(\omega) = \left(\int_{S(\omega)} s_M(\omega', \omega) z(\omega')^\eta d\omega' \right)^{\frac{1}{\eta}}, \quad s_M(\omega', \omega) = \frac{x(\omega', \omega)}{\int_{S(\omega)} x(\omega'', \omega) d\omega''} \quad (7)$$

$$\bar{z}_B(\omega) = \left(\int_{B(\omega)} b_M(\omega, \omega') z(\omega')^\eta d\omega' \right)^{\frac{1}{\eta}}, \quad b_M(\omega, \omega') = \frac{x(\omega, \omega')}{\int_{B(\omega)} x(\omega, \omega'') d\omega''}, \quad (8)$$

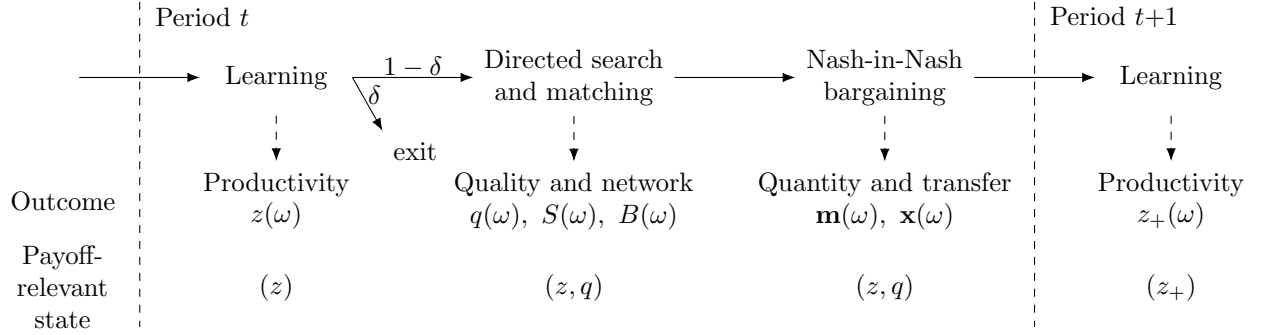
where $s_M(\omega', \omega)$ is share of total intermediate expenditures spent on inputs from supplier $\omega' \in S(\omega)$, and $b_M(\omega, \omega')$ is the share of intermediate sales coming from buyer $\omega' \in B(\omega)$. $x(\omega', \omega)$ denotes the trade volumes between supplier ω' and buyer ω . The term η captures the degree of substitutability across different knowledge sources within suppliers or buyers.

Several properties of this flexible learning technology are worth emphasizing. First, it allows for learning from both suppliers and buyers if λ_S and λ_B are positive in equation (4). Second, the productivity of each trading partner is weighted by its trade share, which is an intuitive notion of importance compared with uniform weights. Third, it generalizes the workhorse linear-in-means specification from the peer effects literature and is flexible to capture varying degrees of substitutability across different knowledge sources, governed by ζ and η . Specifically, they determine which type of supplier or buyer matters the most for learning. When $\zeta, \eta, \lambda_S, \lambda_B$ are all positive, firms disproportionately learn from their most

productive suppliers or buyers as $\eta \rightarrow \infty$.¹⁷ Note that ζ and η drive the sorting between firms due to learning, where firms have more incentives to connect with the partners from whom they learn the most.

Timing of Events. The timeline of events for firm ω in each period is shown in Figure 3. First, it receives a new productivity draw $z(\omega)$ from the distribution determined by the learning technology. Second, it is destroyed with rate δ . Third, conditional on survival, it chooses quality $q(\omega)$ and matches with suppliers and buyers with the same quality, which determines the network $S(\omega)$ and $B(\omega)$. Lastly, given its network, it bargains with its suppliers and buyers over bilateral quantities and transfers (prices are then determined). Firms then transact upon agreed terms. In the following sections, we lay out these decisions in reverse chronological order. Section 3.3 describes the bargaining process given the network. Section 3.4 characterizes optimal quality choice and network formation.

Figure 3: Event timeline for firm ω , closed economy



Notes: $\mathbf{m}(\omega)$ and $\mathbf{x}(\omega)$ collect all quantities and transfers relevant to firm ω . Specifically, $\mathbf{m}(\omega) = \{m(\omega', \omega)\}_{\omega' \in S(\omega)} \cup \{m(\omega, \omega')\}_{\omega' \in B(\omega) \cup \{H\}}$ and $\mathbf{x}(\omega) = \{x(\omega', \omega)\}_{\omega' \in S(\omega)} \cup \{x(\omega, \omega')\}_{\omega' \in B(\omega) \cup \{H\}}$.

3.3 Bargaining Given Network of Firms

Given a network structure, firms simultaneously bargain with each of their trading partners over a contract that specifies a bilateral quantity and a bilateral transfer. Bargaining follows the Nash-in-Nash protocol, where each supplier-buyer pair bargains, taking the bargaining outcomes of other pairs as given. This is natural as each trading relationship is atomistic when firms match with a continuum of suppliers and buyers.

Consider the bargaining problem between a buyer $\omega \in B(\omega')$ and a supplier $\omega' \in S(\omega)$. The pair bargains over bilateral quantity $m(\omega', \omega)$ and transfer $x(\omega', \omega)$ that maximizes

¹⁷Our learning technology is similar to Crews (2023) and Boucher et al. (2024), who characterize how own outcome is affected by peer outcomes, aggregated using a CES function.

the bargaining power weighted average of supplier and buyer surpluses, common in Nash bargaining:

$$\max_{m,x} \kappa \log \underbrace{V_S(m, x; \omega', \omega)}_{\text{supplier surplus}} + (1 - \kappa) \log \underbrace{V_B(m, x; \omega', \omega)}_{\text{buyer surplus}},$$

where $\kappa \in [0, 1]$ is the bargaining power of suppliers. The supplier and buyer surpluses are

$$\begin{aligned} V_S(m, x; \omega', \omega) &= \underbrace{x - c(\omega')m}_{\text{supplier gains from trade}} + \underbrace{\nu_B(\omega; \omega')x}_{\text{supplier gains from learning}}, \\ V_B(m, x; \omega', \omega) &= \underbrace{\frac{\sigma}{\sigma - 1} d(\omega)^{\frac{1}{\sigma}} m^{\frac{\sigma-1}{\sigma}} - x}_{\text{buyer gains from trade}} + \underbrace{\nu_S(\omega'; \omega)x}_{\text{buyer gains from learning}}. \end{aligned}$$

Surpluses consist of two components: gains from trade and gains from learning. Gains from trade are the changes in static profits associated with the trading relationship. From the supplier's perspective, the gains from trade are the transfer x it receives from the buyer net from the cost of production $c(\omega')m$, with $c(\omega')$ being its marginal cost. From the buyer's perspective, the gains from trade are the cost saved $d(\omega)^{\frac{1}{\sigma}} m^{\frac{\sigma-1}{\sigma}}$, with $d(\omega)$ being the demand shifter, net from the transfer x it pays the supplier.¹⁸

Gains from learning are the changes in expected future value from the trading relationship through shifting the mean of the distribution of future productivity. We define *learning rate* ν as the gains from learning divided by the bilateral trade volume. Therefore, $\nu_B(\omega; \omega')$, for example, represents how much supplier ω' learns from buyer ω per value of goods traded. Formally, $\nu_B(\omega; \omega')$ can be written as:

$$\begin{aligned} \nu_B(\omega; \omega') &= \beta(1 - \delta) \left(\int_{\mathbb{Z}} V(z_+) \frac{\partial g(z_+; \log \bar{z}_+(\omega'))}{\partial \log \bar{z}_+(\omega')} dz_+ \right) \\ &\quad \times \frac{\gamma \lambda_B}{\eta} \left(\frac{\bar{z}_B(\omega')}{\bar{z}_+(\omega')} \right)^\zeta \left(\frac{z(\omega)}{\bar{z}_B(\omega')} \right)^\eta \left(\int_{B(\omega')} x(\omega', \omega'') d\omega'' \right)^{-1}. \end{aligned} \quad (9)$$

We present the derivations in Appendix B.1 and provide the expression for the learning rate of buyer ω from supplier ω' , $\nu_S(\omega'; \omega)$, in equation (B6) in Appendix B.1. The first line of equation (9), captures how marginal change in the mean of the distribution g for supplier ω' , $\log \bar{z}_+(\omega')$, shifts its expected future value. $V(z_+)$ is the value of a firm with productivity

¹⁸As the contract terms of other pairs are taken as given, a supplier-buyer pair is only able to adjust labor in the case of bargaining failure. Specifically, the supplier needs to fire redundant workers to adjust for lost buyer demand, and the buyer needs to hire additional workers to replace lost supplier inputs. Therefore, supplier marginal cost and buyer demand shifter are closely tied to the payoffs from labor adjustments. In Appendix B.1, we provide explicit expressions for marginal costs and demand shifters given our production technology.

z_+ at the beginning of the period (defined below). The change in expected future value is discounted at rate $\beta(1 - \delta)$. When V is increasing in z , marginal increases in $\log \bar{z}_+(\omega')$ lead to higher expected future value.

The second line of equation (9), using the chain rule, captures how the trading relationship with buyer ω shifts $\log \bar{z}_+(\omega')$ for supplier ω' . This term consists of four components. First, it is increasing in persistence, γ , increasing in exposure, λ_B , and decreasing in substitutability, η . Second, it is increasing in average buyer productivity, $\bar{z}_B(\omega')$, relative to the mean, $\bar{z}_+(\omega')$, when $\zeta > 0$. Third, it is increasing in the productivity of buyer ω , $z(\omega)$, relative to average buyer productivity, $\bar{z}_B(\omega')$, when $\eta > 0$. Lastly, it is decreasing in total sales to all buyers, $\int_{B(\omega')} x(\omega', \omega'') d\omega''$, as the supplier relies less on a single buyer for learning when its total sales to all buyers are higher. Taken together, this setup reflects the novel feature that gains from learning within supplier-buyer relationships are internalized through bargaining, and these gains depend crucially on the learning technology and the network structure.

For transactions with final good firms, we assume intermediate good firms have full bargaining power and make take-it-or-leave-it offers. This resembles monopolistic competitive pricing when firms sell to final demand. The marginal cost of final good firms equals the price index P . Denote the demand shifter for final good firms as D . As intermediate good firms do not learn from final good firms, the corresponding learning rates are zero.

Bilateral Bargaining Solutions. Proposition 1 below characterizes the optimal quantity and transfer from solving the bilateral bargaining problem.

Proposition 1. *Given supplier marginal cost, $c(\omega')$, buyer demand shifter, $d(\omega)$, as well as learning rates, $\nu_B(\omega; \omega')$ and $\nu_S(\omega'; \omega)$, optimal quantity, $m(\omega', \omega)$, and transfer, $x(\omega', \omega)$, between supplier $\omega' \in B(\omega)$ and buyer $\omega \in S(\omega')$ satisfy*

$$m(\omega', \omega) = \left(\frac{\mu}{1 + \nu_B(\omega; \omega')} c(\omega') \right)^{-\sigma} \left(\frac{\mu}{1 - \nu_S(\omega'; \omega)} \right)^{\sigma} d(\omega), \quad (10)$$

$$x(\omega', \omega) = \left(\frac{\mu}{1 + \nu_B(\omega; \omega')} c(\omega') \right)^{1-\sigma} \left(\frac{\mu}{1 - \nu_S(\omega'; \omega)} \right)^{\sigma} d(\omega), \quad (11)$$

where $\mu = \kappa \frac{\sigma}{\sigma-1} + 1 - \kappa$. The implied price, defined as $p(\omega', \omega) = \frac{x(\omega', \omega)}{m(\omega', \omega)}$, is given by

$$p(\omega', \omega) = \frac{\mu}{1 + \nu_B(\omega; \omega')} c(\omega') = \mu(\omega', \omega) c(\omega'), \quad (12)$$

where $\mu(\omega', \omega)$ is the bilateral markup. Furthermore, given demand shifter for final good

firms D , optimal quantity, $m(\omega, H)$, and transfer, $x(\omega, H)$, satisfy

$$m(\omega, H) = \left(\frac{\sigma}{\sigma-1} c(\omega) \right)^{-\sigma} \left(\frac{\sigma}{\sigma-1} \right)^{\sigma} D, \quad (13)$$

$$x(\omega, H) = \left(\frac{\sigma}{\sigma-1} c(\omega) \right)^{1-\sigma} \left(\frac{\sigma}{\sigma-1} \right)^{\sigma} D, \quad (14)$$

with implied price $p(\omega, H) = \frac{x(\omega, H)}{m(\omega, H)} = \frac{\sigma}{\sigma-1} c(\omega)$.

Proof. See Appendix B.1.2. □

The solutions take a familiar form from a standard CES system. The key difference is that learning rates ν_S and ν_B affect relationship-specific contact terms. On the one hand, how much the supplier learns from the buyer, $\nu_B(\omega; \omega')$, affects the bilateral markup $\mu(\omega', \omega)$. Note that $\mu(\omega', \omega)$ is decreasing in $\nu_B(\omega; \omega')$: suppliers charge a lower markup to buyers when they learn more from them.¹⁹ On the other hand, how much the buyer learns from the supplier, $\nu_S(\omega'; \omega)$, affects bilateral demand through the term $\left(\frac{\mu}{1-\nu_S(\omega'; \omega)} \right)^{\sigma}$. Note that bilateral demand is increasing in $\nu_S(\omega'; \omega)$: buyers have higher demand for supplier inputs at any price when they learn more from them. In other words, firms receive more favorable contract terms in the form of lower input markup and additional output demand when their suppliers and buyers learn more from them. Proposition 1 emphasizes that firm-to-firm transactions not only reflect supplier efficiency and buyer size but also contain relationship-specific investment in learning. The rate of return from such investment, captured by the learning rate, is priced into the value of the goods, which in turn affects the value of the relationship for the firm.

Bargaining Equilibrium. Proposition 2 below characterizes the marginal costs and the demand shifters that are consistent with the bilateral bargaining solutions of all pairs.

Proposition 2. *Given network, $\{S(\omega), B(\omega)\}_{\omega \in \Omega}$, firm value, $\{V(z)\}_{z \in \mathbb{Z}}$, demand shifter of final good firms, D , and wage, w , the marginal cost of firm ω , $c(\omega)$, satisfies*

$$c(\omega) = \frac{1}{z(\omega)} \left(\alpha c_M(\omega)^{1-\rho} + w^{1-\rho} \right)^{\frac{1}{1-\rho}}, \quad (15)$$

where the cost index for intermediate inputs, $c_M(\omega)$, is given by

$$c_M(\omega) = \left(\int_{S(\omega)} \frac{\phi(q(\omega))}{n_S(\omega)^{\varphi\sigma}} \left(\frac{\mu}{1-\nu_S(\omega'; \omega)} \right)^{\sigma} \left(\frac{\mu}{1+\nu_B(\omega; \omega')} c(\omega') \right)^{1-\sigma} \right)^{\frac{1}{1-\sigma}}, \quad (16)$$

¹⁹It is possible that firms charge a price below marginal cost when learning rate is high. Similarly, bilateral markups may be higher than μ in cases where learning rates are negative.

and learning rates and trade volumes jointly satisfy equations (9), (B6), (11), and (14). The demand shifter of firm ω , $d(\omega)$, satisfies

$$d(\omega) = \frac{\phi(q(\omega))}{n_S(\omega)^{\varphi\sigma}} \frac{\alpha c_M(\omega)^{\sigma-\rho}}{c(\omega)^{\sigma-\rho}} z(\omega)^{\rho-1} \left(D + \int_{B(\omega)} \left(\frac{1 - \nu_S(\omega; \omega')}{1 + \nu_B(\omega'; \omega)} \right)^{-\sigma} d(\omega') d\omega' \right). \quad (17)$$

Lastly, the price index for final goods, P , satisfies

$$P = \left(\int_{\Omega} \left(\frac{\sigma}{\sigma-1} \right)^{\sigma} \left(\frac{\sigma}{\sigma-1} c(\omega) \right)^{1-\sigma} d\omega \right)^{\frac{1}{1-\sigma}}. \quad (18)$$

Proof. See Appendix B.1.3. □

Again, the marginal costs and the demand shifters are similar to what would obtain from a standard CES system with roundabout production. The key observation from equations (15) to (18) is that learning-induced variations in contract terms shape the cost and demand of a firm, which propagate to connected firms in the network and eventually affect aggregate productive efficiency through the price index P .

We are now ready to define the Nash-in-Nash equilibrium.

Definition 1. Given network, $\{S(\omega), B(\omega)\}_{\omega \in \Omega}$, firm value, $\{V(z)\}_{z \in \mathbb{Z}}$, demand shifter of final good firms, D , and wage, w , a Nash-in-Nash equilibrium (or bargaining equilibrium) is characterized by marginal costs, $\{c(\omega)\}_{\omega \in \Omega}$, demand shifters, $\{d(\omega)\}_{\omega \in \Omega}$, trade volumes, $\{x(\omega, \omega')\}_{\omega \in \Omega, \omega' \in \Omega \cup \{H\}}$, and learning rates, $\{\nu_B(\omega'; \omega), \nu_S(\omega'; \omega)\}_{\omega, \omega' \in \Omega}$ such that the conditions specified in Propositions 1 and 2 hold.

When bargaining reaches an equilibrium, labor demand is also determined:

$$l(\omega) = \frac{w^{-\rho}}{\alpha c_M(\omega)^{\sigma-\rho}} \frac{d(\omega)}{\phi(q(\omega))/n_S(\omega)^{\varphi\sigma}}. \quad (19)$$

We then compute variable profits as net firm-to-firm transfers minus labor costs:

$$\pi^{var}(\omega) = \int_{B(\omega) \cup \{H\}} x(\omega, \omega') d\omega' - \int_{S(\omega)} x(\omega', \omega) d\omega' - wl(\omega). \quad (20)$$

3.4 Network Formation

Entrepreneurs choose the quality of their product to maximize firm value, which, adjusted for the price index, is related to their lifetime consumption. They have idiosyncratic tastes in terms of utility, $\varepsilon_q(\omega)$, over different quality levels. This feature conveniently captures

frictions in network formation: firms with the same productivity may end up matching with trading partners with different quality.

Consider a firm who has chosen quality q . It enters quality market q together with other firms who choose the same quality. For simplicity, we assume each firm exerts unit effort at not cost to search for potential suppliers and buyers in the market.²⁰ Each firm then randomly matches with a continuum of suppliers and buyers in the market. We assume a Cobb-Douglas matching function: $AN_S(q)^aN_B(q)^{1-a}$, where $N_S(q)$ and $N_B(q)$ are the masses of suppliers and buyers in market q , respectively. With roundabout production, each firm serves as both a supplier and a buyer. Because products are symmetric, each firm contributes equally to the mass of suppliers and buyers in a given market. This implies that $N_S(q) = N_B(q) = \int_{\Omega} \mathbb{I}\{q(\omega) = q\}d\omega$ and that market tightness equals one in all markets. As a result, each firm matches with a mass A of suppliers and buyers.

Payoff-Relevant State. To characterize the set of suppliers $S(\omega)$ and buyers $B(\omega)$, one needs to know the identities of all trading partners of firm ω . In our model, the identity of a firm is characterized by its productivity $z(\omega)$ and the vector of idiosyncratic tastes across quality markets $\varepsilon(\omega)$. The set $S(\omega)$ or $B(\omega)$ thus contains the vector of (z, ε) for all suppliers or buyers, which is a high-dimensional object and is infeasible to track in practice. However, given the structure of the model, it is sufficient to track the payoff-relevant state of a firm, characterized by $(z, q) \in \mathbb{Z} \times \mathbb{Q}$. Conditional on choosing quality q , the vector of idiosyncratic tastes $\varepsilon(\omega)$ is irrelevant for payoffs. Firms in quality market q knows the distribution of suppliers and buyers present in that market. Random matching implies that each firm connects with the same distribution of trading partners in terms of their type, and the entire network structure upstream and downstream of each firm is known. Tracking a two-dimensional object makes the model quantitatively feasible to solve. With slight abuse of notation, we also use ω to denote the payoff-relevant state (or type) of a firm.

Supplier and Buyer Sets. We are now ready to characterize $S(\omega)$ and $B(\omega)$ for a firm with type ω . The supplier set is defined as $S(\omega) = \{n_S(\omega'; \omega)\}_{\omega' \in \Omega}$, where $n_S(\omega'; \omega)$ is the

²⁰We abstract from variations in the mass of suppliers and buyers across firms and focus instead on variations in partner quality. One potential concern from this simplification is that it imposes restrictions on firm size when some firms have incentives to exert higher search efforts to match with more suppliers and buyers. To alleviate such concern, when quantifying the model, we remove the love-of-variety effect in sourcing by setting $\varphi = \frac{1}{\sigma}$. In addition, we internally estimate the shifter $\phi(q)$, which partially captures additional size variations conditional on partner quality. That is, size variations explained by the number of partners are instead explained by variations in the intensity of trade with a given set of partners.

mass of suppliers with type ω' given by

$$n_S(\omega'; \omega) = A \frac{\mathbb{I}\{q(\omega') = q(\omega)\} \tilde{h}(\omega')}{\int_{\Omega} \mathbb{I}\{q(\omega'') = q(\omega)\} \tilde{h}(\omega'') d\omega''}. \quad (21)$$

$\tilde{h}(\omega)$ is the distribution of firms with type ω , and the indicator function $\mathbb{I}$ implies that firms only match with suppliers with the same quality. Similarly, the buyer set is defined as $B(\omega) = \{n_B(\omega'; \omega)\}_{\omega' \in \Omega}$, where $n_B(\omega'; \omega)$ is the mass of buyers with type ω' given by

$$n_B(\omega'; \omega) = A \frac{\mathbb{I}\{q(\omega') = q(\omega)\} \tilde{h}(\omega')}{\int_{\Omega} \mathbb{I}\{q(\omega'') = q(\omega)\} \tilde{h}(\omega'') d\omega''}. \quad (22)$$

Optimal Quality and Firm Value. The value of a firm with productivity z choosing quality q , denoted $\tilde{V}(z, q)$, satisfies

$$\tilde{V}(z, q) = \pi^{var}(z, q) + \beta(1 - \delta) \int_{\mathbb{Z}} V(z_+) g(z_+; z, q) dz_+. \quad (23)$$

Assume idiosyncratic tastes ε are independently drawn from a *Gumbel*(0, ς_q) distribution. Following [McFadden \(1972\)](#), the *ex ante* probability of a firm with productivity z choosing quality q is given by

$$Pr(q; z) = \frac{\exp\left(\tilde{V}(z, q)/P\right)^{\frac{1}{\varsigma_q}}}{\int_{\mathbb{Q}} \exp\left(\tilde{V}(z, q')/P\right)^{\frac{1}{\varsigma_q}} dq'}. \quad (24)$$

We can then write the value of a firm with productivity z as

$$V(z) = \int_{\mathbb{Q}} Pr(q; z) \tilde{V}(z, q) dq. \quad (25)$$

Entry. A mass δ of exiting firms is replaced by an equal mass of entrants that draw productivity from a given distribution $h_e(z)$. After drawing productivity, the behavior of entrants is identical to that of incumbents.

3.5 Equilibrium

Definition 2 characterizes a stationary equilibrium. All variables are defined using the payoff-relevant state.

Definition 2. *Given parameters, a stationary equilibrium is characterized by*

1. *Variables at the firm level:* $\{c(\omega), d(\omega), y(\omega), l(\omega), m(\omega, H), x(\omega, H), \pi^{var}(\omega)\}_{\omega \in \Omega}$,
2. *Variables at the pair level:* $\{m(\omega, \omega'), x(\omega, \omega'), \nu_S(\omega'; \omega), \nu_B(\omega'; \omega), n_S(\omega'; \omega), n_B(\omega'; \omega)\}_{\omega, \omega' \in \Omega}$,
3. *Firm value and quality choice:* $V(z), \tilde{V}(z, q), Pr(q; z)$,
4. *Variables in the aggregate:* $h(z), \tilde{h}(z, q), Y, P, D, w$ (normalized to one),

such that:

1. *Bargaining reaches an equilibrium as in Definition 1.*
2. *Network structure satisfies equations (21) and (22).*
3. *Firm value and quality choice satisfy equations (23), (24), (25).*
4. *Final good market clears:*

$$PY = wL + \int_{\Omega} \pi^{var}(\omega) \tilde{h}(\omega) d\omega. \quad (26)$$

The demand shifter for final good firms is $D = \frac{Y}{P-\sigma}$.

5. *Labor market clears:*

$$L = \int_{\Omega} l(\omega) \tilde{h}(\omega) d\omega. \quad (27)$$

6. *The productivity distribution of firms is stationary:*

$$h(z') = (1 - \delta) \int_{\mathbb{Z}} \int_{\mathbb{Q}} g(z'; z, q) Pr(q; z) h(z) dq dz + \delta h_e(z'), \quad \forall z' \in \mathbb{Z}, \quad (28)$$

where the payoff-relevant state distribution is $\tilde{h}(z, q) = Pr(q; z)h(z)$.

3.6 Small Open Economy

We now extend the model to a small open economy, which is the baseline for the quantitative analysis. There is a measure one of homogeneous foreign suppliers and buyers.²¹ Foreign suppliers produce using only foreign labor with productivity z_F . This implies the marginal cost of foreign suppliers equals $c_F = \frac{w_F}{z_F}$, where w_F is the exogenous foreign wage. Foreign buyers source inputs from domestic suppliers and have exogenous foreign demand shifter d_F . The productivity of foreign buyers is also z_F .

²¹We assume foreign firms are homogeneous as we do not observe the identities of domestic firms' foreign suppliers and buyers in the data.

Consider a domestic firm ω that enters the foreign market. It matches with foreign suppliers and produces output using additional foreign inputs. Specifically, the bundle of intermediate inputs is

$$M(\omega) = \left(M_D(\omega)^{\frac{\sigma-1}{\sigma}} + i(\omega)\alpha_F^{\frac{1}{\sigma}} M_F(\omega)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad (29)$$

where $M_D(\omega)$ is the bundle of domestic intermediate inputs defined in equation (5), $i(\omega)$ is an indicator that equals one if firm ω enters the foreign market, α_F is relative weight of foreign inputs. The bundle of foreign inputs $M_F(\omega)$ is given by

$$M_F(\omega) = \left(n_S^*(\omega)^{-\varphi} \int_{S^*(\omega)} \phi(q(\omega))^{\frac{1}{\sigma}} m(\omega', \omega)^{\frac{\sigma-1}{\sigma}} d\omega' \right)^{\frac{\sigma}{\sigma-1}}, \quad (30)$$

where $S^*(\omega)$ is the set of foreign suppliers with the mass of foreign suppliers given by $n_S^*(\omega) = |S^*(\omega)|$. $\phi(q(\omega))$ is the quality shifter, which captures that firms with different quality may have different demand for foreign inputs. By entering the foreign market, the firm also matches with foreign buyers and faces additional foreign demand. The firm-specific foreign demand is given by $\phi(q(\omega))d_F$, where $\phi(q(\omega))$ captures that firms with different quality may have different demand from foreign buyers. Lastly, the firm also learns from foreign suppliers and buyers whose productivity is part of its learning technology:

$$\bar{z}_S(\omega) = \left(\int_{S(\omega)} s_M(\omega', \omega) z(\omega')^\eta d\omega' + \int_{S^*(\omega)} s_M(\omega', \omega) z_F^\eta d\omega' \right)^{\frac{1}{\eta}}, \quad (31)$$

$$\bar{z}_B(\omega) = \left(\int_{B(\omega)} b_M(\omega, \omega') z(\omega')^\eta d\omega' + \int_{B^*(\omega)} b_M(\omega, \omega') z_F^\eta d\omega' \right)^{\frac{1}{\eta}}. \quad (32)$$

The timeline of events for a domestic firm ω in the small open economy is as follows. In each period after productivity and destruction shock are realized, it receives an idiosyncratic fixed cost draw $f(\omega)$ in units of final goods and chooses whether to enter the foreign market or not (Melitz 2003). If it chooses to do so, it connects with all foreign suppliers and foreign buyers in the market, which determines $S^*(\omega)$ and $B^*(\omega)$. After the domestic network is formed, contract terms with foreign firms are bargained in the same way as with other domestic firms. As we do not observe the network structure of foreign firms in the data, we assume foreign firms do not learn from domestic firms: $\nu_B(\omega; F) = 0$ for foreign suppliers and $\nu_S(\omega; F) = 0$ for foreign buyers. In addition, the payoff-relevant state is given by $(z, i, q) \in \mathbb{Z} \times \{0, 1\} \times \mathbb{Q}$, which is augmented to include international trade participation decision. We assume trade is balanced, an additional equilibrium condition used to pin down domestic wage w , where the foreign wage w_F is normalized to one. We present the full setup

in Appendix B.2.

4 Quantification

This section describes the quantification of the model. Section 4.1 lists the parameters determined outside of the model. The remaining parameters are determined in two steps. First, in Section 4.2, we recover firm-level productivity and estimate the learning-technology parameters using data on the network structure. Second, in Section 4.3, we estimate the parameters governing the extensive and intensive margins of the network via simulated method of moments. Finally, Section 4.4 discusses equilibrium properties and model validation.

4.1 Assigned Parameters

We set several parameters outside the model, summarized in Table 1. We set the period length to one year to match the data frequency. The discount rate $\beta = 0.98$ implies an annual interest rate of 2 percent. The firm exit rate $\delta = 0.05$ is computed from the share of firms exiting from one year to the next in the data. For the elasticities of substitution in the production function, we take the estimates from Dhyne et al. (2022), who construct a model of supplier-buyer-specific markups that depend on ρ , σ , and bilateral input shares. The parameters ρ and σ are calibrated by fitting markup-deflated total sales to total variable costs, using a sample of Belgian firms. We let $\rho = 1.67$ and $\sigma = 2.98$, where σ is computed from the average industry-specific elasticity of substitution from Dhyne et al. (2022). Since $\rho < \sigma$, inputs across different suppliers are more substitutable than between intermediates and labor. As discussed in Section 3.4, we set $\varphi = \frac{1}{\sigma}$ to remove love-of-variety across inputs from different suppliers. Because each firm is both a supplier and a buyer, we assume equal bargaining power on both sides, $\kappa = 0.5$. Finally, because the number of trading partners is irrelevant for firm’s productive efficiency when $\varphi = \frac{1}{\sigma}$, we normalize the matching efficiency to $A = 1$.

Table 1: Assigned parameters

Parameter	Description	Value	Source
β	Discount rate	0.98	2 percent interest rate
δ	Firm exit rate	0.05	Data
ρ	EoS between intermediates and labor	1.67	Dhyne et al. (2022)
σ	EoS across inputs from suppliers	2.98	Dhyne et al. (2022)
φ	Degree of love of variety	0.33	$\varphi = \frac{1}{\sigma}$
κ	Supplier bargaining power	0.50	Symmetry
A	Matching efficiency	1.00	Normalization

4.2 Estimating Productivity and Learning Technology Parameters

In general, estimating productivity in the presence of unobserved supply- and demand-side heterogeneity is challenging. This issue is particularly relevant in our context. On the one hand, as firms connect with different suppliers and buyers, variations in supplier efficiency and buyer demand contribute to differences in firm performance measures commonly used to estimate productivity (e.g., value added or total sales). Therefore, estimation strategies without proper controls for such heterogeneities may yield biased estimates of firm productivity. On the other hand, firm-to-firm transactions involve relationship-specific markup and demand that are unobserved in our data. This introduces heterogeneity at the firm-pair level, which is difficult to account for without granular information on bilateral prices and quantities.

Aware of these challenges, we rely on a structural relationship implied by the model to estimate productivity. Given parameters ρ and σ , the productivity of firm ω can be inferred, up to a general equilibrium constant, from its sales to domestic final demand, $x(\omega, H)$, and the labor share of total variable costs, $s(L, \omega)$.²²

$$z(\omega) \propto x(\omega, H)^{\frac{1}{\sigma-1}} s(L, \omega)^{\frac{1}{\rho-1}}.$$

The detailed derivations are presented in Appendix B.3. Intuitively, firms face the same demand shifter when selling to final good firms. Sales to domestic final demand, $x(\omega, H)$, thus contain clean information about firm's marginal cost.²³ Moreover, under the structure of the model, the labor share of total variable costs, $s(L, \omega)$, serves as a sufficient statistic for supplier cost index. Controlling for $s(L, \omega)$ therefore separates supplier cost index from firm productivity in marginal cost. The above equation uses cross-sectional information and can be applied to recover productivity year by year for firms with non-missing $x(\omega, H)$ and $s(L, \omega)$.²⁴

With estimated productivity in hand, we then recover the parameters of the learning technology, characterized by equations (6), (31), and (32), using data on firm-to-firm trans-

²²In the data, sales to domestic final demand, $x(\omega, H)$, is computed from total sales minus sales to domestic buyers and exports. Labor share of total variable costs, $s(L, \omega)$, is computed from labor costs divided by total variable costs, which consist of labor costs, input purchases from domestic suppliers, and imports.

²³One potential concern is that firms across different industries or locations may face distinct domestic final demand shifters. As a robustness check, we demean estimated productivity by NACE4 industry and location. Note that this also removes systematic productivity differences across industries (firms in certain sectors are more productive) and locations (productive firms may cluster in certain areas). Re-estimating the learning technology using the demeaned productivity series yields similar results, as reported in Table C1 in Appendix C.

²⁴Sales to final demand account for 44 percent of total sales in our sample, indicating that these sales represent an important share of firm activity. Nearly all firms sell to domestic final demand, with only about 7.4 percent of observations missing the variable $x(\omega, H)$.

actions. The parameters are estimated using non-linear least squares. Identification relies on the timing assumption that the productivity shock, $\xi_+(\omega)$, is realized *after* firms make their current-period decisions. This assumption is standard in the production function and productivity estimation literature (De Loecker and Syverson 2021) and is consistent with our model.

We present the relevant parameter estimates in Table 2. The estimated firm productivity follows a log-normal distribution with a standard deviation of 0.895 (mean normalized to zero). The productivity distribution for entrants is also log-normal but has a lower mean, $\vartheta_e = -0.400$, and a higher standard deviation, $\varsigma_e = 1.014$.²⁵ Firm productivity depreciates over time, as indicated by a negative constant λ_O and persistence γ below one. The estimated learning parameters, λ_S and λ_B , are both positive, implying that firms learn from both suppliers and buyers. Moreover, the estimates $\zeta = 1.105$ and $\eta = 13.310$ imply that firms disproportionately learn from their top suppliers and buyers, consistent with the empirical evidence documented in Section 2. As discussed in Section 3.2, the sign and magnitude of the estimated parameters λ_S , λ_B , ζ , and η drive sorting among firms, pushing firms to connect with the most productive partners in order to learn from them. In addition, the standard deviation of the idiosyncratic productivity shock is $\varsigma_\xi = 0.358$, which accounts for about $\frac{0.358^2}{0.895^2} \approx 16$ percent of the variance in productivity, with the remainder explained by variation in the systematic component of the learning technology. We also treat the productivity of foreign firms as a parameter and estimate $\log z_F = 1.541$, corresponding to the 95.7th percentile of the domestic productivity distribution. Note that z_F can be interpreted as the weighted average productivity of foreign firms that matters for learning.²⁶ This implies that the average foreign firm is much more productive than the average Belgian firm.

4.3 Simulated Method of Moments

We estimate the remaining parameters governing the extensive and intensive margins of the network through simulated method of moments (SMM). To proceed, we specify functional forms for several objects. The idiosyncratic fixed cost of international trade follows a *Lognormal*(ϑ_f, ς_f) distribution. The quality shifter is assumed to take the form

²⁵Table C2 in Appendix C shows that estimated productivity is positively correlated with various firm performance measures. Figure C1 in Appendix C plots the productivity distributions for all firms and for entrants.

²⁶Foreign productivity z_F is identified from variation in the import share of expenditures and export share of sales across firms, conditional on firms' own productivity and their exposure to domestic trading partners. Here, z_F can be interpreted as a CES aggregator of productivity across foreign firms, with trade shares as weights and elasticity parameter η . Given the magnitude of η , z_F approximates the maximum productivity among all foreign firms.

Table 2: Productivity and learning technology parameters

Parameter	Description	Value
ϑ_e	Entrant log productivity distribution, mean	-0.400
ς_e	Entrant log productivity distribution, s.d.	1.014
λ_O	Constant	-0.045
γ	Persistence	0.971
λ_S	Supplier share	0.005
λ_B	Buyer share	0.002
ζ	Substitutability, outer nest	1.105
η	Substitutability, inner nest	13.310
ς_ξ	Productivity shock, s.d.	0.358
$\log z_F$	Log foreign productivity	1.541

$\phi(q) = \exp(\phi_q q)$. We therefore estimate $\vartheta_f, \varsigma_f, \phi_q$, together with the dispersion of idiosyncratic quality taste shocks, ς_q , the weight of intermediate inputs relative to labor, α , the weight of foreign inputs relative to domestic inputs, α_F , and the exogenous foreign demand shifter, d_F .

These seven parameters are estimated to match four sets of moments:

1. Expenditure share weighted average log productivity of suppliers by productivity quintile (5 moments).
2. Share of firms participating in international trade by productivity quintile (5 moments).
3. Expenditure share on intermediate inputs by productivity quintile (5 moments).
4. Aggregate domestic intermediate input share of total expenditures, aggregate import share of total expenditures, and aggregate export share of total sales (3 moments).

Denote $\chi = (\vartheta_f, \varsigma_f, \varsigma_q, \phi_q, \alpha, \alpha_F, d_F)$ as the vector of parameters to be estimated, $\mathbf{m}^{\text{model}}(\chi)$ as the vector of moments simulated from the model given χ , and $\mathbf{m}^{\text{data}}$ as the corresponding vector of moments in the data. We estimate χ by minimizing the distance between model and data moments:

$$\hat{\chi} = \arg \min_{\chi} (\mathbf{m}^{\text{model}}(\chi) - \mathbf{m}^{\text{data}})' \mathbf{W} (\mathbf{m}^{\text{model}}(\chi) - \mathbf{m}^{\text{data}}), \quad (33)$$

where $\mathbf{W}$ is a diagonal weighting matrix. Moments by productivity quintile (the first three sets of moments) are weighted equally, while aggregate moments (the fourth set of moments) are scaled by a factor of five.

As the number of moments exceeds the number of parameters, the model is overidentified. While we use information from all moments to jointly estimate the parameters, some moments are more closely associated with certain parameters than others. The first set

of moments captures assortative matching in productivity among domestic firms, which is informative for ς_q . The second set of moments captures productivity selection into international trade, which is informative for ϑ_f and ς_f . Note that the first two sets of moments reflect exposure to domestic and foreign productivity across the productivity distribution. The third set of moments captures the intensive margin of trade conditional on the network, which is informative for ϕ_q . The fourth set of moments captures the aggregate importance of domestic intermediate inputs, imports, and exports, which is informative for α , α_F , and d_F , respectively.

Solution Algorithm. We describe the algorithm used to numerically solve for an equilibrium. The productivity space $\mathbb{Z}$ is discretized into quintiles of a log-normal distribution with mean 0 and standard deviation 0.895. The quality space $\mathbb{Q}$ is discretized into five equally spaced grid points over the interval $[0, 1]$. This yields a payoff-relevant state space with cardinality $5 \times 2 \times 5 = 50$. We characterize the decisions of firms for these 50 types.

The solution algorithm, detailed in Appendix B.4, consists of three parts. The first and key step involves solving for a bargaining equilibrium in each period, given an initial guess for the distribution of firms, $\{\tilde{h}(\omega)\}_{\omega \in \Omega}$, the domestic final demand shifter, D , the domestic wage, w , and the firm value function, $\{V(z)\}_{z \in \mathbb{Z}}$. Specifically, we solve for a fixed point in the distributions of marginal costs, $\{c(\omega)\}_{\omega \in \Omega}$, demand shifters, $\{d(\omega)\}_{\omega \in \Omega}$, and firm-to-firm trade volumes, $\{x(\omega, \omega')\}_{\omega, \omega' \in \Omega}$. The fixed-point algorithm has a nested structure: the inner loop solves for the fixed point of firm-to-firm trade volumes given marginal costs and demand shifters, while the outer loop iterates on marginal costs and demand shifters until the bargaining equilibrium conditions are satisfied.

Given the solution to the bargaining equilibrium, the second part of the algorithm solves for policy functions for international trade participation, $\{Pr(i; z)\}_{i \in \{0, 1\}, z \in \mathbb{Z}}$, and quality choice, $\{Pr(q; z, i)\}_{q \in \mathbb{Q}, z \in \mathbb{Z}, i \in \{0, 1\}}$, as well as firm value, $\{V(z)\}_{z \in \mathbb{Z}}$, conditional on $\{\tilde{h}(\omega)\}_{\omega \in \Omega}$, D , and w . This step is implemented using standard value function iteration. In the third and final step, the algorithm solves for the stationary distribution of firms, $\{\tilde{h}(\omega)\}_{\omega \in \Omega}$, the domestic final demand shifter, D , and the domestic wage, w , such that all equilibrium conditions hold. Although we do not provide a formal proof of equilibrium uniqueness, we verify numerically that, given the estimated parameters, the algorithm converges to the same equilibrium from sensible initial guesses.

Estimation Results and Model Fit. Table 3a reports the estimated parameter values, and Table 3b presents the data and model moments. Even though the number of parameters is less than the number of targeted moments, the estimated model fits the extensive and intensive margins of the network reasonably well, given the discretized state space. Several

observations are worth noting. First, we estimate large average fixed costs of international trade participation, along with substantial dispersion across firms. This aligns with the fact that, although more productive firms select into international trade, a nontrivial share of less productive Belgian firms also trade with foreign partners. Second, the model successfully reproduces the increasing and convex relationship between a firm’s own productivity and the average productivity of its suppliers — reflecting positive assortative matching (PAM) in productivity. In the model, both the supermodularity of the production technology and the submodularity of the learning technology incentivize firms to choose the highest quality and connect with the most productive partners. Dispersion in idiosyncratic tastes for quality, however, introduces frictions that counteract these forces, particularly by preventing less productive firms from choosing the highest quality, thereby generating PAM in productivity. Third, a quality shifter ϕ_q of 0.235 implies that trade volumes increase by 26 percent when moving from the lowest to the highest quality, all else equal. Lastly, in the bottom panel of Table 3b, we report the share of economic activity accounted for by firms in the top quintile. The model fits the observed concentration of large firms well, providing further evidence that it captures key features of the Belgian economy.

We perform several checks to assess identification with respect to the chosen moments in Appendix C. First, we perturb each parameter around the estimated point, holding other parameters fixed. Figure C2 in Appendix C shows that deviations in either direction increase the mean squared error of moments, implying local identification. Second, we report the sensitivity measure following Andrews et al. (2017) in Figure C3 in Appendix C and find that the parameters are generally sensitive to our chosen moments.

4.4 Equilibrium Properties and Model Validation

We discuss several properties of the equilibrium. Figure 4a plots the quality choice probabilities by productivity. For the least productive firms, quality choice is nearly random. More productive firms are more likely to enter the highest-quality market and match with other productive suppliers and buyers, generating PAM in productivity. This pattern is also reflected in the average quality, which increases with firm productivity, as shown by the red line in the figure. On average, firms choose a quality level close to the 65th percentile of the quality distribution, with the least and most productive firms choosing quality levels near the 51st and 98th percentiles, respectively.

Figure 4b shows the log bilateral markup and demand associated with suppliers (blue lines) and buyers (red lines) across the productivity distribution. For each firm type, we compute the expenditure- or sales-share-weighted average log bilateral markup and demand

Table 3: Simulated method of moments: estimated parameters and model fit**(a)** Estimated parameters

Parameter	Description	Value
ϑ_f	Fixed cost of international trade participation, mean	3.476
ς_f	Fixed cost of international trade participation, s.d.	4.240
ς_q	Quality taste shock, dispersion	2.168
ϕ_q	Quality shifter	0.235
α	Relative weight of intermediate inputs	0.345
α_F	Relative weight of foreign inputs	2.445
d_F	Foreign demand shifter	0.787

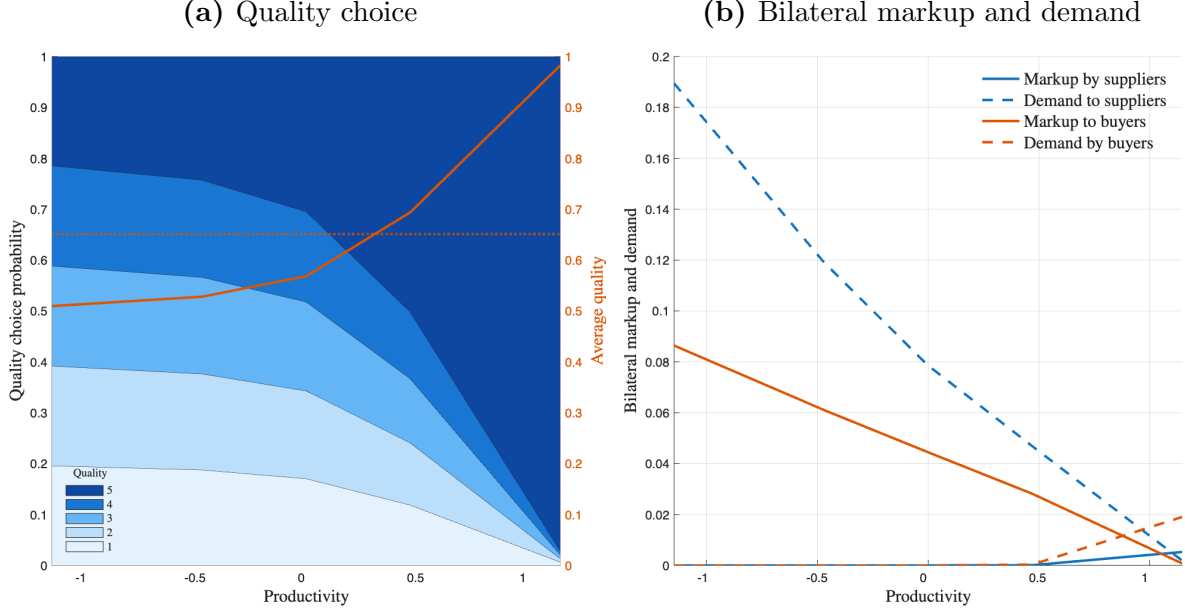
(b) Model fit: targeted and untargeted moments

Target moments by productivity quintile		Q1	Q2	Q3	Q4	Q5
Expenditure share weighted average productivity of domestic suppliers	Data	0.43	0.45	0.50	0.59	0.88
	Model	0.43	0.45	0.49	0.65	1.01
Share of firms participating in international trade	Data	0.22	0.20	0.21	0.26	0.42
	Model	0.17	0.21	0.25	0.32	0.44
Intermediate input share of total expenditures	Data	0.46	0.55	0.60	0.65	0.72
	Model	0.50	0.52	0.54	0.60	0.74
Targeted aggregate moments		Data			Model	
Domestic intermediate input share of total expenditures		0.46			0.45	
Import share of total expenditures		0.33			0.32	
Export share of total sales		0.20			0.21	
Untargeted moments		Data			Model	
Share of total sales of firms in the top quintile		0.87			0.85	
Share of sales to domestic buyers of firms in the top quintile		0.84			0.86	
Share of inputs from domestic suppliers of firms in the top quintile		0.83			0.86	
Share of labor costs of firms in the top quintile		0.79			0.82	

Notes: The table reports the results from the SMM estimation. Panel (a) reports the estimated parameters from solving the minimization problem defined in equation (33), and Panel (b) reports targeted and selected untargeted moments from the data and the model. For the targeted moments used in estimation, moments by productivity quintile are weighted equally, while aggregate moments are scaled by a factor of five. The mean squared error across moments is 0.0013.

across all suppliers and buyers. We then average these values across firm types at each productivity level. Log bilateral markups are scaled by $1 - \sigma$ and log bilateral demand by σ , so that marginal increases in both have the same effect on trade volumes. For less productive firms, they connect with more productive trading partners and have higher rates of learning from them. As a result, these firms have greater demand for suppliers' inputs (dashed blue line) and charge lower markups to buyers (solid red line). However, because

Figure 4: Equilibrium properties



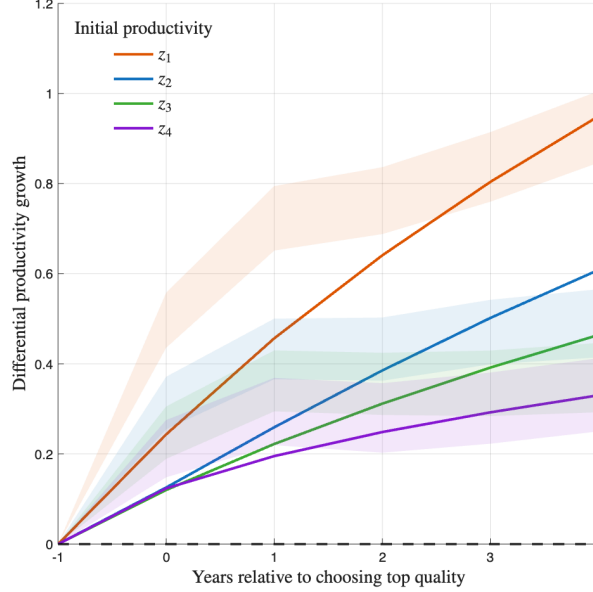
Notes: Figure 4a plots the quality choice probability by productivity. For a given productivity level, the size of the shaded areas (left axis) represents the average probability of choosing each quality across firm types, while the red line (right axis) shows the average quality level across firm types. Figure 4b plots the average log bilateral markup and demand with suppliers and buyers by productivity. For each firm type, we compute the average link-type-specific log markup or log demand, weighted by the trade share associated with that link type. We then take the average across firm types at each productivity level. Average log markup is scaled by $1 - \sigma$, and average log demand is scaled by σ .

their trading partners do not learn from them, these firms do not benefit from favorable contract terms from their partners, reflected by the solid blue and dashed red lines being zero. Quantitatively, for the least productive firms, learning-induced additional trade volumes are 19 percent with suppliers and 9 percent with buyers. By contrast, more productive firms tend to connect with trading partners of similar productivity. Since knowledge is not perfectly substitutable, they still learn from each other, though at a lower rate. For the most productive firms, learning-induced extra trade volumes are about 3 percent with both suppliers and buyers.²⁷

We further validate the model by examining firms' productivity growth after exposure to

²⁷Figure 4b suggests that average markups to buyers increase with firm productivity, while average markups by suppliers are relatively flat across the productivity distribution. We verify this pattern in the data. As our data do not allow us to estimate pair-specific markups, we use the accounting markup (total sales divided by total variable costs) as a measure of firm-level markup, which also corresponds to the average markup to buyers. We then compute the expenditure-share-weighted average firm-level markups across suppliers. Figure C4 in Appendix C plots the average markups by suppliers and to buyers across the productivity distribution and confirms the patterns hold in the data. Note that empirical markups may capture sources of market power that are outside the model. Nevertheless, learning-induced variations in markups in our model may partly explain the observed markup patterns across the productivity distribution.

Figure 5: Productivity after choosing top quality



Notes: The figure plots differential productivity growth after exposure to top partners in the model and the data, following the event study design in Section 2.3. In the model, the event is defined as choosing the highest quality, and treated firm types are those that choose the highest quality in period t but did not do so in period $t - 1$. The results are indicated by solid lines. In the data, the event is defined as connecting with both top suppliers and top buyers for the first time. The 95th confidence intervals are indicated by shaded areas.

top suppliers and buyers. In the model, exposure to top partners is approximated by choosing the highest quality, the market for which contains the majority of the most productive firms, as shown in Figure 4a. Following the event study design in Section 2.3, we define the event in the model as choosing the highest quality. Treated firm types are those that choose the highest quality in period t but did not do so in period $t - 1$. We group them by their productivity in period $t - 1$, simulate their productivity after the event, and compute the differential productivity growth relative to those with the highest pre-event productivity.²⁸

As shown by the lines in Figure 5, firms experience persistent differential productivity growth after the event. To interpret the results, note that in the model, the productivity effect in event year 0 is driven by selection where more productive firms are more likely to choose the highest quality and experience the event. However, the productivity trajectory after event year 0 is driven by learning. Learning raises the expected productivity in the next period, such that firms are more likely to connect with and learn from productive partners in

²⁸Specifically, the model generates a transition matrix for the vector of payoff-relevant states, denoted $\Psi = (\psi(\omega_-, \omega))_{\omega_-, \omega \in \Omega}$, where ω is the current state and ω_- is the previous state. Treated firm types are those with $q(\omega) = q_5$ and $q(\omega_-) < q_5$. Given the current distribution of firm types, $\tilde{h}(\omega)$, we compute the distribution of (ω_-, ω) by simulating backwards using Ψ , from which we can compute the distribution of treated firm types by pre-event productivity z_- . We then obtain the post-event productivity for different groups by simulating forward using Ψ .

subsequent periods. In other words, current exposure to productive partners places firms on a higher long-term growth path. Moreover, since the intensity of learning increases with the productivity gap given the submodular learning technology, initially less productive firms benefit disproportionately more from exposure to top partners, driving the widening gaps across different groups.

We also compare these patterns to those in the data. To this end, we replicate the event study design in Section 2.3 using the model-consistent productivity measure. Here, we define the event as connecting with both top suppliers and top buyers for the first time, corresponding to choosing top quality in the model. We then plot the 95th confidence intervals for estimates from the data as shaded areas in Figure 5. We find the model-generated differential productivity trajectories across groups of treated firms generally align with those in the data. Within a given group, the pattern in the data shows a slightly higher degree of concavity. Interpreting through the lens of the model, this suggests that productivity-based selection into connecting with top partners in event year 0 seems to be more stringent in the data. Nevertheless, the long-run productivity effects become closer between the two cases, suggesting that the model captures convergence in productivity over time. Taken together, although our model offers a parsimonious representation of the data, it is able to reproduce the event study patterns, even though they were not directly targeted in estimation.

5 Quantitative Implications

In this section, we examine the quantitative implications of the model through a series of counterfactual exercises. Section 5.1 quantifies the importance of learning by considering a counterfactual scenario in which firms do not learn from their suppliers or buyers. Section 5.2 then applies the model to analyze the effects of reducing international trade participation costs and to assess the role of learning in shaping the gains from trade.

5.1 Value of Learning

We first examine the role of learning in shaping firm-level and aggregate outcomes. To do so, we conduct a counterfactual exercise in which firms do not learn from their suppliers or buyers (i.e., $\lambda_S = \lambda_B = 0$), while keeping all other parameters at their baseline estimates. Table 4a reports the welfare changes under this no learning scenario. Welfare is measured as the sum of the present discounted value of consumption for all workers and entrepreneurs. The contribution of learning to welfare is substantial: aggregate welfare declines by about 19 percent in the absence of learning, indicating that learning accounts for roughly one-fifth

of total consumption. Entrepreneurs experience a welfare loss of 25 percent, as firms are directly affected by the absence of learning. Less productive firms are disproportionately affected, as they benefit more from learning, consistent with the results shown in Figure C6 in Appendix C. Workers experience a relatively smaller welfare loss of 16 percent that arises through general equilibrium adjustments in the domestic wage and the price index.

Table 4: Value of learning: welfare and firm outcomes

(a) Welfare changes, percent

Aggregate welfare	Entrepreneur welfare	Worker welfare
-18.93	-25.12	-16.18

(b) Firm outcome changes, percentage point

Productivity growth	International trade	Quality	Average log productivity
-2.38	-5.72	-3.10	-14.04

(c) Entrepreneur welfare change decomposition

Productivity growth	Network adjustment	Bilateral markup and demand	General equilibrium
-10.94	-11.42	-1.25	-4.00

Notes: The table summarizes welfare and firm outcomes in the counterfactual with no learning. Table 4a reports welfare changes in percent. Table 4b reports changes in selected firm outcomes in percentage points. Table 4c reports the decomposition of entrepreneur welfare change, with the methodology described in Appendix B.5 and additional details shown in Figure C7 in Appendix C.

Firms perform significantly worse in the no learning scenario. The absence of learning directly reduces firms’ productivity growth. Table 4b shows that the average productivity growth rate across firms (conditional on survival to the next period) decreases by more than 2 percentage points. Given an average productivity growth rate of about 1.35 percent in the baseline economy, this implies that firms, on average, experience negative productivity growth without learning. Moreover, firms adjust their networks: the share of firms participating in international trade falls by nearly 6 percentage points, and firms downgrade quality by around 3 percentage points. These results suggest that learning not only affects productivity growth but also alters firms’ incentives to form productive relationships. In equilibrium, the average log productivity across firms is about 14 percent lower. As a result, the economy becomes less knowledgeable overall.

Entrepreneur Welfare Change Decomposition. How do changes in firm performance translate into entrepreneur welfare? In Appendix B.5, we present a decomposition of the en-

entrepreneur welfare change up to a first-order approximation and summarize it into four main components: productivity growth, network adjustment, bilateral markup and demand, and general equilibrium. The productivity growth component captures how changes in learning affect firms' productivity paths over time, which in turn affect firm value and entrepreneur welfare. The network adjustment component captures factors related to international trade participation (including the associated fixed costs), quality, and matching with suppliers and buyers in a given market. Specifically, the match component reflects not only changes in the types of directly connected suppliers and buyers but also adjustments in matches for firms further upstream and downstream. Similarly, the bilateral markup and demand component captures changes in learning-induced contract terms accumulated along supply chains that shape welfare. Finally, the general equilibrium component captures factors related to the productivity distribution, price index, factor prices (i.e., domestic wage and foreign marginal cost), as well as final demand shifters (i.e., domestic final demand shifter and foreign demand shifter).

Table 4c reports the four main components from the decomposition of entrepreneur welfare change in the no-learning counterfactual. The detailed decomposition into various sub-components is shown in Figure C7 in Appendix C. We discuss the four main components in turn. First, the productivity growth component accounts for 10.94 percentage points of the total entrepreneur welfare loss. As expected, firms' growth prospects deteriorate substantially without learning, reducing firm value. Slower productivity growth is therefore a key determinant of the welfare value of learning for entrepreneurs.

Second, the network adjustment component accounts for 11.42 percentage points, making it even more important than the productivity growth component. Within this component, lower international trade participation explains 1.11 percentage points, and quality downgrading explains 1.08 percentage points. In addition, when suppliers and buyers are less likely to form productive relationships, firms on average match with worse partners within a given market, and the resulting deterioration in matching along the supply chain accounts for 9.39 percentage points. This suggests that the value of learning for firms arises primarily from the incentives to form productive relationships, and that productive efficiency gains from these relationships accumulate throughout the network.

Third, the bilateral markup and demand component contributes 1.25 percentage points. The relatively small contribution of this component may partly reflect the equilibrium PAM and the submodular learning technology. Under PAM, economic activity is concentrated among highly productive firms, which tend to have relatively lower learning rates with one another. With submodular learning, less productive firms disproportionately learn from the most productive partners with higher associated learning rates, but they account for a smaller share of overall economic activity. In an alternative economy with weaker PAM or a

less submodular learning technology, firms could either be more likely to connect with productive partners or learn from less productive partners, such that learning-induced changes in bilateral markup and demand would play a more important role.

Lastly, the general equilibrium component explains the remaining 4.00 percentage points of the total loss. Within this component, 7.46 percentage points are attributed to a thinner right tail of the productivity distribution, and 9.01 percentage points are due to a higher price index reflecting increased marginal costs across firms. These effects are offset by a lower domestic wage because of lower labor productivity, and by stronger domestic final demand because of eased competition.

Taken together, the decomposition results reveal that learning enhances firm value not only by raising productivity growth but also by improving network formation and other aspects of firm operations that jointly shape aggregate productivity in the economy.

5.2 Reductions in International Trade Participation Costs

We apply the model to study the effects of international trade promotion policies. In Belgium, government institutions such as the Belgian Foreign Trade Agency are dedicated to facilitating international trade by providing information about foreign markets and helping Belgian firms connect with foreign suppliers and buyers. To illustrate the effects of such policies, we consider a shock $\Delta\vartheta_f = -0.1$, which corresponds to a 10 percent reduction in the unconditional mean of the fixed costs of international trade.

Effects of the Shock. The effects of the shock in the baseline economy are reported in column (1) of Table 5. The shock leads to substantial welfare improvements: aggregate welfare rises by 1.18 percent, with entrepreneur welfare increasing by 1.67 percent and worker welfare by 0.96 percent. Firms also perform better on average, exhibiting faster productivity growth, a higher likelihood of participating in international trade, quality upgrading, and higher average productivity. Network adjustment is the main driver of the welfare gains for entrepreneurs. This occurs not only through the direct effects of greater trade participation resulting from lower fixed costs (1.04 percentage points) but also through domestic network adjustments, including quality upgrading and improved matching with suppliers and buyers (0.67 percentage points), as shown in column (1) of Table C4 in Appendix C. In other words, domestic network adjustments amplify the effects of increased international trade participation. Other components play a relatively modest role in explaining the entrepreneur welfare gains. Perhaps surprisingly, productivity growth contributes negatively to the gains. This reflects a shift in firms' exposure toward less productive domestic partners in lower-quality markets. As domestic firms upgrade quality in response to the shock, high-productivity firms

Table 5: Reductions in international trade participation costs: welfare and firm outcomes

	Baseline (1)	No learning (2)	Difference (3)
<i>Panel A: Welfare changes, percent</i>			
Aggregate welfare	1.18	1.09	-0.09
Entrepreneur welfare	1.67	1.58	-0.09
Worker welfare	0.96	0.89	-0.06
<i>Panel B: Firm outcome changes, percentage point</i>			
Productivity growth	0.01	-0.00	-0.01
International trade	0.83	0.66	-0.17
Quality	0.10	0.07	-0.03
Average log productivity	0.05	0.00	-0.05
<i>Panel C: Entrepreneur welfare change decomposition</i>			
Productivity growth	-0.02	0.00	0.02
Network adjustment	1.72	1.66	-0.06
Bilateral markup and demand	0.01	0.00	-0.01
General equilibrium	-0.03	-0.08	-0.04

Notes: The table summarizes changes in various welfare and firm outcomes from reductions in international trade participation costs. Panel A reports welfare changes in percent. Panel B reports changes in selected firm outcomes in percentage points. Panel C reports the decomposition of entrepreneur welfare change, with the methodology described in Appendix B.5 and additional details shown in Table C4 in Appendix C. Column (1) reports the effects in the baseline economy, column (2) those in the economy without learning, and column (3) the differences between the two.

move into top-quality markets, leaving fewer of them in the lower-quality segment, as shown in Figure C8 in Appendix C. Consequently, firms operating in lower-quality markets tend to match with less productive partners and experience slower productivity growth.²⁹ Overall, these results highlight the quantitative importance of endogenous network formation, particularly domestic network adjustments, in shaping the effects of trade policies on firms and the broader economy.

Gains from Trade and Learning. To assess the extent to which the welfare gains from trade are driven by learning, we examine the effects of the same shock in an alternative economy without learning. The results are reported in column (2) of Table 5, with differences

²⁹The results suggest that, holding the distribution of firm types fixed at the baseline, average productivity growth decreases following the shock. To reconcile this with the slight increase in average productivity growth in Panel B of Table 5, note that the latter also reflects a shift in the composition of firms toward higher-growth types. For instance, more firms participate in international trade after the shock, and these firms tend to have higher productivity growth rates.

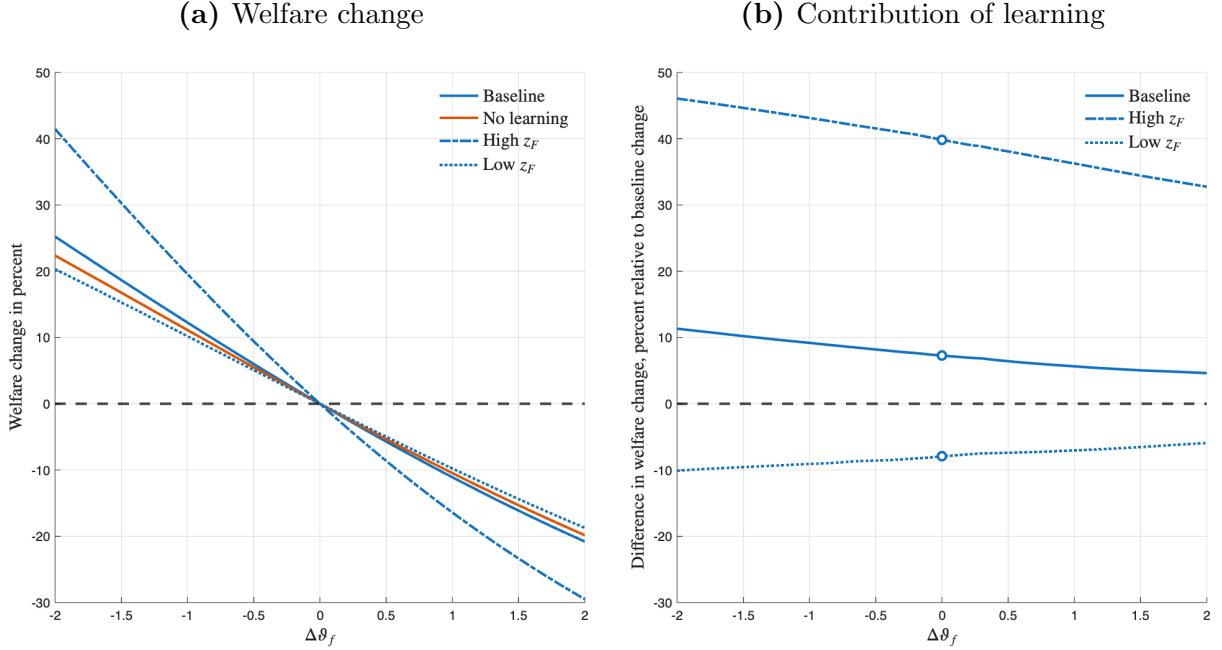
relative to the baseline shown in column (3). We estimate that the shock raises aggregate welfare by 1.09 percent in the no-learning economy, an effect 0.09 percentage points smaller than in the baseline. Put differently, learning accounts for roughly 7.6 percent of the welfare gains from trade. Moreover, in a world without learning, improvements in firm performance are smaller across multiple dimensions. Firms invest less in productive relationships at both the extensive and intensive margins when they do not internalize potential learning opportunities. This reduction in investment explains the missing entrepreneur welfare gains from trade: network adjustment accounts for 0.06 percentage points of the difference, while bilateral markup and demand account for an additional 0.01 percentage points. These findings further highlight how the key mechanisms of the model shape the welfare gains from trade.

We further explore how learning shapes gains from trade across different degrees of openness by varying the size of the shock. Figure 6a plots welfare changes in both the baseline economy and the no-learning economy under different shock magnitudes. We find that learning amplifies the welfare effects across a wide range of shocks. Figure 6b shows the contribution of learning to welfare by calculating the difference in welfare gains between the two economies as a percentage of the welfare gains in the baseline. Across the range of positive shocks considered, learning accounts for more than 7 percent of the welfare gains from trade, and its contribution rises with the degree of openness (i.e., as $\Delta\vartheta_f$ becomes smaller). Learning becomes disproportionately more valuable when greater openness increases the importance of foreign firms, which are on average more productive than domestic ones, as sources of learning. Accounting for learning is thus quantitatively important when evaluating the welfare effects of trade policies.

Gains from Trade and Foreign Productivity. We also examine how the gains from trade depend on the productivity of foreign partners. To this end, we consider two cases in which log foreign productivity is either one log point higher or one log point lower relative to the baseline level. The foreign marginal cost and demand shifter are fixed at their baseline values and do not vary with foreign productivity. This ensures that the resulting welfare effects are driven solely by differences in learning rather than by changes in productive efficiency. Naturally, varying foreign productivity alone would generate no effect in an economy without learning.

Gains from trade are further amplified when foreign partners are more productive, as shown by the dash-dotted lines in Figures 6a and 6b. Quantitatively, across the range of positive shocks, learning accounts for more than 40 percent of the gains from trade. As the productivity gap between foreign and domestic firms widens, foreign productivity becomes a more important source of learning and growth. These gains from better learning opportunities are further reinforced as firms adjust both their foreign and domestic networks.

Figure 6: Reductions in international trade participation costs: varying $\Delta\vartheta_f$ and z_F



Notes: The figure shows welfare changes under different shock magnitudes across several economies. Figure 6a plots welfare changes in the baseline economy (solid blue line), the economy without learning (solid red line), the economy with foreign productivity $\Delta \log z_F = 1$ (dash-dotted blue line), and the economy with foreign productivity $\Delta \log z_F = -1$ (dotted blue line). Figure 6b displays the differences in welfare changes relative to those in the economy without learning, expressed as a percentage of the welfare changes in the baseline economy. Undefined points are indicated as hollow circles. The shaded areas represent the range of gains from trade across within the interval of foreign productivity levels.

In contrast, when foreign partners are sufficiently unproductive, accounting for learning dampens the welfare gains from trade, as depicted by the dotted lines in Figures 6a and 6b. For the same range of positive shocks, the gains from trade would be overestimated by at least 8 percent without learning. With less productive foreign partners, firms lose the additional incentives to connect with foreign suppliers and buyers through learning, and these effects are further transmitted through adjustments in domestic networks. Moreover, greater exposure to unproductive foreign partners crowds out learning from productive domestic ones, leading to slower productivity growth and lower average productivity.³⁰ The results demonstrate that the productivity gap between foreign and domestic firms is crucial in shaping the learning, network, and overall gains from trade.

These results have several policy implications. On the one hand, policymakers should account for learning when evaluating the effects of trade policies. Through the lens of the model, the contribution of learning to welfare can be substantial, reflecting its effects on

³⁰Table C3 in Appendix C reports changes in welfare and firm outcomes under the shock $\Delta\vartheta_f = -0.1$ for cases with high and low foreign productivity. Table C4 in Appendix C provides a detailed decomposition of entrepreneur welfare changes for these two cases.

productivity growth and network adjustment. On the other hand, policymakers should consider the productivity of foreign partners and the potential for learning from them. Trade policies are more welfare-enhancing when they promote trade with more productive foreign partners, but may generate undesirable crowding-out effects when directed toward less productive ones.

6 Conclusion

Using comprehensive data on firm-to-firm transactions in Belgium, we document key evidence consistent with learning from both suppliers and buyers, particularly from the most productive ones. To study the aggregate implications of such firm learning, we develop a novel framework featuring firm productivity dynamics, endogenous network formation, and bargaining. A theoretical prediction is that when bilateral learning surpluses are internalized through bargaining, learning increases bilateral trade volumes. Specifically, suppliers charge lower markups when they learn more from buyers, and buyers demand more inputs when they learn more from suppliers. Quantifying the model for the Belgian economy, we find that the welfare value of learning is substantial: aggregate welfare decreases by about 19 percent in the absence of learning. Learning not only benefits firms through productivity growth but also shifts their incentives to form productive relationships. The resulting economy-wide adjustments in network structure contribute significantly to firm value. When applying the model to study reductions in international trade participation costs, we find that welfare gains from trade are underestimated by more than 7 percent without learning. The contribution of learning depends on the productivity gap between foreign and domestic firms. Opening up to trade with more productive foreign partners amplifies the gains from trade through productivity catch-up and network upgrading, while opening up to trade with less productive partners dampens the gains through crowding out domestic learning and reducing investment in productive relationships. These results suggest that trade policies should account for learning and promote trade with the most productive foreign partners.

Our theoretical framework can be extended in several directions to derive a broader set of policy implications. First, with more granular data on the identities of foreign suppliers and buyers, it would be possible to examine the interactions between heterogeneous domestic and foreign firms and to study trade policy effects targeting specific types of foreign partners, providing a more nuanced interpretation of our quantitative predictions. With additional information on the network structures of foreign firms, our small open economy model could also be adapted to study the global economy, where trade policies propagate through complex international supply chains. Second, potential learning opportunities are important considerations in the design of industrial policies. For example, policies that attract produc-

tive firms to specific regions (e.g., domestic and foreign direct investment subsidies) often cite learning benefits to local firms. Through the lens of our model, these learning benefits may be amplified by adjustments in input-output and geographic linkages. In another example, the existence of learning externalities often justifies policy interventions to address market failures. To the extent that firms internalize these learning benefits, the effectiveness of such policies may differ from standard predictions. Third, our framework can also be applied to study the implications of antitrust and competition policies, which usually weigh the potential negative effects of market power against efficiency gains that arise from the concentration of large firms. Our results suggest that productive large firms generate additional learning opportunities for the economy, potentially altering the cost-benefit trade-offs underlying such policies. More broadly, the effectiveness of these policies depends critically on the underlying learning technology, the endogenous network structure, and the extent to which firms internalize learning benefits. Accounting for these factors may alter the optimal design of policies for the efficient allocation of government resources, pointing to several promising directions for future research.

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Learning in Firm-to-Firm Trade

Online Appendix

Xianglong Kong Emmanuel Dhyne

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A Data and Additional Empirical Results

Table A1: Summary statistics

Variable	Mean	Std. dev.
Average number of firms per year	74,559.17	-
Total sales, million euro	8.76	172.01
Employment	26.01	284.64
Number of suppliers	45.86	70.59
Number of buyers	45.86	215.70
Importer share	0.21	-
Exporter share	0.14	-
R&D share	0.26	-
Labor productivity, log total sales per employment	12.11	0.98
Average supplier labor productivity, weighted by trade share	12.80	0.69
Average buyer labor productivity, weighted by trade share	12.55	0.75
Labor productivity growth	0.00	0.38

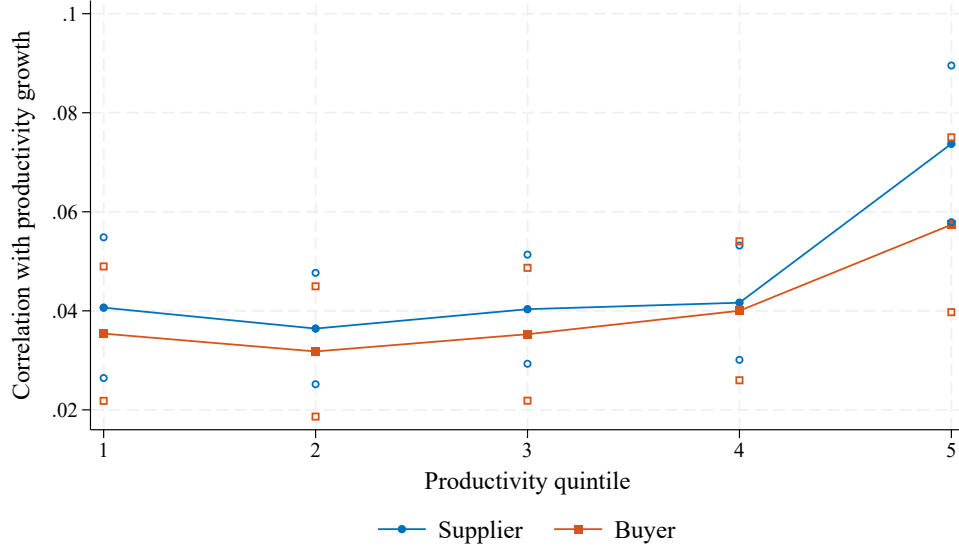
Notes: The table reports the summary statistics of selected variables in our analysis sample. Sample restrictions are described in Section 2.1.

Table A2: Results from estimating equations (1) and (2)

	(1)	(2)	(3)	(4)
Y_{jt}^S	0.025 (0.001)		0.010 (0.001)	
$Y_{jt,\text{non-top}}^S$		0.071 (0.005)		0.028 (0.006)
$Y_{jt,\text{top}}^S$		0.094 (0.005)		0.037 (0.006)
Y_{jt}^B	0.012 (0.001)		0.003 (0.001)	
$Y_{jt,\text{non-top}}^B$		0.037 (0.006)		-0.002 (0.006)
$Y_{jt,\text{top}}^B$		0.048 (0.006)		0.002 (0.006)
Importing	0.045 (0.002)	0.045 (0.002)	-0.017 (0.003)	-0.017 (0.003)
Exporting	0.043 (0.002)	0.042 (0.002)	0.005 (0.003)	0.005 (0.003)
R&D	0.000 (0.001)	0.000 (0.001)	0.000 (0.002)	0.000 (0.002)
Y_{jt}	0.745 (0.002)	0.745 (0.002)	0.346 (0.002)	0.346 (0.002)
Industry-year FE	✓	✓	✓	✓
Location-year FE	✓	✓	✓	✓
Firm FE			✓	✓
Observations	818805	818805	809172	809172
R^2	0.656	0.656	0.751	0.751

Notes: Standard errors are in parentheses.

Figure A1: Results from estimating equation (2), by productivity quintile



Notes: Mean estimates are indicated as the solid dots, while the 95th confidence intervals are indicated as the hollow dots.

Table A3: Results from estimating equation (2), long-run correlations

	Y_{jt+1} (1)	Y_{jt+2} (2)	Y_{jt+3} (3)	Y_{jt+4} (4)	Y_{jt+5} (5)
$Y_{jt,\text{non-top}}^S$	0.071 (0.005)	0.068 (0.006)	0.076 (0.007)	0.073 (0.008)	0.077 (0.009)
$Y_{jt,\text{top}}^S$	0.094 (0.005)	0.094 (0.006)	0.104 (0.007)	0.103 (0.008)	0.110 (0.009)
$Y_{jt,\text{non-top}}^B$	0.037 (0.006)	0.043 (0.007)	0.052 (0.008)	0.054 (0.009)	0.066 (0.010)
$Y_{jt,\text{top}}^B$	0.048 (0.006)	0.057 (0.007)	0.067 (0.008)	0.070 (0.009)	0.082 (0.010)
Importing	0.045 (0.002)	0.070 (0.003)	0.088 (0.003)	0.100 (0.004)	0.121 (0.004)
Exporting	0.042 (0.002)	0.045 (0.003)	0.051 (0.004)	0.055 (0.004)	0.051 (0.005)
R&D	0.000 (0.001)	0.000 (0.002)	-0.003 (0.002)	-0.005 (0.003)	-0.006 (0.003)
Y_{jt}	0.745 (0.002)	0.686 (0.002)	0.642 (0.002)	0.608 (0.002)	0.579 (0.003)
Industry-year FE	✓	✓	✓	✓	✓
Location-year FE	✓	✓	✓	✓	✓
Observations	818805	780346	707162	639859	577328
R^2	0.656	0.586	0.539	0.504	0.476

Notes: Standard errors are in parentheses.

Table A4: Results from estimating equation (2), by productivity quintile of self

	(1)	(2)	(3)	(4)	(5)
$Y_{jt,\text{non-top}}^S$	0.073 (0.013)	0.044 (0.008)	0.035 (0.009)	0.039 (0.010)	0.090 (0.015)
$Y_{jt,\text{top}}^S$	0.091 (0.014)	0.058 (0.009)	0.045 (0.009)	0.048 (0.010)	0.108 (0.015)
$Y_{jt,\text{non-top}}^B$	0.030 (0.015)	0.029 (0.009)	0.046 (0.010)	0.029 (0.011)	0.063 (0.014)
$Y_{jt,\text{top}}^B$	0.036 (0.015)	0.033 (0.009)	0.048 (0.010)	0.035 (0.011)	0.076 (0.014)
Importing	0.076 (0.007)	0.043 (0.004)	0.033 (0.004)	0.028 (0.004)	0.021 (0.004)
Exporting	0.037 (0.009)	0.007 (0.005)	0.012 (0.005)	0.026 (0.004)	0.039 (0.005)
R&D	0.013 (0.005)	0.001 (0.003)	0.002 (0.003)	-0.005 (0.003)	-0.005 (0.003)
Y_{jt}	0.352 (0.005)	0.782 (0.010)	0.814 (0.011)	0.843 (0.008)	0.892 (0.004)
Industry-year FE	✓	✓	✓	✓	✓
Location-year FE	✓	✓	✓	✓	✓
Observations	154686	168064	167605	162925	160344
R^2	0.153	0.111	0.109	0.146	0.571

Notes: Standard errors are in parentheses.

Table A5: Results from estimating equation (2), by age group of self

	Young (1)	Old (2)	Missing (3)
$Y_{jt,\text{non-top}}^S$	0.050 (0.015)	0.072 (0.013)	0.074 (0.006)
$Y_{jt,\text{top}}^S$	0.069 (0.016)	0.089 (0.013)	0.098 (0.006)
$Y_{jt,\text{non-top}}^B$	0.048 (0.019)	0.034 (0.015)	0.036 (0.006)
$Y_{jt,\text{top}}^B$	0.060 (0.019)	0.042 (0.016)	0.048 (0.006)
Importing	0.051 (0.009)	0.056 (0.006)	0.044 (0.002)
Exporting	0.079 (0.013)	0.059 (0.009)	0.039 (0.003)
R&D	0.013 (0.004)	0.002 (0.004)	-0.004 (0.002)
Y_{jt}	0.665 (0.004)	0.750 (0.004)	0.753 (0.002)
Industry-year FE	✓	✓	✓
Location-year FE	✓	✓	✓
Observations	77563	93740	645209
R^2	0.615	0.677	0.664

Notes: Standard errors are in parentheses. Age is defined as the number of years since the first year of a firm's appearance in the data. Age is missing for firms that appear in the first year of the data.

Table A6: Results from estimating equation (2), by characteristics of partners

	Old firm (1)	Old link (2)	Int'l trade (3)	R&D (4)
$Y_{jt,\text{non-top},g=0}^S$	0.025 (0.002)	0.074 (0.005)	0.072 (0.005)	0.077 (0.005)
$Y_{jt,\text{non-top},g=1}^S$	0.031 (0.002)	0.080 (0.006)	0.075 (0.005)	0.069 (0.005)
$Y_{jt,\text{top},g=0}^S$	0.021 (0.001)	0.050 (0.003)	0.053 (0.003)	0.070 (0.004)
$Y_{jt,\text{top},g=1}^S$	0.029 (0.002)	0.068 (0.004)	0.084 (0.005)	0.070 (0.004)
$Y_{jt,\text{non-top},g=0}^B$	0.013 (0.002)	0.038 (0.006)	0.028 (0.006)	0.034 (0.006)
$Y_{jt,\text{non-top},g=1}^B$	0.016 (0.003)	0.034 (0.005)	0.037 (0.006)	0.037 (0.005)
$Y_{jt,\text{top},g=0}^B$	0.011 (0.002)	0.028 (0.004)	0.021 (0.003)	0.033 (0.004)
$Y_{jt,\text{top},g=1}^B$	0.015 (0.002)	0.032 (0.004)	0.040 (0.005)	0.037 (0.004)
Importing	0.045 (0.002)	0.046 (0.002)	0.043 (0.002)	0.045 (0.002)
Exporting	0.042 (0.002)	0.044 (0.002)	0.040 (0.002)	0.042 (0.002)
R&D	0.000 (0.001)	-0.002 (0.001)	0.000 (0.001)	0.000 (0.001)
Y_{jt}	0.745 (0.002)	0.744 (0.002)	0.744 (0.002)	0.745 (0.002)
Industry-year FE	✓	✓	✓	✓
Location-year FE	✓	✓	✓	✓
Observations	818805	818805	818805	818805
R^2	0.656	0.656	0.656	0.656

Notes: Standard errors are in parentheses.

Table A7: Results from estimating equation (2), alternative productivity measures

	Baseline (1)	OP (2)	LP (3)	ACF (4)	Wooldridge (5)	Model (6)
$Y_{jt,\text{non-top}}^S$	0.071 (0.005)	0.046 (0.003)	0.054 (0.004)	0.042 (0.003)	0.050 (0.004)	0.049 (0.003)
$Y_{jt,\text{top}}^S$	0.094 (0.005)	0.068 (0.003)	0.078 (0.004)	0.063 (0.003)	0.074 (0.004)	0.068 (0.003)
$Y_{jt,\text{non-top}}^B$	0.037 (0.006)	0.026 (0.003)	0.050 (0.004)	0.015 (0.003)	0.045 (0.004)	0.019 (0.003)
$Y_{jt,\text{top}}^B$	0.048 (0.006)	0.049 (0.003)	0.077 (0.004)	0.036 (0.003)	0.070 (0.004)	0.033 (0.003)
Importing	0.045 (0.002)	0.091 (0.002)	0.110 (0.002)	0.076 (0.002)	0.105 (0.002)	0.063 (0.001)
Exporting	0.042 (0.002)	0.083 (0.002)	0.091 (0.002)	0.075 (0.002)	0.090 (0.002)	0.016 (0.001)
R&D	0.000 (0.001)	0.034 (0.001)	0.045 (0.001)	0.025 (0.001)	0.042 (0.001)	0.034 (0.001)
Y_{jt}	0.745 (0.002)	0.733 (0.002)	0.753 (0.002)	0.723 (0.002)	0.748 (0.002)	0.862 (0.001)
Industry-year FE	✓	✓	✓	✓	✓	✓
Location-year FE	✓	✓	✓	✓	✓	✓
Observations	818805	832591	832591	832591	832591	942643
R^2	0.656	0.656	0.696	0.632	0.685	0.817

Notes: Standard errors are in parentheses.

Table A8: Event study, table of estimated coefficients

	Supplier event				Buyer event			
	Min-20th	20th-40th	40th-60th	60th-80th	Min-20th	20th-40th	40th-60th	60th-80th
$s = -4$	-0.110 (0.083)	0.090 (0.084)	0.039 (0.080)	0.011 (0.071)	-0.030 (0.038)	-0.021 (0.031)	0.043 (0.030)	0.068 (0.033)
$s = -3$	-0.077 (0.070)	-0.024 (0.070)	0.034 (0.070)	0.009 (0.064)	-0.054 (0.034)	-0.048 (0.027)	0.005 (0.027)	0.036 (0.029)
$s = -2$	-0.008 (0.068)	0.004 (0.060)	-0.006 (0.054)	0.017 (0.051)	-0.080 (0.030)	-0.023 (0.022)	-0.005 (0.022)	0.045 (0.023)
$s = -1$	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)
$s = 0$	0.583 (0.061)	0.371 (0.058)	0.209 (0.057)	0.176 (0.054)	0.474 (0.030)	0.251 (0.024)	0.206 (0.024)	0.134 (0.026)
$s = 1$	0.760 (0.064)	0.480 (0.062)	0.343 (0.057)	0.231 (0.055)	0.610 (0.031)	0.305 (0.026)	0.251 (0.025)	0.181 (0.027)
$s = 2$	0.849 (0.068)	0.527 (0.067)	0.370 (0.067)	0.282 (0.061)	0.693 (0.034)	0.408 (0.029)	0.327 (0.028)	0.236 (0.030)
$s = 3$	0.838 (0.070)	0.574 (0.068)	0.444 (0.069)	0.280 (0.062)	0.754 (0.035)	0.446 (0.030)	0.340 (0.029)	0.230 (0.030)
$s = 4$	0.874 (0.073)	0.541 (0.070)	0.460 (0.070)	0.189 (0.066)	0.819 (0.036)	0.491 (0.032)	0.408 (0.031)	0.237 (0.033)

Notes: Standard errors are in parentheses.

Table A9: Event study, supplier event, alternative productivity measures

	Baseline (1)	OP (2)	LP (3)	ACF (4)	Wooldridge (5)	Model (6)
<i>Panel A: Quintile 1 of pre-event productivity</i>						
$-4 \leq s \leq -2$	-0.065 (0.058)	0.084 (0.102)	0.032 (0.104)	0.063 (0.093)	0.050 (0.103)	-0.208 (0.139)
$s = 0$	0.583 (0.061)	0.996 (0.102)	0.944 (0.107)	0.972 (0.096)	0.959 (0.107)	1.109 (0.130)
$1 \leq s \leq 4$	0.830 (0.061)	1.356 (0.105)	1.283 (0.106)	1.320 (0.100)	1.314 (0.107)	1.411 (0.136)
<i>Panel B: Quintile 2 of pre-event productivity</i>						
$-4 \leq s \leq -2$	0.024 (0.058)	0.037 (0.100)	0.090 (0.110)	0.140 (0.100)	0.097 (0.111)	0.237 (0.103)
$s = 0$	0.371 (0.058)	0.613 (0.099)	0.661 (0.107)	0.658 (0.094)	0.673 (0.104)	0.434 (0.142)
$1 \leq s \leq 4$	0.531 (0.058)	0.799 (0.107)	0.687 (0.112)	0.807 (0.102)	0.701 (0.114)	0.585 (0.131)
<i>Panel C: Quintile 3 of pre-event productivity</i>						
$-4 \leq s \leq -2$	0.022 (0.056)	0.119 (0.112)	0.181 (0.098)	0.165 (0.096)	0.159 (0.105)	0.058 (0.105)
$s = 0$	0.209 (0.057)	0.472 (0.118)	0.424 (0.107)	0.558 (0.111)	0.404 (0.112)	0.262 (0.111)
$1 \leq s \leq 4$	0.404 (0.057)	0.579 (0.117)	0.632 (0.108)	0.729 (0.104)	0.595 (0.112)	0.512 (0.097)
<i>Panel D: Quintile 4 of pre-event productivity</i>						
$-4 \leq s \leq -2$	0.012 (0.050)	0.136 (0.080)	0.132 (0.083)	0.083 (0.086)	0.161 (0.078)	0.027 (0.060)
$s = 0$	0.176 (0.054)	0.414 (0.094)	0.509 (0.100)	0.313 (0.095)	0.516 (0.097)	0.227 (0.087)
$1 \leq s \leq 4$	0.245 (0.052)	0.560 (0.091)	0.469 (0.095)	0.445 (0.092)	0.496 (0.092)	0.260 (0.084)

Notes: Standard errors are in parentheses.

Table A10: Event study, buyer event, alternative productivity measures

	Baseline (1)	OP (2)	LP (3)	ACF (4)	Wooldridge (5)	Model (6)
<i>Panel A: Quintile 1 of pre-event productivity</i>						
$-4 \leq s \leq -2$	-0.055 (0.028)	-0.059 (0.036)	-0.004 (0.035)	-0.094 (0.036)	-0.005 (0.035)	-0.112 (0.028)
$s = 0$	0.474 (0.030)	0.612 (0.037)	0.629 (0.039)	0.616 (0.036)	0.640 (0.039)	0.560 (0.032)
$1 \leq s \leq 4$	0.719 (0.029)	0.935 (0.039)	0.945 (0.040)	0.939 (0.038)	0.944 (0.040)	0.831 (0.035)
<i>Panel B: Quintile 2 of pre-event productivity</i>						
$-4 \leq s \leq -2$	-0.031 (0.022)	-0.035 (0.030)	0.010 (0.029)	-0.078 (0.029)	0.013 (0.029)	-0.044 (0.024)
$s = 0$	0.251 (0.024)	0.372 (0.032)	0.398 (0.034)	0.355 (0.029)	0.398 (0.034)	0.349 (0.030)
$1 \leq s \leq 4$	0.413 (0.025)	0.620 (0.033)	0.642 (0.036)	0.590 (0.031)	0.624 (0.035)	0.471 (0.033)
<i>Panel C: Quintile 3 of pre-event productivity</i>						
$-4 \leq s \leq -2$	0.014 (0.022)	-0.025 (0.030)	-0.001 (0.030)	-0.042 (0.029)	-0.005 (0.030)	0.001 (0.023)
$s = 0$	0.206 (0.024)	0.311 (0.032)	0.346 (0.034)	0.316 (0.030)	0.346 (0.034)	0.276 (0.030)
$1 \leq s \leq 4$	0.332 (0.024)	0.553 (0.032)	0.559 (0.035)	0.533 (0.030)	0.546 (0.035)	0.371 (0.033)
<i>Panel D: Quintile 4 of pre-event productivity</i>						
$-4 \leq s \leq -2$	0.049 (0.024)	-0.046 (0.031)	0.009 (0.033)	-0.064 (0.029)	0.010 (0.033)	0.010 (0.026)
$s = 0$	0.134 (0.026)	0.199 (0.033)	0.241 (0.037)	0.205 (0.031)	0.259 (0.035)	0.240 (0.033)
$1 \leq s \leq 4$	0.221 (0.025)	0.409 (0.034)	0.435 (0.037)	0.393 (0.031)	0.431 (0.036)	0.315 (0.036)

Notes: Standard errors are in parentheses.

B Model Appendix

B.1 Model Solutions

From equation (4), it is convenient to write $l(\omega)$ in terms of $y(\omega)$ and $m(\omega)$:

$$l(\omega) = \left(\left(\frac{y(\omega)}{z(\omega)} \right)^{\frac{\rho-1}{\rho}} - \alpha^{\frac{1}{\rho}} m(\omega)^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}. \quad (\text{B1})$$

Also write $m(\omega)$ in terms of $y(\omega)$ and $l(\omega)$:

$$m(\omega) = \alpha^{\frac{1}{1-\rho}} \left(\left(\frac{y(\omega)}{z(\omega)} \right)^{\frac{\rho-1}{\rho}} - l(\omega)^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}. \quad (\text{B2})$$

Given equation (B1), compute the partial derivative of $l(\omega)$ with respect to the marginal supplier ω' ordered to be the last element in $S(\omega)$:

$$\begin{aligned} \frac{\partial l(\omega)}{\partial n_S(\omega)} &= \frac{\partial l(\omega)}{\partial m(\omega)} \frac{\partial m(\omega)}{\partial n_S(\omega)} \\ &= -\frac{\sigma}{\sigma-1} \alpha^{\frac{1}{\rho}} \left(\left(\frac{y(\omega)}{z(\omega)} \right)^{\frac{\rho-1}{\rho}} - \alpha^{\frac{1}{\rho}} m(\omega)^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}-1} m(\omega)^{\frac{1}{\sigma}-\frac{1}{\rho}} n_S(\omega)^{-\varphi} \phi(q(\omega))^{\frac{1}{\sigma}} m(\omega', \omega)^{\frac{\sigma-1}{\sigma}} \\ &= -\frac{\sigma}{\sigma-1} \alpha^{\frac{1}{\rho} + \frac{1}{1-\rho}(\frac{1}{\sigma}-\frac{1}{\rho})} l(\omega)^{\frac{1}{\sigma}} \left(\left(\frac{y(\omega)}{z(\omega)l(\omega)} \right)^{\frac{\rho-1}{\rho}} - 1 \right)^{\frac{\rho}{\rho-1}(\frac{1}{\sigma}-\frac{1}{\rho})} n_S(\omega)^{-\varphi} \phi(q(\omega))^{\frac{1}{\sigma}} m(\omega', \omega)^{\frac{\sigma-1}{\sigma}}, \end{aligned} \quad (\text{B3})$$

where the third equality uses equation (B2) to eliminate $m(\omega)$. Also compute the partial derivative of $l(\omega)$ with respect to the marginal buyer ω' ordered to be the last element in $B(\omega)$:

$$\begin{aligned} \frac{\partial l(\omega)}{\partial n_B(\omega)} &= \frac{\partial l(\omega)}{\partial y(\omega)} \frac{\partial y(\omega)}{\partial n_B(\omega)} \\ &= \left(\left(\frac{y(\omega)}{z(\omega)} \right)^{\frac{\rho-1}{\rho}} - \alpha^{\frac{1}{\rho}} m(\omega)^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}-1} z(\omega)^{-\frac{\rho-1}{\rho}} y(\omega)^{-\frac{1}{\rho}} m(\omega, \omega') \\ &= z(\omega)^{-1} \left(\frac{y(\omega)}{z(\omega)l(\omega)} \right)^{-\frac{1}{\rho}} m(\omega, \omega'), \end{aligned} \quad (\text{B4})$$

where the second equality uses the market clearing condition for ω , and the third equality uses equation (B2) to eliminate $m(\omega)$.

B.1.1 Learning Rate

Recall that learning rate are defined as:

$$\begin{aligned}\nu_B(\omega'; \omega) &\equiv \frac{1}{x(\omega, \omega')} \beta(1 - \delta) \int_{\mathbb{Z}} V(z') \frac{\partial g(z'; \omega)}{\partial n_B(\omega)} dz', \\ \nu_S(\omega'; \omega) &\equiv \frac{1}{x(\omega', \omega)} \beta(1 - \delta) \int_{\mathbb{Z}} V(z') \frac{\partial g(z'; \omega)}{\partial n_S(\omega)} dz'.\end{aligned}$$

Define $\bar{z}_+(\omega)$ such that $\log z_+(\omega) = \log \bar{z}_+(\omega) + \xi_+(\omega)$. Using the chain rule, compute partial derivatives inside the integral:

$$\begin{aligned}\frac{\partial g(z'; \omega)}{\partial n_B(\omega)} &= \frac{\partial g(z'; \omega)}{\partial \log \bar{z}_+(\omega)} \frac{\partial \log \bar{z}_+(\omega)}{\partial n_B(\omega)} \\ \frac{\partial g(z'; \omega)}{\partial n_S(\omega)} &= \frac{\partial g(z'; \omega)}{\partial \log \bar{z}_+(\omega)} \frac{\partial \log \bar{z}_+(\omega)}{\partial n_S(\omega)}.\end{aligned}$$

When $\xi_+(\omega)$ is normal with mean zero and variance ς_ξ^2 , future productivity $z_+(\omega)$ is log-normal where $\log z_+(\omega)$ has mean $\log \bar{z}_+(\omega)$ and variance ς_ξ^2 . The partial derivative $\frac{\partial g(z'; \omega)}{\partial \log \bar{z}_+(\omega)}$ is known. It remains to compute $\frac{\partial \log \bar{z}_+(\omega)}{\partial n_B(\omega)}$ and $\frac{\partial \log \bar{z}_+(\omega)}{\partial n_S(\omega)}$. Given the functional form of the learning technology in equation (6),

$$\begin{aligned}\frac{\partial \log \bar{z}_+(\omega)}{\partial n_B(\omega)} &= \frac{\partial \log \bar{z}_+(\omega)}{\partial \log \bar{z}_B(\omega)} \frac{\partial \log \bar{z}_B(\omega)}{\partial n_B(\omega)} = \frac{\gamma \lambda_B}{\eta} \left(\frac{\bar{z}_B(\omega)}{\bar{z}_+(\omega)} \right)^\varsigma \left(\frac{z(\omega')}{\bar{z}_B(\omega)} \right)^\eta \frac{x(\omega, \omega')}{\int_{B(\omega)} x(\omega, \omega'') d\omega''} \\ \frac{\partial \log \bar{z}_+(\omega)}{\partial n_S(\omega)} &= \frac{\partial \log \bar{z}_+(\omega)}{\partial \log \bar{z}_S(\omega)} \frac{\partial \log \bar{z}_S(\omega)}{\partial n_S(\omega)} = \frac{\gamma \lambda_S}{\eta} \left(\frac{\bar{z}_S(\omega)}{\bar{z}_+(\omega)} \right)^\varsigma \left(\frac{z(\omega')}{\bar{z}_S(\omega)} \right)^\eta \frac{x(\omega', \omega)}{\int_{S(\omega)} x(\omega'', \omega) d\omega''}.\end{aligned}$$

We assume that each trading relationship is small such that it does not affect other relationships and firm-level outcomes. Taken together, the learning rates are given by:

$$\nu_B(\omega'; \omega) = \beta(1 - \delta) \int_{\mathbb{Z}} V(z') \frac{\partial g(z'; \omega)}{\partial \log \bar{z}_+(\omega)} \frac{\gamma \lambda_B}{\eta} \left(\frac{\bar{z}_B(\omega)}{\bar{z}_+(\omega)} \right)^\varsigma \left(\frac{z(\omega')}{\bar{z}_B(\omega)} \right)^\eta \left(\int_{B(\omega)} x(\omega', \omega'') d\omega'' \right)^{-1} dz' \quad (\text{B5})$$

$$\nu_S(\omega'; \omega) = \beta(1 - \delta) \int_{\mathbb{Z}} V(z') \frac{\partial g(z'; \omega)}{\partial \log \bar{z}_+(\omega)} \frac{\gamma \lambda_S}{\eta} \left(\frac{\bar{z}_S(\omega)}{\bar{z}_+(\omega)} \right)^\varsigma \left(\frac{z(\omega')}{\bar{z}_S(\omega)} \right)^\eta \left(\int_{S(\omega)} x(\omega'', \omega) d\omega'' \right)^{-1} dz'. \quad (\text{B6})$$

Suppose $\lambda_B \geq 0$, $\lambda_S \geq 0$, $\eta > 0$, and in equilibrium, firm value is increasing in productivity (i.e., $V'(z) \geq 0$). Then, given a firm ω , learning rates are increasing in the productivity of its suppliers or buyers.

B.1.2 Proof of Proposition 1

Proof. Recall that the supplier surplus is:

$$V_S(m, x; \omega', \omega) = x - c(\omega')m + \nu_B(\omega; \omega')x,$$

and the buyer surplus is:

$$V_B(m, x; \omega', \omega) = -w \frac{\partial l(\omega)}{\partial n_S(\omega)} - x + \nu_S(\omega'; \omega)x.$$

Write

$$-w \frac{\partial l(\omega)}{\partial n_S(\omega)} = \frac{\sigma}{\sigma - 1} d(\omega)^{\frac{1}{\sigma}} m^{\frac{\sigma-1}{\sigma}},$$

where from equation (B3), define $d(\omega)$ as

$$d(\omega)^{\frac{1}{\sigma}} = \alpha^{\frac{1}{\rho} + \frac{1}{1-\rho}(\frac{1}{\sigma} - \frac{1}{\rho})} l(\omega)^{\frac{1}{\sigma}} \left(\left(\frac{y(\omega)}{z(\omega)l(\omega)} \right)^{\frac{\rho-1}{\rho}} - 1 \right)^{\frac{\rho}{\rho-1}(\frac{1}{\sigma} - \frac{1}{\rho})} n_S(\omega)^{-\varphi} \phi(q(\omega))^{\frac{1}{\sigma}} w \quad (\text{B7})$$

Solve the bargaining problem:

$$\max_{m, x} \kappa \log(x - c(\omega')m + \nu_B(\omega; \omega')x) + (1 - \kappa) \log \left(\frac{\sigma}{\sigma - 1} d(\omega)^{\frac{1}{\sigma}} m^{\frac{\sigma-1}{\sigma}} - x + \nu_S(\omega'; \omega)x \right)$$

The first-order conditions are given by:

$$[x] : \frac{\kappa}{V_S(m, x; \omega', \omega)} (1 + \nu_B(\omega; \omega')) = \frac{1 - \kappa}{V_B(m, x; \omega', \omega)} (1 - \nu_S(\omega'; \omega)) \quad (\text{B8})$$

$$[m] : \frac{\kappa}{V_S(m, x; \omega', \omega)} c(\omega') = \frac{1 - \kappa}{V_B(m, x; \omega', \omega)} d(\omega)^{\frac{1}{\sigma}} m^{-\frac{1}{\sigma}} \quad (\text{B9})$$

(B8) and (B9) imply:

$$\frac{1 - \nu_S(\omega'; \omega)}{1 + \nu_B(\omega; \omega')} c(\omega') = d(\omega)^{\frac{1}{\sigma}} m^{-\frac{1}{\sigma}}$$

Re-arrange and obtain an expression for $m(\omega', \omega)$:

$$m(\omega', \omega) = \left(\frac{1 - \nu_S(\omega'; \omega)}{1 + \nu_B(\omega; \omega')} c(\omega') \right)^{-\sigma} d(\omega) \quad (\text{B10})$$

Equation (B8) can be written as:

$$\begin{aligned} \frac{1}{\kappa} (1 - \nu_S(\omega'; \omega)) x &= \frac{\sigma}{\sigma - 1} d(\omega)^{\frac{1}{\sigma}} m^{\frac{\sigma-1}{\sigma}} + \frac{1 - \kappa}{\kappa} \frac{1 - \nu_S(\omega'; \omega)}{1 + \nu_B(\omega; \omega')} c(\omega') m \\ &= \left(\frac{\sigma}{\sigma - 1} + \frac{1 - \kappa}{\kappa} \right) \left(\frac{1 - \nu_S(\omega'; \omega)}{1 + \nu_B(\omega; \omega')} c(\omega') \right)^{1-\sigma} d(\omega), \end{aligned}$$

where the second equality uses equation (B10). Re-arrange and obtain an expression for $x(\omega', \omega)$:

$$x(\omega', \omega) = \left(\frac{\mu}{1 + \nu_B(\omega; \omega')} c(\omega') \right)^{1-\sigma} \left(\frac{\mu}{1 - \nu_S(\omega'; \omega)} \right)^{\sigma} d(\omega) \quad (\text{B11})$$

where $\mu = \kappa \frac{\sigma}{\sigma-1} + 1 - \kappa$. Note that equation (B10) can be written as:

$$m(\omega', \omega) = \left(\frac{\mu}{1 + \nu_B(\omega; \omega')} c(\omega') \right)^{-\sigma} \left(\frac{\mu}{1 - \nu_S(\omega'; \omega)} \right)^{\sigma} d(\omega) \quad (\text{B12})$$

Define bilateral price $p(\omega', \omega)$ as:

$$p(\omega', \omega) = \frac{x(\omega', \omega)}{m(\omega', \omega)} = \frac{\mu}{1 + \nu_B(\omega; \omega')} c(\omega') \equiv \mu(\omega', \omega) c(\omega'). \quad (\text{B13})$$

It remains to characterize quantities and trade volumes to final good firms. Firm ω makes a take-it-or-leave-it offer to final good firms and has full bargaining power. This implies that:

$$m(\omega, H) = \left(\frac{\sigma}{\sigma - 1} c(\omega) \right)^{-\sigma} \left(\frac{\sigma}{\sigma - 1} \right)^{\sigma} D \quad (\text{B14})$$

$$x(\omega, H) = \left(\frac{\sigma}{\sigma - 1} c(\omega) \right)^{1-\sigma} \left(\frac{\sigma}{\sigma - 1} \right)^{\sigma} D \quad (\text{B15})$$

$$p(\omega, H) = \frac{x(\omega, H)}{m(\omega, H)} = \frac{\sigma}{\sigma - 1} c(\omega). \quad (\text{B16})$$

This concludes the proof. □

B.1.3 Proof of Proposition 2

Proof. First, derive the equation for the vector of marginal costs. Note that the total variable costs of firm ω is

$$\begin{aligned} tvc(\omega) &= \int_{S(\omega)} x(\omega', \omega) d\omega' + wl(\omega) \\ &= \int_{S(\omega)} \left(\frac{\mu}{1 + \nu_B(\omega; \omega')} c(\omega') \right)^{1-\sigma} \left(\frac{\mu}{1 - \nu_S(\omega'; \omega)} \right)^{\sigma} d(\omega) d\omega' + wl(\omega), \end{aligned}$$

where the second equality uses equations (B11). To proceed, note that the marginal cost of firm ω satisfies:

$$c(\omega) = wz(\omega)^{-1} \left(\frac{y(\omega)}{z(\omega)l(\omega)} \right)^{-\frac{1}{\rho}} \implies l(\omega) = \left(\frac{z(\omega)c(\omega)}{w} \right)^{\rho} \frac{y(\omega)}{z(\omega)}.$$

Plug $l(\omega)$ into equation (B7) to write $d(\omega)$ in terms of $y(\omega)$:

$$\begin{aligned} d(\omega) &= \alpha^{\frac{\sigma}{\rho} + \frac{1}{1-\rho}(1-\frac{\sigma}{\rho})} l(\omega) \left(\left(\frac{y(\omega)}{z(\omega)l(\omega)} \right)^{\frac{\rho-1}{\rho}} - 1 \right)^{\frac{\rho}{\rho-1}(1-\frac{\sigma}{\rho})} n_S(\omega)^{-\varphi\sigma} \phi(q(\omega)) w^{\sigma} \\ &= \alpha^{\frac{\sigma-1}{\rho-1}} \left(\frac{z(\omega)c(\omega)}{w} \right)^{\rho} \frac{y(\omega)}{z(\omega)} \left(\left(\frac{z(\omega)c(\omega)}{w} \right)^{1-\rho} - 1 \right)^{\frac{\rho-\sigma}{\rho-1}} n_S(\omega)^{-\varphi\sigma} \phi(q(\omega)) w^{\sigma}. \end{aligned}$$

Define the cost index for intermediate inputs as:

$$c_M(\omega) = \left(n_S(\omega)^{-\varphi\sigma} \int_{S(\omega)} \phi(q(\omega)) \left(\frac{\mu}{1 - \nu_S(\omega'; \omega)} \right)^{\sigma} \left(\frac{\mu}{1 + \nu_B(\omega; \omega')} c(\omega') \right)^{1-\sigma} \right)^{\frac{1}{1-\sigma}}. \quad (\text{B17})$$

By definition, marginal cost satisfies $c(\omega) = \frac{\partial tvc(\omega)}{\partial y(\omega)}$. Therefore,

$$c(\omega) = c_M(\omega)^{1-\sigma} \alpha^{\frac{\sigma-1}{\rho-1}} \left(\frac{z(\omega)c(\omega)}{w} \right)^{\rho} \frac{1}{z(\omega)} \left(\left(\frac{z(\omega)c(\omega)}{w} \right)^{1-\rho} - 1 \right)^{\frac{\rho-\sigma}{\rho-1}} w^{\sigma} + \left(\frac{z(\omega)c(\omega)}{w} \right)^{\rho} \frac{1}{z(\omega)} w \quad (\text{B18})$$

Claim 1. *The marginal cost of firm ω is given by:*

$$c(\omega) = \frac{1}{z(\omega)} \left(\alpha c_M(\omega)^{1-\rho} + w^{1-\rho} \right)^{\frac{1}{1-\rho}}, \quad (\text{B19})$$

where $c_M(\omega)$ is defined in equation (B17).

Proof. We prove the claim by verifying that equation (B19) satisfies equation (B18). It is equivalent to show that the following equation holds:

$$\left(\frac{z(\omega)c(\omega)}{w}\right)^{1-\rho} - 1 = \alpha^{\frac{\sigma-1}{\rho-1}} \left(\left(\frac{z(\omega)c(\omega)}{w}\right)^{1-\rho} - 1\right)^{\frac{\rho-\sigma}{\rho-1}} \left(\frac{c_M(\omega)}{w}\right)^{1-\sigma}.$$

Equation (B19) implies that

$$\left(\frac{z(\omega)c(\omega)}{w}\right)^{1-\rho} - 1 = \alpha \left(\frac{c_M(\omega)}{w}\right)^{1-\rho}.$$

Plug it into the equation above:

$$\begin{aligned} rhs &= \alpha^{\frac{\sigma-1}{\rho-1}} \left(\left(\frac{z(\omega)c(\omega)}{w}\right)^{1-\rho} - 1\right)^{\frac{\rho-\sigma}{\rho-1}} \left(\frac{c_M(\omega)}{w}\right)^{1-\sigma} \\ &= \alpha^{\frac{\sigma-1}{\rho-1}} \left(\alpha \left(\frac{c_M(\omega)}{w}\right)^{1-\rho}\right)^{\frac{\rho-\sigma}{\rho-1}} \left(\frac{c_M(\omega)}{w}\right)^{1-\sigma} \\ &= \alpha \left(\frac{c_M(\omega)}{w}\right)^{1-\rho} \\ &= lhs. \end{aligned}$$

Therefore, the marginal cost $c(\omega)$ given by equation (B19) indeed satisfies equation (B18). $\square$

Second, derive the equation for the vector of demand shifters. Given equation (B19),

$$d(\omega) = n_S(\omega)^{-\varphi\sigma} \phi(q(\omega)) \frac{\alpha c_M(\omega)^{\sigma-\rho}}{\alpha c_M(\omega)^{1-\rho} + w^{1-\rho}} c(\omega) y(\omega).$$

Using the market clearing condition for ω ,

$$y(\omega) = \int_{B(\omega) \cup \{H\}} m(\omega, \omega') d\omega' = c(\omega)^{-\sigma} D + c(\omega)^{-\sigma} \int_{B(\omega)} \left(\frac{1 - \nu_S(\omega; \omega')}{1 + \nu_B(\omega'; \omega)}\right)^{-\sigma} d(\omega') d\omega'$$

Therefore,

$$d(\omega) = \frac{\phi(q(\omega))}{n_S(\omega)^{\varphi\sigma}} \frac{\alpha c_M(\omega)^{\sigma-\rho}}{c(\omega)^{\sigma-\rho}} z(\omega)^{\rho-1} \left(D + \int_{B(\omega)} \left(\frac{1 - \nu_S(\omega; \omega')}{1 + \nu_B(\omega'; \omega)}\right)^{-\sigma} d(\omega') d\omega'\right) \quad (\text{B20})$$

This concludes the proof. $\square$

Labor Demand. Given marginal costs and demand shifters, write labor demand $l(\omega)$ as

$$l(\omega) = \frac{w^{-\rho}}{\alpha c_M(\omega)^{1-\rho} + w^{1-\rho}} c(\omega) y(\omega) = \frac{w^{-\rho}}{\alpha c_M(\omega)^{\sigma-\rho}} \frac{d(\omega)}{\phi(q(\omega))/n_S(\omega)^{\varphi\sigma}}. \quad (\text{B21})$$

Price Index. Derive the marginal cost for final good firms, which is also the price index for final goods, P .

$$P = \frac{\partial}{\partial Y} \int_{\Omega} x(\omega, H) d\omega = \frac{\partial}{\partial Y} \int_{\Omega} \left(\frac{\sigma}{\sigma-1} c(\omega) \right)^{1-\sigma} \left(\frac{\sigma}{\sigma-1} \right)^{\sigma} \frac{Y}{P^{-\sigma}} d\omega,$$

which implies that

$$P = \left(\int_{\Omega} \left(\frac{\sigma}{\sigma-1} \right)^{\sigma} \left(\frac{\sigma}{\sigma-1} c(\omega) \right)^{1-\sigma} d\omega \right)^{\frac{1}{1-\sigma}}. \quad (\text{B22})$$

B.1.4 Efficient Bilateral Bargaining Outcomes

Consider the problem of a mediator that directly maximizes joint surplus between supplier $\omega' \in S(\omega)$ and buyer $\omega \in B(\omega')$.

$$\max_{m,x} \frac{\sigma}{\sigma-1} d(\omega)^{\frac{1}{\sigma}} m^{\frac{\sigma-1}{\sigma}} - c(\omega')m + \nu_B(\omega; \omega')x + \nu_S(\omega'; \omega)x$$

subject to incentive compatibility constraints:

$$\begin{aligned} x - c(\omega')m + \nu_B(\omega; \omega')x &\geq 0 \\ \frac{\sigma}{\sigma-1} d(\omega)^{\frac{1}{\sigma}} m^{\frac{\sigma-1}{\sigma}} - x + \nu_S(\omega'; \omega)x &\geq 0 \end{aligned}$$

Trade volume x must lie in between the bounds given by the two incentive compatibility constraints. Let

$$x = \iota \frac{\frac{\sigma}{\sigma-1}}{1 - \nu_S(\omega'; \omega)} d(\omega)^{\frac{1}{\sigma}} m^{\frac{\sigma-1}{\sigma}} + (1 - \iota) \frac{1}{1 + \nu_B(\omega; \omega')} c(\omega')m, \quad \iota \in [0, 1].$$

Notice that the optimization object is linear in x . Consider three cases:

1. If $\nu_B(\omega; \omega') + \nu_S(\omega'; \omega) > 0$, x is chosen to be the upper bound such that buyer incentive compatibility constraint binds (i.e., $\iota = 1$).
2. If $\nu_B(\omega; \omega') + \nu_S(\omega'; \omega) < 0$, x is chosen to be the lower bound such that supplier incentive compatibility constraint binds (i.e., $\iota = 0$).

3. If $\nu_B(\omega; \omega') + \nu_S(\omega'; \omega) = 0$, the optimization object does not depend on x , and any $\iota \in [0, 1]$ is optimal.

Plug x into the optimization object:

$$\begin{aligned} \max_m \quad & \left(1 + \iota \frac{\nu_B(\omega; \omega') + \nu_S(\omega'; \omega)}{1 - \nu_S(\omega'; \omega)} \right) \frac{\sigma}{\sigma - 1} d(\omega)^{\frac{1}{\sigma}} m^{\frac{\sigma-1}{\sigma}} \\ & - \left(1 - (1 - \iota) \frac{\nu_B(\omega; \omega') + \nu_S(\omega'; \omega)}{1 + \nu_B(\omega; \omega')} \right) c(\omega') m \end{aligned}$$

The first-order condition with respect to m is given by:

$$\frac{1}{1 - \nu_S(\omega'; \omega)} d(\omega)^{\frac{1}{\sigma}} m^{-\frac{1}{\sigma}} = \frac{1}{1 + \nu_B(\omega; \omega')} c(\omega')$$

This implies the optimal $m^{VI}(\omega', \omega)$ and $x^{VI}(\omega', \omega)$ satisfy:

$$m^{VI}(\omega', \omega) = \left(\frac{1 - \nu_S(\omega'; \omega)}{1 + \nu_B(\omega; \omega')} c(\omega') \right)^{-\sigma} d(\omega) \quad (\text{B23})$$

$$x^{VI}(\omega', \omega) = \left(\iota \frac{\sigma}{\sigma - 1} + 1 - \iota \right) \left(\frac{c(\omega')}{1 + \nu_B(\omega; \omega')} \right)^{1-\sigma} \left(\frac{1}{1 - \nu_S(\omega'; \omega)} \right)^{\sigma} d(\omega) \quad (\text{B24})$$

Compare the solutions with those from the bargaining equilibrium:

$$m^{VI}(\omega', \omega) = m^{BE}(\omega', \omega), \quad x^{VI}(\omega', \omega) = \frac{\iota \frac{\sigma}{\sigma-1} + 1 - \iota}{\kappa \frac{\sigma}{\sigma-1} + 1 - \kappa} x^{BE}(\omega', \omega).$$

This concludes the proof.

B.2 Small Open Economy

In this section, we provide the full setup of the small open economy model. For a firm ω , the production technology is characterized by equations (4) and (29). The learning technology is characterized by equations (6), (31), and (32). At the bargaining stage, the contract terms between domestic and foreign firms essentially follow Proposition 1. Note that the marginal cost of a foreign supplier is $c_F = \frac{w(F)}{z_S(F)}$ and the demand shifter of a foreign buyer is d_F . We assume that foreign firms do not learn from domestic firms. Therefore, $\nu_B(\omega; \omega') = 0$ for any domestic buyer ω and foreign supplier ω' as well as $\nu_S(\omega; \omega') = 0$ for any domestic supplier ω and foreign buyer ω' .

In a bargaining equilibrium, the marginal cost of a domestic firm ω is characterized by

equation (15) with the cost index for intermediate inputs modified as follows:

$$c_M(\omega) = \left(n_S(\omega)^{-\varphi\sigma} \int_{S(\omega)} \phi(q(\omega)) \left(\frac{\mu}{1 - \nu_S(\omega'; \omega)} \right)^\sigma \left(\frac{\mu}{1 + \nu_B(\omega; \omega')} c(\omega') \right)^{1-\sigma} + i(\omega) c_M^*(\omega)^{1-\sigma} \right)^{\frac{1}{1-\sigma}} \quad (\text{B25})$$

$$c_M^*(\omega) = \left(n_S^*(\omega)^{-\varphi\sigma} \int_{S^*(\omega)} \phi(F, q(\omega)) \left(\frac{\mu}{1 - \nu_S(\omega'; \omega)} \right)^\sigma (\mu c_F)^{1-\sigma} d\omega' \right)^{\frac{1}{1-\sigma}}. \quad (\text{B26})$$

$c_M^*(\omega)$ is the cost index for foreign inputs. The demand shifter of a domestic firm ω is characterized by:

$$d(\omega) = z(\omega)^{\rho-1} \frac{\alpha c_M(\omega)^{\sigma-\rho}}{c(\omega)^{\sigma-\rho}} \left(D + \int_{B(\omega)} \left(\frac{1 - \nu_S(\omega; \omega')}{1 + \nu_B(\omega'; \omega)} \right)^{-\sigma} \phi(q(\omega), q(\omega')) d(\omega') d\omega' + i(\omega) \int_{B^*(\omega)} \left(\frac{1}{1 + \nu_B(\omega'; \omega)} \right)^{-\sigma} \phi(q(\omega), F) d_F d\omega' \right). \quad (\text{B27})$$

The variable profits of a domestic firm ω are given by:

$$\begin{aligned} \pi^{var}(\omega) &= x(\omega, H) + \int_{B(\omega)} x(\omega, \omega') d\omega' + i(\omega) \int_{B^*(\omega)} x(\omega, \omega') d\omega' \\ &\quad - \int_{S(\omega)} x(\omega', \omega) d\omega' - i(\omega) \int_{S^*(\omega)} x(\omega', \omega) d\omega' - w l(\omega). \end{aligned} \quad (\text{B28})$$

The formation of domestic networks essentially follows the setup in Section 3.4. Note that the sufficient statistic for firm type is now $\omega = (z, i, q)$: whether a firm participates in the foreign market becomes an additional state variable and is payoff-relevant. The value of a firm at the bargaining stage, denoted $\tilde{V}(z, i, q)$, satisfies

$$\tilde{V}(z, i, q) = \pi^{var}(z, i, q) + \beta(1 - \delta) \int_{\mathbb{Z}} V(z') g(z'; z, i, q) dz'. \quad (\text{B29})$$

Denote $\hat{V}(z, i)$ as the expected value of a firm with productivity z after making the decision to enter the foreign market i . We can write $\hat{V}(z, i)$ as

$$\hat{V}(z, i) = \int_{\mathbb{Q}} Pr(q; z, i) \left(-Pf(q) + \tilde{V}(z, q, i) \right) dq, \quad (\text{B30})$$

where $Pr(q; z, i)$ is the probability that a firm at state (z, i) chooses quality q :

$$Pr(q; z, i) = \frac{\exp\left(\frac{\hat{V}(z, i, q) - Pf(q)}{P\zeta_f}\right)}{\int_{\mathbb{Q}} \exp\left(\frac{\hat{V}(z, i, q') - Pf(q')}{P\zeta_f}\right) dq'}. \quad (\text{B31})$$

It remains to characterize the decision to enter the foreign market and the formation of foreign networks. Each firm receives a draw of fixed costs f_F and decide whether to enter the foreign market. Firms choose to enter the foreign market if

$$\hat{V}(z, 1) - \hat{V}(z, 0) \geq Pf_F.$$

Assume f_F is distribution log-normal with mean μ_F and variance ζ_F^2 . Denote $\bar{f}_F(z) = \frac{\hat{V}(z, 1) - \hat{V}(z, 0)}{P}$ as the cutoff level of fixed costs for firms with productivity z . The probability of firms with productivity z entering the foreign market is

$$Pr(1; z) = \Phi\left(\frac{\log \bar{f}_F(z) - \mu_F}{\zeta_F}\right), \quad (\text{B32})$$

where Φ is the cumulative distribution function of the standard normal distribution.

For firms that enter the foreign market, they randomly match with foreign suppliers and buyers. The mass of domestic firms is $M_D = \int_{\mathbb{Z}} Pr(1; z)h(z)dz$. Given the mass of foreign firms M_F and a Cobb-Douglas matching function, each domestic firm randomly matches with $n_S^*(\omega) = A\left(\frac{M_F}{M_D}\right)^a$ foreign suppliers and $n_B^*(\omega) = A\left(\frac{M_F}{M_D}\right)^{1-a}$ foreign buyers. Therefore, $S^*(\omega) = [0, n_S^*(\omega)]$ and $B^*(\omega) = [0, n_B^*(\omega)]$.

Finally, write the expected value of a firm with productivity z , $V(z)$, as

$$V(z) = Pr(1; z) \left(-P\mathbb{E}_{f_F} [f_F | f_F \leq \bar{f}_F(z)] + \hat{V}(z, 1) \right) + Pr(0; z)\hat{V}(z, 0), \quad (\text{B33})$$

where

$$\mathbb{E}_{f_F} [f_F | f_F \leq \bar{f}_F(z)] = \exp\left(\mu_F + \frac{\zeta_F^2}{2}\right) \frac{\Phi\left(\frac{\log \bar{f}_F(z) - \mu_F - \zeta_F^2}{\zeta_F}\right)}{\Phi\left(\frac{\log \bar{f}_F(z) - \mu_F}{\zeta_F}\right)}$$

is the expected fixed cost conditional on entering the foreign market.

We provide the definition of a stationary equilibrium in the small open economy.

Definition 3. *Given parameters, a stationary equilibrium is a set of quantities $\{y(\omega), m(\omega, H), l(\omega)\}$ transfers $p(\omega, H), c(\omega), d(\omega, H), x(\omega, \omega'), \nu_S(\omega'; \omega), \nu_B(\omega'; \omega)$, prices $\{w(H), P(H)\}$ costs, demand shifters, learning rates, profits, value, firm distribution, such that:*

1. Bargaining reaches an equilibrium.
2. Firm value evolves according to equation (23) and (25).
3. Intermediate good markets clear:

$$y(\omega) = m(\omega, H) + \int_{B(\omega)} m(\omega, \omega') d\omega' + i(\omega) \int_{B^*(\omega)} m(\omega, \omega') d\omega', \quad \forall \omega \in \Omega. \quad (\text{B34})$$

4. Final good market clears:

$$PY = wL + \int_{\Omega} \pi^{var}(\omega) d\omega \quad (\text{B35})$$

5. Labor market clears:

$$L = \int_{\Omega} l(\omega) d\omega. \quad (\text{B36})$$

6. Trade is balanced:

$$\int_{\Omega} i(\omega) \int_{B^*(\omega)} x(\omega, \omega') d\omega' d\omega = \int_{\Omega} i(\omega) \int_{S^*(\omega)} x(\omega', \omega) d\omega' d\omega. \quad (\text{B37})$$

7. The productivity distribution of firms is stationary:

$$h(z') = (1 - \delta) \int_{\mathbb{Z}} \sum_{i=0}^1 \int_{\mathbb{Q}} g(z'; z, i, q) \underbrace{Pr(q; z, i) Pr(i; z) h(z)}_{h(z, i, q)} dq dz + \delta M h^E(z'), \quad \forall z' \in \mathbb{Z}. \quad (\text{B38})$$

B.3 Estimating Productivity

Proposition 2 implies the following closed form expressions for expenditure shares.

Proposition 3. *For firm ω , the expenditure shares on input from supplier ω' and labor are*

$$s(\omega', \omega) = \frac{\alpha c_M(\omega)^{\sigma-\rho}}{\alpha c_M(\omega)^{1-\rho} + w^{1-\rho}} \frac{\phi(q(\omega))}{n_S(\omega)^{\varphi\sigma}} \left(\frac{\mu}{1 - \nu_S(\omega'; \omega)} \right)^{\sigma} \left(\frac{\mu}{1 + \nu_B(\omega; \omega')} c(\omega') \right)^{1-\sigma} \quad (\text{B39})$$

$$s(L, \omega) = \frac{w^{1-\rho}}{\alpha c_M(\omega)^{1-\rho} + w^{1-\rho}}. \quad (\text{B40})$$

Proposition 3 will be useful in estimating productivity. Notice that equations (B19) and

(B40) imply that the marginal cost can be written as:

$$c(\omega) = \frac{1}{z(\omega)} (\alpha c_M(\omega)^{1-\rho} + w^{1-\rho})^{\frac{1}{1-\rho}} = \frac{w}{z(\omega)} s(L, \omega)^{\frac{1}{\rho-1}},$$

which gives the following expression for $z(\omega)$:

$$z(\omega) = \frac{w}{c(\omega)} s(L, \omega)^{\frac{1}{\rho-1}}.$$

From equation (B15), sales to final good firms contain information on marginal costs:

$$c(\omega) = \left(\frac{\sigma - 1}{\sigma} \frac{x(\omega, H)}{D} \right)^{\frac{1}{1-\sigma}}.$$

Taken together, we use the following structural relationship to back out productivity up to a general equilibrium constant that depends on wage w and final demand shifter D :

$$z(\omega) \propto x(\omega, H)^{\frac{1}{\sigma-1}} s(L, \omega)^{\frac{1}{\rho-1}}. \quad (\text{B41})$$

B.4 Solution Algorithm

B.5 Welfare Decomposition

We measure welfare of workers and entrepreneurs as their present value of consumption. Formally, aggregate welfare ($\mathcal{U}$), entrepreneur welfare ($\mathcal{V}$), and worker welfare ($\mathcal{W}$) can be expressed as:

$$\mathcal{U} = \mathcal{V} + \mathcal{W}, \quad \mathcal{V} = \int_{\mathbb{Z}} \frac{V(z)}{P} h(z) dz, \quad \mathcal{W} = \frac{wL/(1-\beta)}{P}.$$

As all agents have linear utility and have constant marginal utility of consumption, aggregate welfare is simply the sum of the welfare of individual agents.

For any change in model fundamentals, we employ the hat algebra technique following Dekle et al. (2008) to compute the change in welfare. We are particularly interested in how entrepreneur welfare reacts to changes in fundamentals and the sources of these changes. To this end, we decompose entrepreneur welfare change into multiple components.

Entrepreneur Welfare. The welfare of entrepreneurs with productivity z is $\frac{V(z)}{P}$.

1. Decompose $V(z)$: Recall that

$$\begin{aligned}
V(z) &= \sum_{i=0}^1 \int_{\mathbb{Q}} Pr(i; z) Pr(q; z, i) \tilde{V}(z, i, q) dq \\
&= \sum_{i=0}^1 \int_{\mathbb{Q}} \underbrace{Pr(i; z) Pr(q; z, i)}_{\text{network}} \left(\underbrace{\underbrace{-f(z, i)}_{\text{fixed costs}} + \underbrace{\pi^{var}(z, i, q)}_{\text{variable profits}}}_{\text{static}} + \underbrace{\beta(1 - \delta) \int_{\mathbb{Z}} \underbrace{V(z')}_{\text{future value}} \underbrace{g(z'; z, i, q)}_{\text{growth}} dz'}_{\text{dynamic}} \right) dq
\end{aligned}$$

Denote

$$\varpi_V(i, q; z) = \frac{Pr(i; z) Pr(q; z, i) \tilde{V}(z, i, q)}{V(z)}.$$

Changes in $V(z)$ are given by

$$\hat{V}(z) = \sum_{i=0}^1 \int_{\mathbb{Q}} \varpi_V(i, q; z) \hat{P}r(i; z) \hat{P}r(q; z, i) \hat{\tilde{V}}(z, i, q) dq$$

For small changes, use a first-order approximation to write

$$d \log V(z) = \sum_{i=0}^1 \int_{\mathbb{Q}} \varpi_V(i, q; z) \left(d \log Pr(i; z) + d \log Pr(q; z, i) + d \log \tilde{V}(z, i, q) \right) dq \quad (\text{B42})$$

The component $d \log Pr(i; z)$ captures investment in connecting with foreign partners, and $d \log Pr(q; z, i)$ captures investment in connecting with domestic partners.

2. Decompose $\tilde{V}(z, i, q)$: Denote

$$\begin{aligned}
\varpi_{\tilde{V},f}(z, i, q) &= \frac{-f(z, i)}{\tilde{V}(z, i, q)} \\
\varpi_{\tilde{V},\pi}(z, i, q) &= \frac{\pi^{var}(z, i, q)}{\tilde{V}(z, i, q)} \\
\varpi_{\tilde{V},v}(z, i, q) &= \frac{v(z, i, q)}{\tilde{V}(z, i, q)} = \frac{\beta(1 - \delta) \int_{\mathbb{Z}} V(z') g(z'; z, i, q) dz'}{\tilde{V}(z, i, q)}
\end{aligned}$$

Changes in $\tilde{V}(z, i, q)$ are given by

$$\hat{\tilde{V}}(z, i, q) = \varpi_{\tilde{V},f}(z, i, q) \hat{f}(z, i) + \varpi_{\tilde{V},\pi}(z, i, q) \hat{\pi}^{var}(z, i, q) + \varpi_{\tilde{V},v}(z, i, q) \hat{v}(z, i, q)$$

For small changes, use a first-order approximation to write

$$\begin{aligned} d \log \tilde{V}(z, i, q) &= \varpi_{\tilde{V},f}(z, i, q) d \log f(z, i) + \varpi_{\tilde{V},\pi}(z, i, q) d \log \pi^{var}(z, i, q) \\ &\quad + \varpi_{\tilde{V},v}(z, i, q) d \log v(z, i, q) \end{aligned} \quad (\text{B43})$$

3. Decompose $v(z, i, q)$: Denote

$$\varpi_v(z'; z, i, q) = \frac{\beta(1 - \delta)V(z')g(z'; z, i, q)}{v(z, i, q)}$$

Changes in $v(z, i, q)$ are given by

$$\hat{v}(z, i, q) = \int_{\mathbb{Z}} \varpi_v(z'; z, i, q) \hat{V}(z') \hat{g}(z'; z, i, q) dz'$$

For small changes, use a first-order approximation to write

$$d \log v(z, i, q) = \int_{\mathbb{Z}} \varpi_v(z'; z, i, q) (d \log V(z') + d \log g(z'; z, i, q)) dz' \quad (\text{B44})$$

4. Present value representation: Combine equations (B42), (B43), and (B44) to obtain a recursive system for $d \log V(z)$:

$$\begin{aligned} d \log V(z) &= \underbrace{\sum_{i=0}^1 \int_{\mathbb{Q}} \varpi_V(i, q; z) (d \log Pr(i; z) + d \log Pr(q; z, i)) dq}_{\text{network, } d \log Pr(z)} \\ &\quad + \underbrace{\sum_{i=0}^1 \int_{\mathbb{Q}} \varpi_V(i, q; z) \varpi_{\tilde{V},f}(z, i, q) d \log f(z, i) dq}_{\text{fixed costs, } d \log f(z)} \\ &\quad + \underbrace{\sum_{i=0}^1 \int_{\mathbb{Q}} \varpi_V(i, q; z) \varpi_{\tilde{V},\pi}(z, i, q) d \log \pi^{var}(z, i, q) dq}_{\text{variable profits, } d \log \pi^{var}(z)} \\ &\quad + \underbrace{\sum_{i=0}^1 \int_{\mathbb{Q}} \varpi_V(i, q; z) \varpi_{\tilde{V},v}(z, i, q) \left(\int_{\mathbb{Z}} \varpi_v(z'; z, i, q) d \log g(z'; z, i, q) dz' \right) dq}_{\text{growth, } d \log g(z)} \\ &\quad + \underbrace{\int_{\mathbb{Z}} \left(\sum_{i=0}^1 \int_{\mathbb{Q}} \varpi_V(i, q; z) \varpi_{\tilde{V},v}(z, i, q) \varpi_v(z'; z, i, q) dq \right) d \log V(z') dz'}_{\text{transition matrix, } \psi(z, z')} \end{aligned}$$

In matrix form, write

$$d \log V = (\mathbb{I}_{\mathbb{Z}} - \Psi)^{-1} (d \log Pr + d \log f + d \log \pi^{var} + d \log g)$$

where $\Psi = (\psi(z, z'))$ is the transition matrix. Each term on the right-hand side represents the present value of changes in a given component holding other components fixed.

The change in welfare of entrepreneurs with productivity z is simply $d \log \frac{V(z)}{P} = d \log V(z) - d \log P$. Aggregate entrepreneur welfare is given by

$$\mathcal{V} = \int_{\mathbb{Z}} h(z) \frac{V(z)}{P} dz$$

Denote $\varpi_{\mathcal{V}}(z) = \frac{h(z)V(z)/P}{\mathcal{V}}$. Changes in aggregate entrepreneur welfare are given by

$$d \log \mathcal{V} = \int_{\mathbb{Z}} \varpi_{\mathcal{V}}(z) (d \log h(z) + d \log V(z) - d \log P) dz$$

One can also decompose entrepreneur welfare with productivity z and international trade participation i , $\hat{V}(z, i)$:

In the remainder of this section, we further decompose changes in variable profits and price index.

Marginal Cost. Recall that marginal cost of a firm at state $\omega = (z, i, q)$ is given by

$$c(\omega) = \frac{1}{z(\omega)} (\alpha c_M(\omega)^{1-\rho} + w^{1-\rho})^{\frac{1}{1-\rho}}$$

where

$$\begin{aligned} c_M(\omega) &= (c_D(\omega)^{1-\sigma} + i(\omega) \alpha_F c_F(\omega)^{1-\sigma})^{\frac{1}{1-\sigma}} \\ c_D(\omega) &= \left(\int_{\mathbb{X}} \frac{n_S(\omega'; \omega)}{n_S(\omega)^{\varphi\sigma}} \phi(q(\omega)) \left(\frac{\mu}{1 - \nu_S(\omega'; \omega)} \right)^{\sigma} \left(\frac{\mu}{1 + \nu_B(\omega; \omega')} \right)^{1-\sigma} c(\omega')^{1-\sigma} d\omega' \right)^{\frac{1}{1-\sigma}} \\ c_F(\omega) &= \left(n_S(F; \omega)^{1-\varphi\sigma} \phi(q(\omega)) \left(\frac{\mu}{1 - \nu_S(F; \omega)} \right)^{\sigma} \mu^{1-\sigma} c_F^{1-\sigma} \right)^{\frac{1}{1-\sigma}}. \end{aligned}$$

In changes,

$$\hat{c}(\omega) = (s(M, \omega) \hat{c}_M(\omega)^{1-\rho} + s(L, \omega) \hat{w}^{1-\rho})^{\frac{1}{1-\rho}},$$

where $s(M, \omega)$ and $s(L, \omega)$ are expenditure shares on intermediate inputs and labor in the baseline equilibrium. The change in the cost index for intermediate inputs is given by

$$\begin{aligned}\hat{c}_M(\omega)^{1-\sigma} &= \int_{\mathbb{X}} s_M(\omega', \omega) \frac{\hat{n}_S(\omega'; \omega)}{\hat{n}_S(\omega)^{\varphi\sigma}} \left(\frac{\widehat{1}}{1 - \nu_S(\omega'; \omega)} \right)^\sigma \left(\frac{\widehat{1}}{1 + \nu_B(\omega; \omega')} \right)^{1-\sigma} \hat{c}(\omega')^{1-\sigma} d\omega' \\ &\quad + s_M(F, \omega) \hat{n}_S(F; \omega)^{1-\varphi\sigma} \left(\frac{\widehat{1}}{1 - \nu_S(F; \omega)} \right)^\sigma \hat{c}_F^{1-\sigma}\end{aligned}$$

where $s_M(\omega', \omega)$ and $s_M(F, \omega)$ are the domestic and foreign supplier shares of total intermediate input expenditures in the baseline equilibrium. For small changes, we can use a first-order approximation around the baseline equilibrium:

$$d \log c(\omega) \approx s(M, \omega) d \log c_M(\omega) + s(L, \omega) d \log w$$

where

$$\begin{aligned}d \log c_M(\omega) &\approx \int_{\mathbb{X}} s_M(\omega', \omega) \left(\frac{1}{1 - \sigma} d \log \frac{n_S(\omega'; \omega)}{n_S(\omega)^{\varphi\sigma}} + \frac{1}{1 - \sigma} d \log \frac{(1 - \nu_S(\omega'; \omega))^{-\sigma}}{(1 + \nu_B(\omega; \omega'))^{1-\sigma}} + d \log c(\omega') \right) d\omega' \\ &\quad + s_M(F, \omega) \left(\frac{1}{1 - \sigma} d \log n_S(F; \omega)^{1-\varphi\sigma} + \frac{1}{1 - \sigma} d \log (1 - \nu_S(F; \omega))^{-\sigma} + d \log c_F \right)\end{aligned}$$

In equilibrium, $n_S(\omega) = A$ for all $\omega \in \mathbb{X}$. When $\varphi = \frac{1}{\sigma}$, $n_S(F; \omega)^{1-\varphi\sigma} = 1$. We can simplify the above expression to:

$$\begin{aligned}d \log c(\omega) &\approx \underbrace{\int_{\mathbb{X}} s(\omega', \omega) d \log c(\omega') d\omega' + s(F, \omega) d \log c_F + s(L, \omega) d \log w}_{\text{supplier cost}} \\ &\quad + \underbrace{\frac{1}{1 - \sigma} \int_{\mathbb{X}} s(\omega', \omega) d \log n_S(\omega'; \omega) d\omega'}_{\text{match}} \\ &\quad + \underbrace{\frac{1}{1 - \sigma} \left(\underbrace{\int_{\mathbb{X}} s(\omega', \omega) d \log \frac{(1 - \nu_S(\omega'; \omega))^{-\sigma}}{(1 + \nu_B(\omega; \omega'))^{1-\sigma}} d\omega'}_{\text{domestic contract terms}} + \underbrace{s(F, \omega) d \log (1 - \nu_S(F; \omega))^{-\sigma}}_{\text{foreign contract terms}} \right)}_{\text{contract terms}}\end{aligned}$$

Notice that the equation above is a recursive system in $d \log c(\omega)$. We can compute the total contribution of each component using the Leontief inverse of the firm-to-firm expenditure

share matrix:

$$d \log c = (\mathbb{I}_{\mathbb{X}} - \Sigma)^{-1} (\text{factor cost} + \text{match} + \text{contract terms}),$$

where $\Sigma = (s(\omega', \omega))_{\omega, \omega' \in \mathbb{X}}$ is the firm-to-firm expenditure share matrix.

Demand Shifter. Recall that the demand shifter of a firm at state ω is:

$$\begin{aligned} d(\omega) &= n_S(\omega)^{-\varphi\sigma} \frac{\alpha c_M(\omega)^{\sigma-\rho}}{c(\omega)^{\sigma-\rho}} z(\omega)^{\rho-1} \left(D + \int_{\mathbb{X}} n_B(\omega'; \omega) \left(\frac{1 - \nu_S(\omega; \omega')}{1 + \nu_B(\omega'; \omega)} \right)^{-\sigma} \phi(q(\omega)) d(\omega') d\omega' \right. \\ &\quad \left. + i(\omega) n_B(F; \omega) \left(\frac{1}{1 + \nu_B(F; \omega)} \right)^{-\sigma} \phi(q(\omega)) d_F \right) \end{aligned}$$

In changes:

$$\begin{aligned} \hat{d}(\omega) &= \frac{\hat{c}_M(\omega)^{\sigma-\rho}}{\hat{c}(\omega)^{\sigma-\rho}} \left(\frac{D}{D(\omega)} \hat{D} + \int_{\mathbb{X}} \frac{D(\omega, \omega')}{D(\omega)} \hat{n}_B(\omega'; \omega) \left(\frac{1 - \widehat{\nu_S(\omega; \omega')}}{1 + \nu_B(\omega'; \omega)} \right)^{-\sigma} \hat{d}(\omega') d\omega' \right. \\ &\quad \left. + \frac{D(\omega, F)}{D(\omega)} \hat{n}_B(F; \omega) \left(\frac{1}{1 + \nu_B(F; \omega)} \right)^{-\sigma} \hat{d}_F \right), \end{aligned}$$

where

$$\begin{aligned} D(\omega) &= D + \int_{\mathbb{X}} n_B(\omega'; \omega) \left(\frac{1 - \nu_S(\omega; \omega')}{1 + \nu_B(\omega'; \omega)} \right)^{-\sigma} \phi(q(\omega)) d(\omega') d\omega' \\ &\quad + i(\omega) n_B(F; \omega) \left(\frac{1}{1 + \nu_B(F; \omega)} \right)^{-\sigma} \phi(q(\omega)) d_F \\ D(\omega, \omega') &= n_B(\omega'; \omega) \left(\frac{1 - \nu_S(\omega; \omega')}{1 + \nu_B(\omega'; \omega)} \right)^{-\sigma} \phi(q(\omega)) d(\omega') \\ D(\omega, F) &= i(\omega) n_B(F; \omega) \left(\frac{1}{1 + \nu_B(F; \omega)} \right)^{-\sigma} \phi(q(\omega)) d_F \end{aligned}$$

For small changes, we can use a first-order approximation:

$$\begin{aligned}
d \log d(\omega) &\approx (\sigma - \rho) d \log c_M(\omega) - (\sigma - \rho) d \log c(\omega) \\
&+ \underbrace{\frac{D}{D(\omega)} d \log D + \int_{\mathbb{X}} \frac{D(\omega, \omega')}{D(\omega)} d \log d(\omega') d\omega' + \frac{D(\omega, F)}{D(\omega)} d \log d_F}_{\text{buyer demand}} \\
&+ \underbrace{\int_{\mathbb{X}} \frac{D(\omega, \omega')}{D(\omega)} d \log n_B(\omega'; \omega) d\omega'}_{\text{match}} \\
&+ \underbrace{\int_{\mathbb{X}} \frac{D(\omega, \omega')}{D(\omega)} d \log \frac{(1 - \nu_S(\omega; \omega'))^{-\sigma}}{(1 + \nu_B(\omega'; \omega))^{-\sigma}} d\omega'}_{\text{domestic contract terms}} + \underbrace{\frac{D(\omega, F)}{D(\omega)} d \log \frac{1}{(1 + \nu_B(F; \omega))^{-\sigma}}}_{\text{foreign contract terms}} \\
&\underbrace{\hspace{10em}}_{\text{contract terms}}
\end{aligned}$$

Notice that

$$\begin{aligned}
d \log c_M(\omega) - d \log c(\omega) &= \frac{1}{s(M, \omega)} d \log c(\omega) - \frac{s(L, \omega)}{s(M, \omega)} d \log w - d \log c(\omega) \\
&= \frac{s(L, \omega)}{s(M, \omega)} (d \log c(\omega) - d \log w)
\end{aligned}$$

Similarly, we can compute the total contribution of each component using the Leontief inverse of the firm-to-firm demand share matrix.

Variable Profits. Recall that the variable profits of a firm ω is:

$$\begin{aligned}
\pi^{var}(\omega) &= \underbrace{\left(1 - \frac{1}{\mu(\omega, H)}\right) x(\omega, H)}_{\pi^{var}(\omega, H)} + \underbrace{\int_{\mathbb{X}} n_B(\omega'; \omega) \left(1 - \frac{1}{\mu(\omega, \omega')}\right) x(\omega, \omega') d\omega'}_{\pi^{var}(\omega, \omega')} \\
&+ \underbrace{i(\omega) n_B(F; \omega) \left(1 - \frac{1}{\mu(\omega, F)}\right) x(\omega, F)}_{\pi^{var}(\omega, F)}
\end{aligned}$$

The change in variable profits is:

$$\begin{aligned}
\hat{\pi}^{var}(\omega) &= \frac{\pi^{var}(\omega, H)}{\pi^{var}(\omega)} \hat{x}(\omega, H) + \int_{\mathbb{X}} \frac{\pi^{var}(\omega, \omega')}{\pi^{var}(\omega)} \hat{n}_B(\omega'; \omega) \left(1 - \widehat{\frac{1}{\mu(\omega, \omega')}}\right) \hat{x}(\omega, \omega') d\omega' \\
&+ \frac{\pi^{var}(\omega, F)}{\pi^{var}(\omega)} \hat{n}_B(F; \omega) \left(1 - \widehat{\frac{1}{\mu(\omega, F)}}\right) \hat{x}(\omega, F)
\end{aligned}$$

For small changes,

$$\begin{aligned}
d \log \pi^{var}(\omega) &\approx \frac{\pi^{var}(\omega, H)}{\pi^{var}(\omega)} d \log x(\omega, H) \\
&+ \int_{\mathbb{X}} \frac{\pi^{var}(\omega, \omega')}{\pi^{var}(\omega)} \left[d \log n_B(\omega'; \omega) + d \log \left(1 - \frac{1}{\mu(\omega, \omega')} \right) + d \log x(\omega, \omega') \right] d\omega' \\
&+ \frac{\pi^{var}(\omega, F)}{\pi^{var}(\omega)} \left[d \log n_B(F; \omega) + d \log \left(1 - \frac{1}{\mu(\omega, F)} \right) + d \log x(\omega, F) \right]
\end{aligned}$$

where

$$\begin{aligned}
d \log x(\omega, H) &= (1 - \sigma) d \log c(\omega) + d \log D \\
d \log x(\omega, \omega') &= (1 - \sigma) d \log c(\omega) + d \log \frac{(1 - \nu_S(\omega; \omega'))^{-\sigma}}{(1 + \nu_B(\omega'; \omega))^{1-\sigma}} + d \log d(\omega') \\
d \log x(\omega, F) &= (1 - \sigma) d \log c(\omega) + d \log \frac{1}{(1 + \nu_B(F; \omega))^{1-\sigma}} + d \log d_F
\end{aligned}$$

We have derived the total contribution of each component to changes in marginal cost and demand shifter. Plug them into the equation above to obtain the total contribution of each component to changes in variable profits.

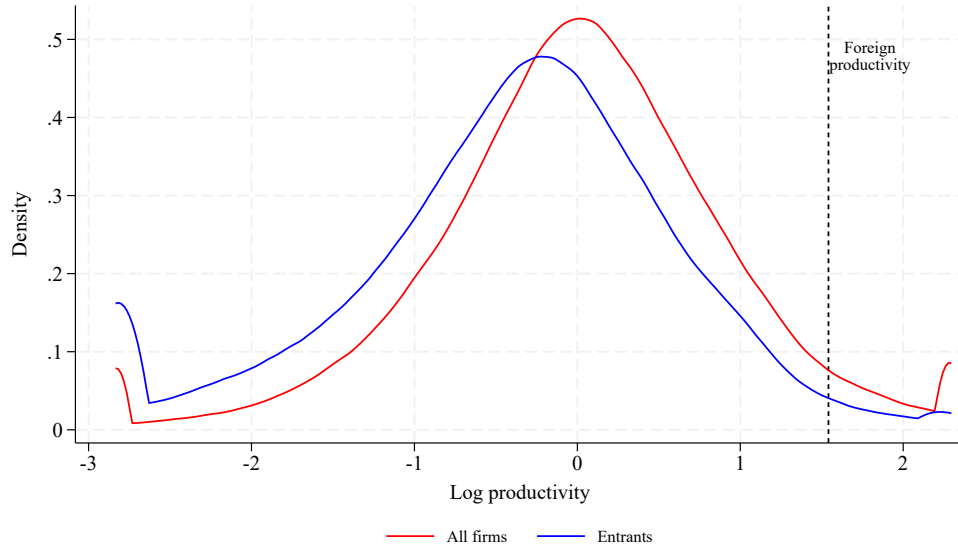
C Additional Quantitative Results

Table C1: Productivity and learning technology parameters, demeaned productivity

Parameter	Name	Value
ϑ_e	Entrant log productivity distribution, mean	-0.43
ς_e	Entrant log productivity distribution, s.d.	0.97
λ_O	Constant	-0.02
γ	Persistence	0.92
λ_S	Supplier share	0.002
λ_B	Buyer share	0.002
ζ	Substitutability, outer nest	1.21
η	Substitutability, inner nest	197.84
ς_ξ	Productivity shock, s.d.	0.54
$\log z_F$	Log foreign productivity	1.29

Notes: Parameters are estimated using productivity demeaned by NACE4 industries and locations.

Figure C1: Productivity distribution



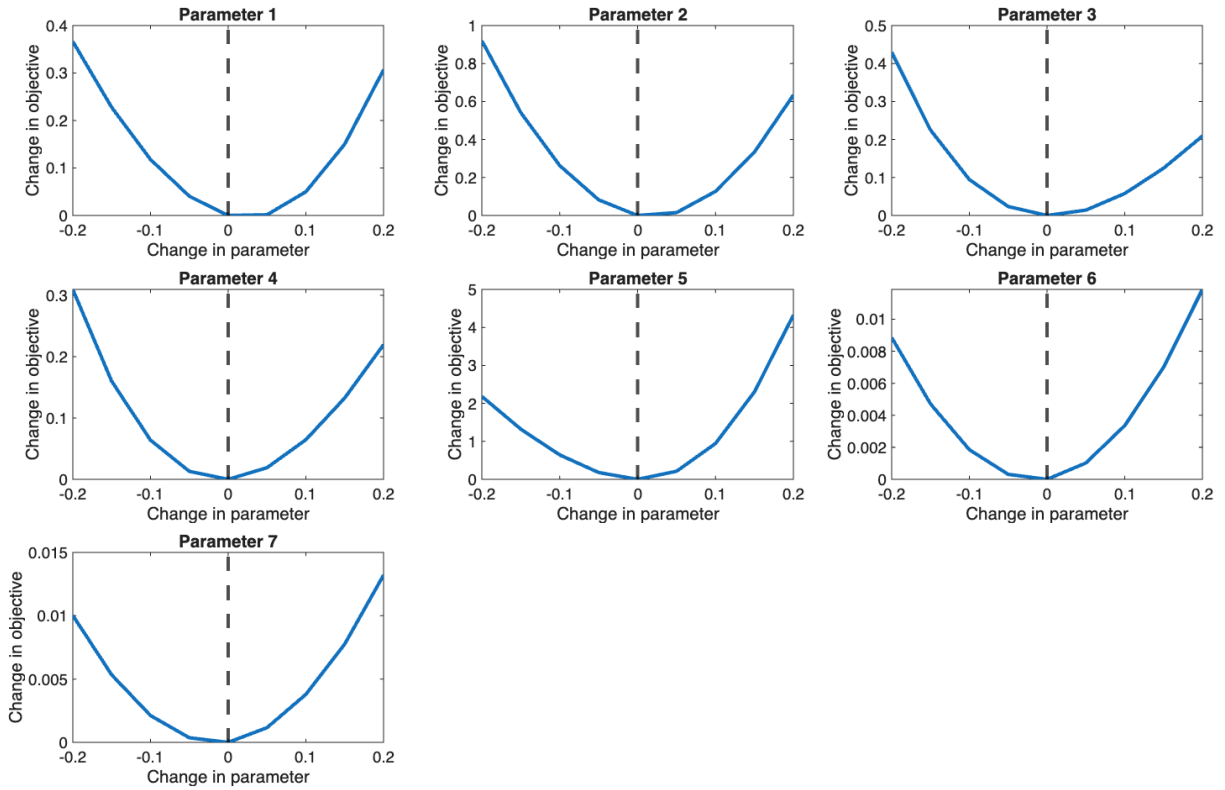
Notes: The figure plots the distribution of estimated productivity for all firms (red line) and entrants (blue line). Estimated productivity for foreign suppliers and buyers is indicated by the dashed vertical line.

Table C2: Correlation between estimated productivity and firm outcomes

Variable	Correlation with estimated productivity
Log total sales	0.48
Log value added	0.67
Log variable profits	0.51
Log employment	0.74
Log indegree	0.28
Log outdegree	0.21
Importer	0.20
Exporter	0.18
R&D	0.17

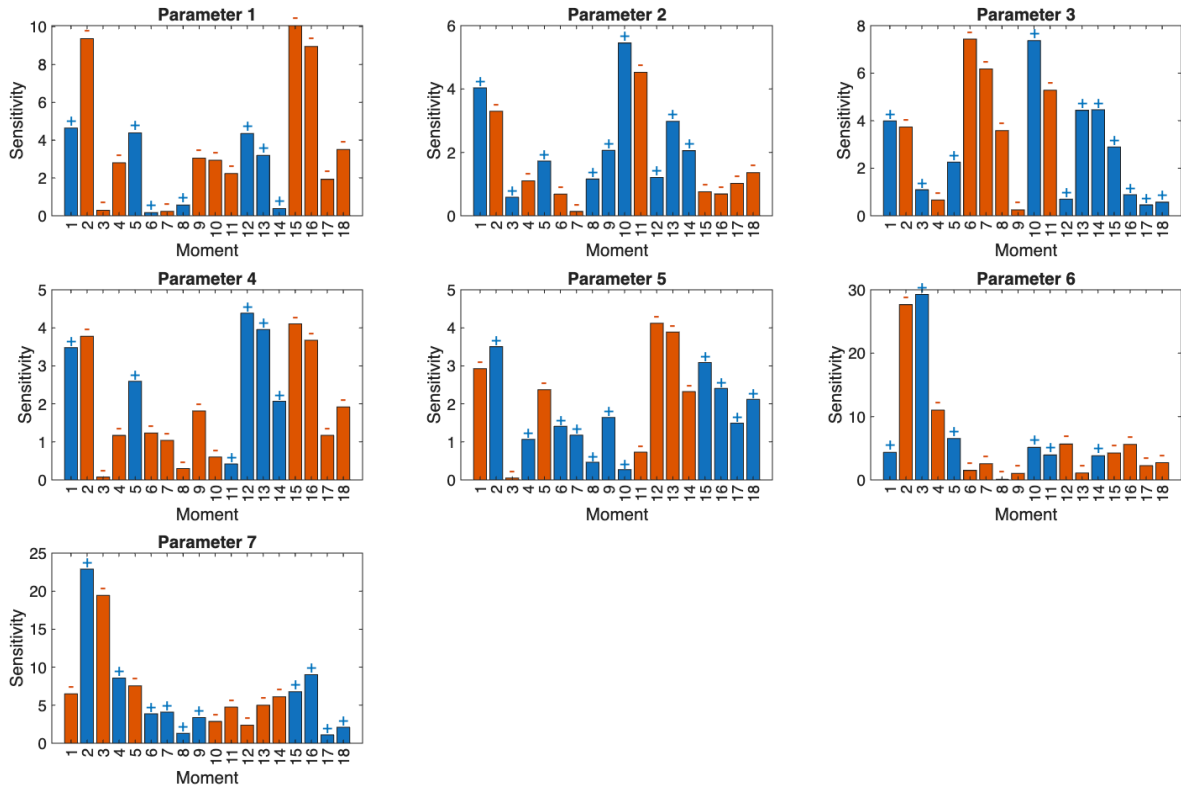
Notes: This table reports the correlation between estimated productivity and various firm outcomes in the data.

Figure C2: Perturbing parameters around the estimated point



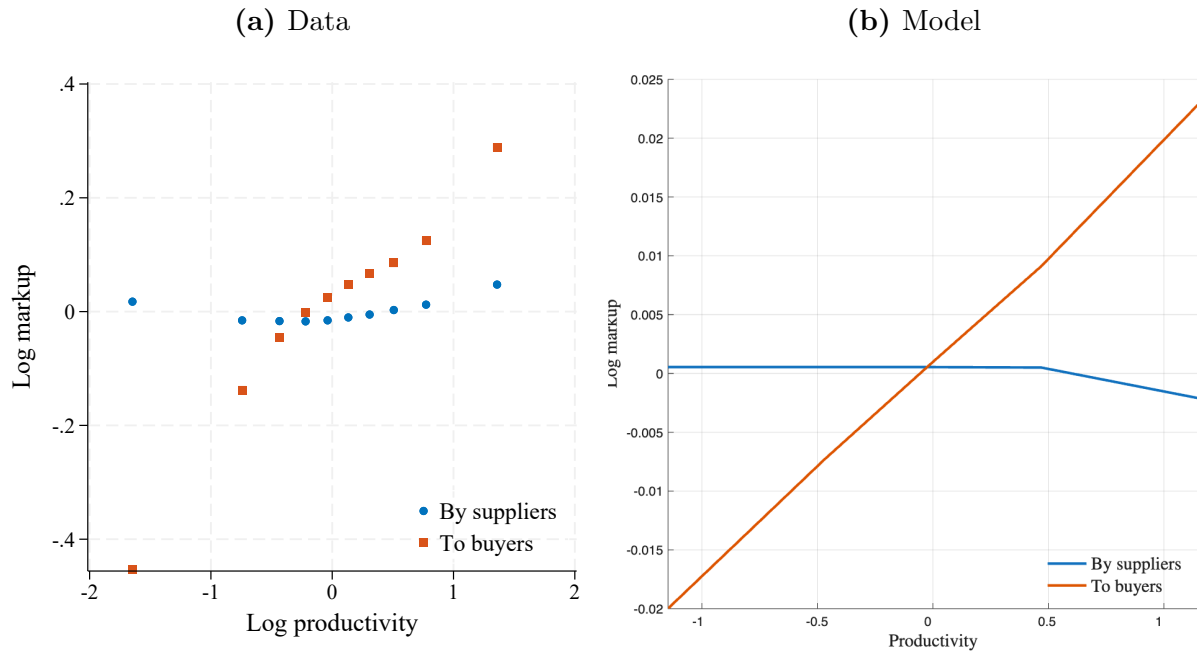
Notes: The figure plots the change in SMM objective function value when perturbing each parameter around the estimated point while holding other parameters fixed.

Figure C3: Sensitivity of parameters to moments



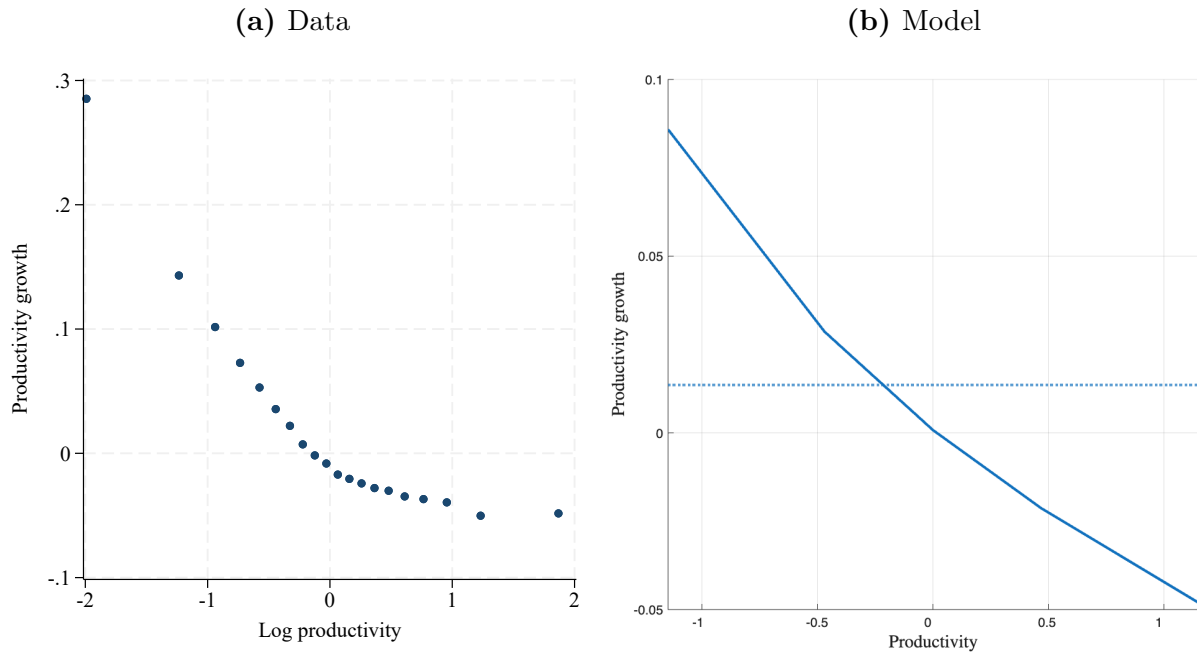
Notes: The figure plots the sensitivity of each estimated parameter to each moment used in estimation, following Andrews et al. (2017).

Figure C4: Markup and productivity, data and model



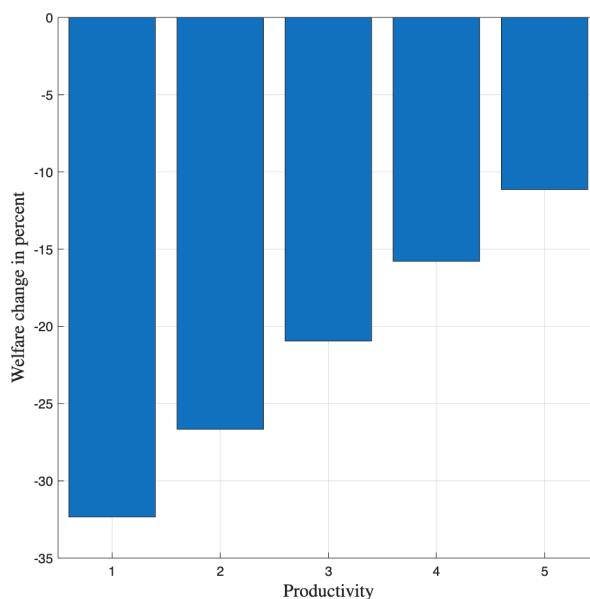
Notes: The figure plots the relationship between average markup by suppliers or to buyers and productivity in the data (Panel (a)) and in the model (Panel (b)). Markups are computed from total sales divided by total variable costs.

Figure C5: Productivity growth, data and model



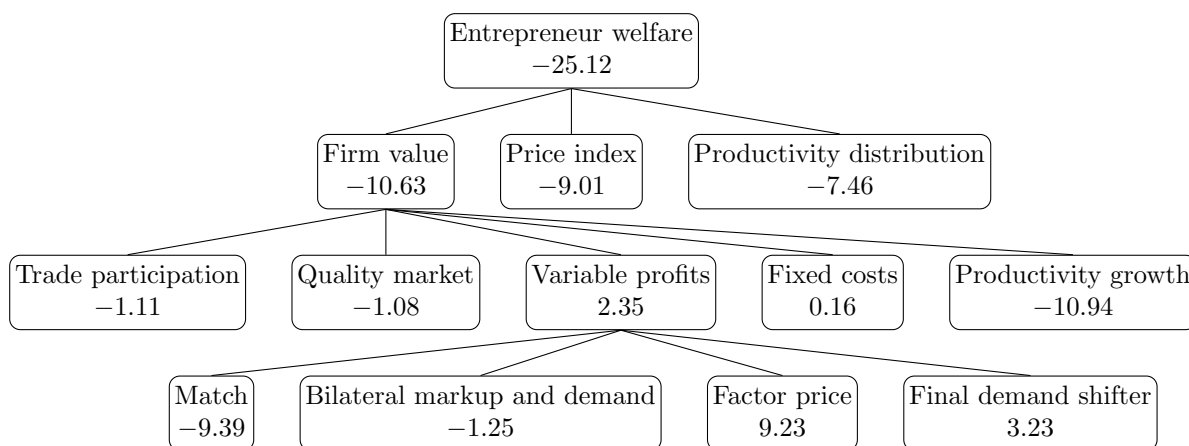
Notes: The figure plots the relationship between productivity growth and productivity in the data (Panel (a)) and in the model (Panel (b)).

Figure C6: Value of learning: heterogeneous welfare effects by productivity



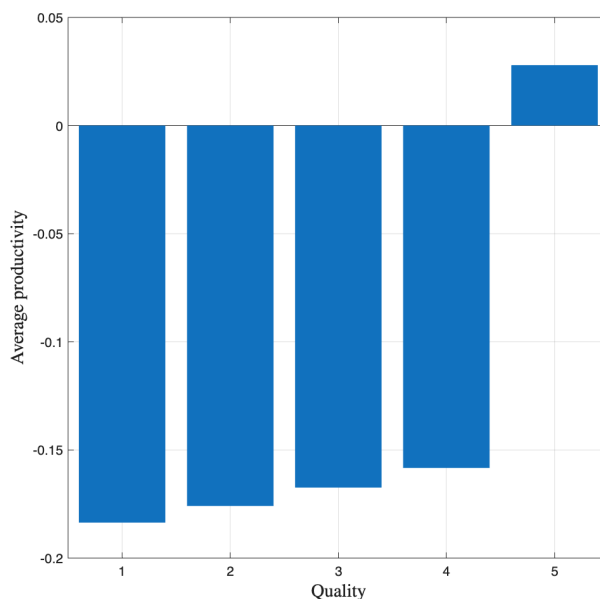
Notes: The figure plots the change in entrepreneur welfare by productivity in the counterfactual with no learning.

Figure C7: Value of learning: entrepreneur welfare change decomposition details



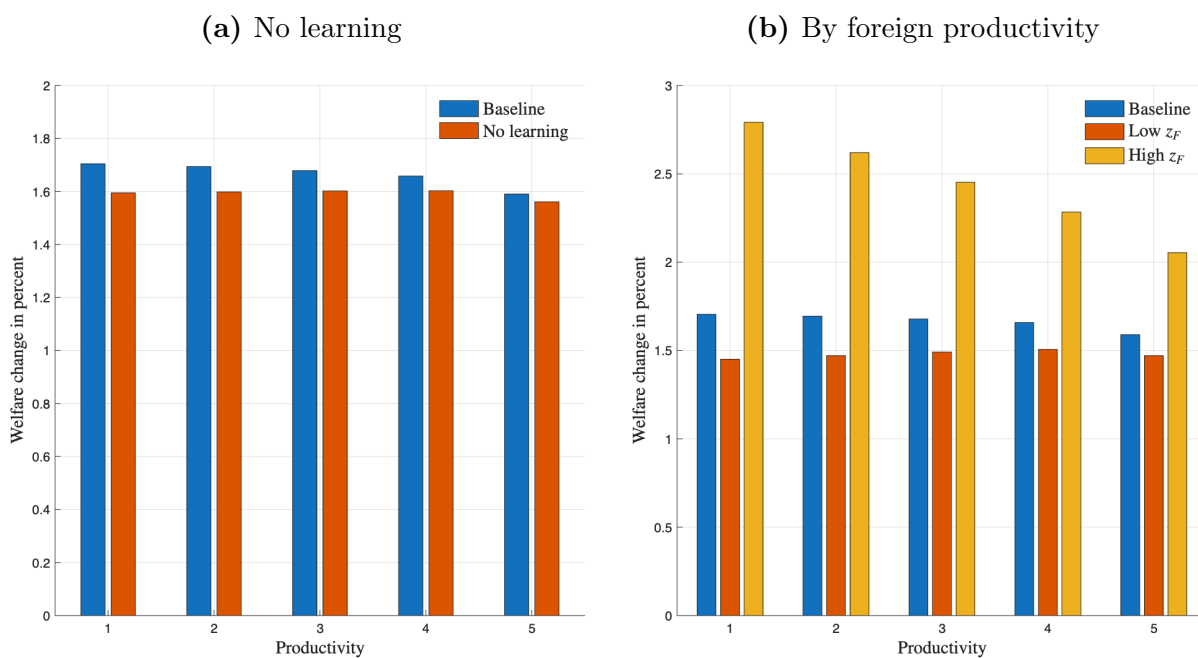
Notes: The figure presents a decomposition of the change in entrepreneur welfare in the counterfactual with no learning.

Figure C8: Reductions in international trade participation costs: average productivity by quality



Notes: The figure plots the change in average productivity of firms with each quality in the counterfactual with reductions in international trade participation costs.

Figure C9: Reductions in international trade participation costs: heterogeneous welfare effects by productivity



Notes: The figure plots the change in entrepreneur welfare by productivity in the counterfactual with reductions in international trade participation costs. Panel (a) compares the baseline changes with those in the case with no learning. Panel (b) compares the baseline changes with those under different levels of foreign productivity.

Table C3: Reductions in international trade participation costs: changes in welfare and firm outcomes

	Baseline (1)	No learning (2)	Low z_F (3)	High z_F (4)
<i>Panel A: Welfare changes, percent</i>				
Aggregate welfare	1.18	1.09	1.01	1.83
Entrepreneur welfare	1.67	1.58	1.46	2.52
Worker welfare	0.96	0.89	0.81	1.48
<i>Panel B: Firm outcome changes, percentage point</i>				
Productivity growth	0.01	-0.00	-0.01	0.05
International trade	0.83	0.66	0.63	1.02
Quality	0.10	0.07	0.09	0.22
Average log productivity	0.05	0.00	-0.04	0.46
<i>Panel C: Entrepreneur welfare change decomposition</i>				
Productivity growth	-0.02	0.00	-0.02	-0.02
Network adjustment	1.72	1.66	1.52	2.39
Bilateral markup and demand	0.01	0.00	0.01	0.01
General equilibrium	-0.03	-0.08	-0.06	0.11

Notes: The table presents the changes in welfare and firm outcomes in the counterfactual with reductions in international trade participation costs. Column (1) presents the baseline results. Column (2) presents results with no learning. Columns (3) and (4) present results under low and higher foreign productivity, respectively.

Table C4: Reductions in international trade participation costs: entrepreneur welfare change decomposition details

	Baseline (1)	No learning (2)	Low z_F (3)	High z_F (4)
Entrepreneur welfare	1.67	1.58	1.46	2.52
(1) Firm value	1.21	1.18	1.12	1.61
(1.1) International trade	0.92	0.87	0.85	1.22
(1.2) Quality	0.09	0.09	0.09	0.11
(1.3) Variable profits	0.10	0.10	0.09	0.16
(1.3.1) Match	0.58	0.57	0.47	0.92
(1.3.2) Bilateral markup and demand	0.01	0.00	0.01	0.01
(1.3.3) Factor price	-0.56	-0.55	-0.49	-0.83
(1.3.4) Final demand shifter	0.07	0.08	0.09	0.04
(1.4) Fixed costs	0.12	0.13	0.12	0.14
(1.5) Productivity growth	-0.02	0.00	-0.02	-0.02
(2) Price index	0.43	0.39	0.36	0.67
(3) Productivity distribution	0.03	0.00	-0.02	0.22

Notes: The table presents a decomposition of the change in entrepreneur welfare in the counterfactual with reductions in international trade participation costs. Column (1) presents the baseline results. Column (2) presents results with no learning. Columns (3) and (4) present results under low and higher foreign productivity, respectively.